

Ocean carbon cycle: Past, Present and Future

Jack Middelburg

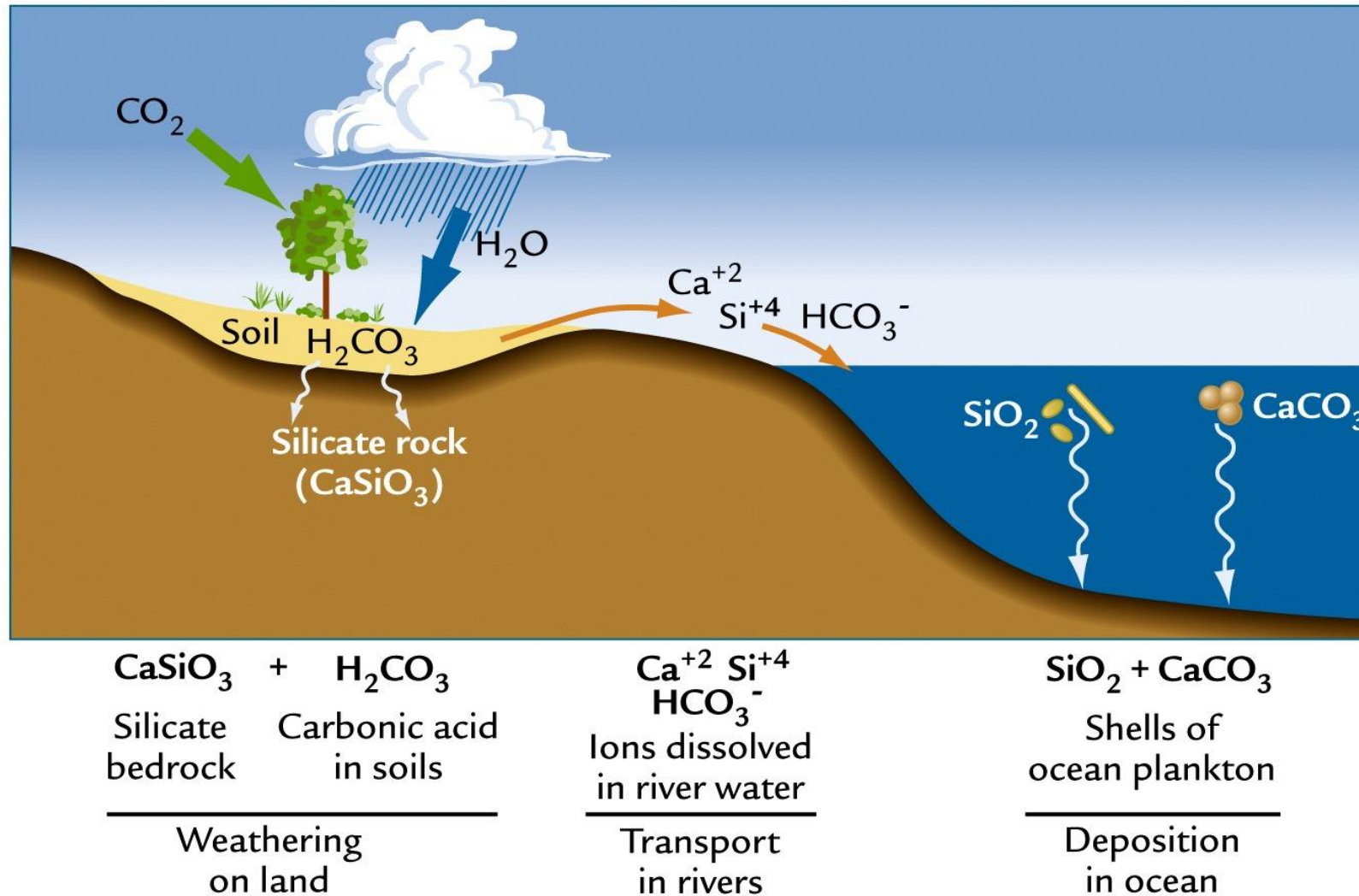


What you should know after this class

- Long-term carbon cycle
- Global and ocean carbon balance
- Carbon pumps to ocean interior
- Our carbon perturbation
- Long-term fate of anthropogenic carbon dioxide
- Ocean acidification

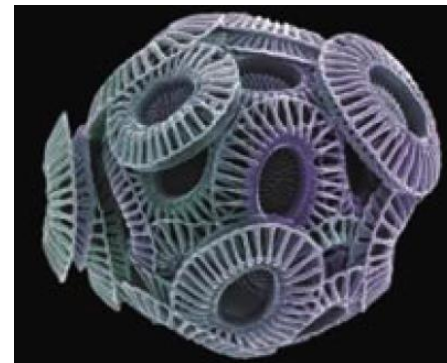


Weathering on continents and mineral formation in ocean

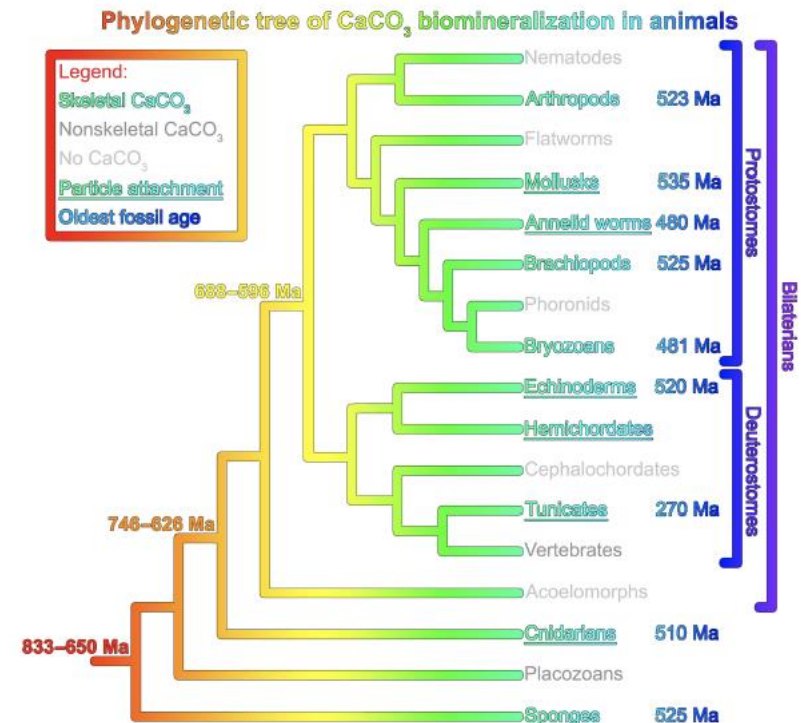
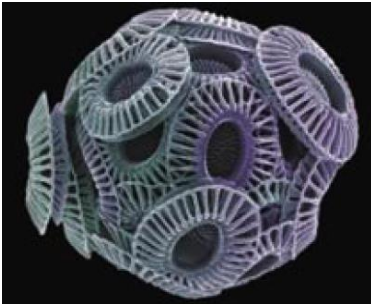


Long-term C cycle

- Weathering on land balanced by carbonate precipitation in ocean:
 - $\text{CaSiO}_3 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{SiO}_2$
- Rivers transport Ca^{2+} and HCO_3^-
- CaCO_3 precipitates in surface water:
 - $\text{Ca}^{2+} + 2 \text{HCO}_3^- \rightarrow \text{CaCO}_3(\text{s}) + \text{CO}_2 + \text{H}_2\text{O}$
 - **Precipitation releases CO_2**
- Precipitation is biologically controlled



Carbonate minerals in the sea are produced by organisms: animals, algae, microbes...



Major Calcifiers

- **Open ocean:**

- Coccolithophores (calcite)
- Foraminifera (calcite, few aragonite)
- Pteropods (aragonite)
- Deep-sea corals (aragonite)
- Fish (high Mg-calcite)

- **Coastal ocean:**

- Corals (aragonite)
- Bivalves (calcite, few aragonite)
- Coralline Algae
- Epiphytes on macrophytes
-

Solubility of CaCO_3

- Equilibrium reaction: $\text{Ca}^{2+} + \text{CO}_3^{2-} \leftrightarrow \text{CaCO}_3 (\text{s})$
- Solubility product in seawater: $K_{\text{sp}} = ([\text{Ca}^{2+}]_{\text{sat}} [\text{CO}_3^{2-}]_{\text{sat}})$

Degree of saturation of seawater

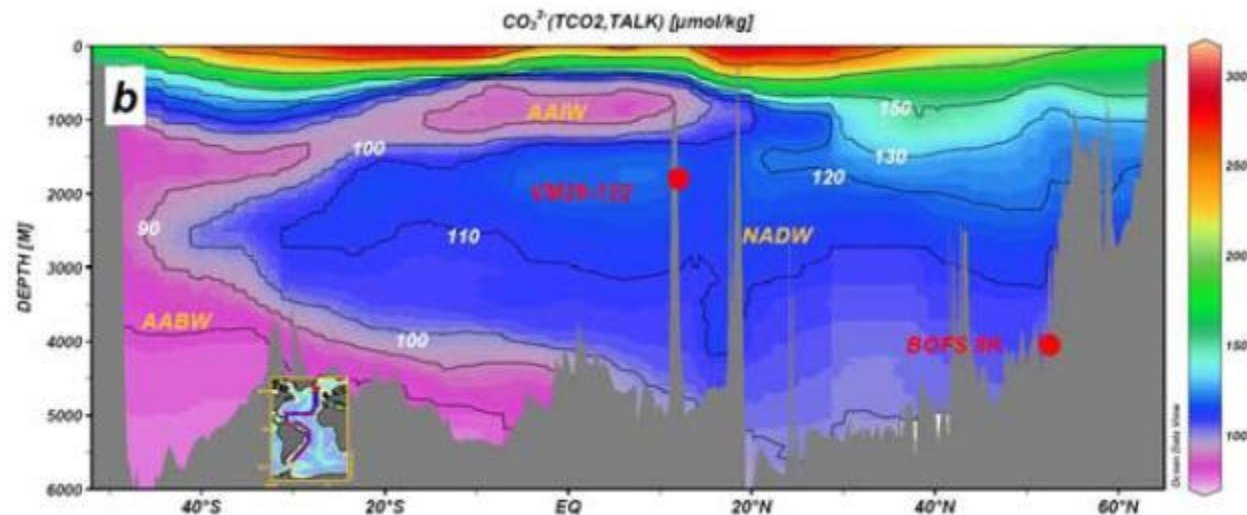
- $\Omega = ([\text{Ca}^{2+}][\text{CO}_3^{2-}])/K_{\text{sp}}$
- $\Omega < 1$: dissolution, $\Omega > 1$ precipitation possible

Solubility product is not constant

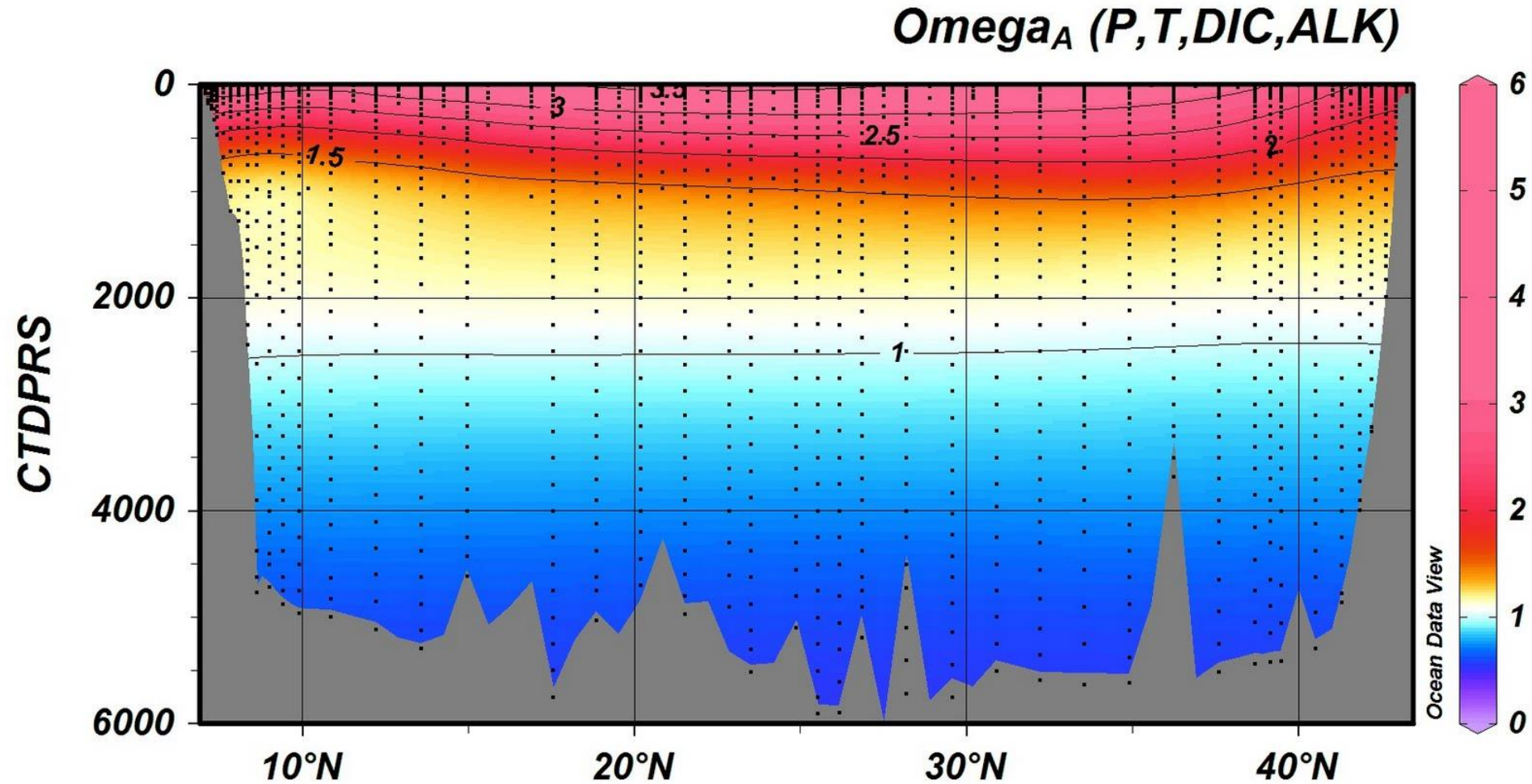
- Solubility product in seawater: $K_{sp} = ([Ca^{2+}]_{sat}[CO_3^{2-}]_{sat})$
- Solubility product depends on environmental factors such as:
 - Salinity (more soluble in seawater than freshwater)
 - Temperature (colder more soluble, higher K_{sp})
 - Pressure (higher more soluble, higher K_{sp})
 - Mineral form (Calcite vs. Aragonite)

Saturation of CaCO_3 decreases with water depth

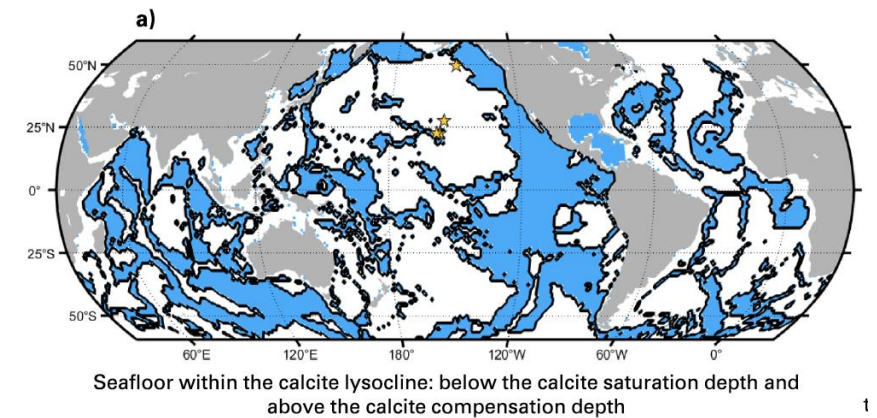
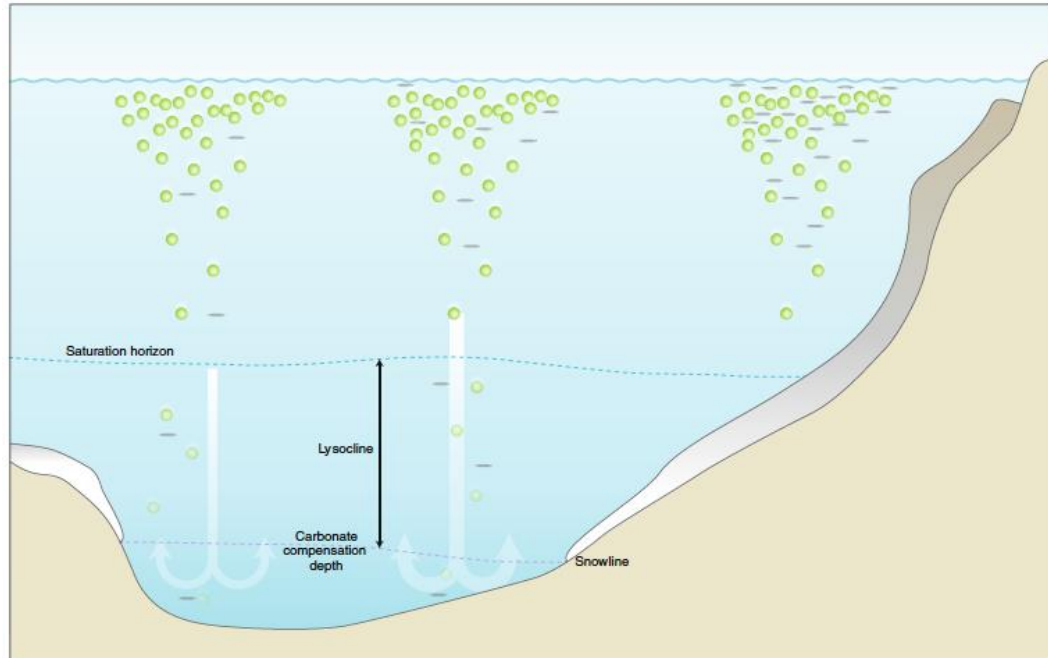
- Higher K_{sp} because of low T and high P
- Calcium concentration rather uniform with depth at about 10 mM (or 400 ppm)
- Carbonate ion concentration decreases with depth because of organic matter breakdown (oxidation) releasing CO_2



Saturation (omega) decreases with depth



Saturation Horizon and CCD



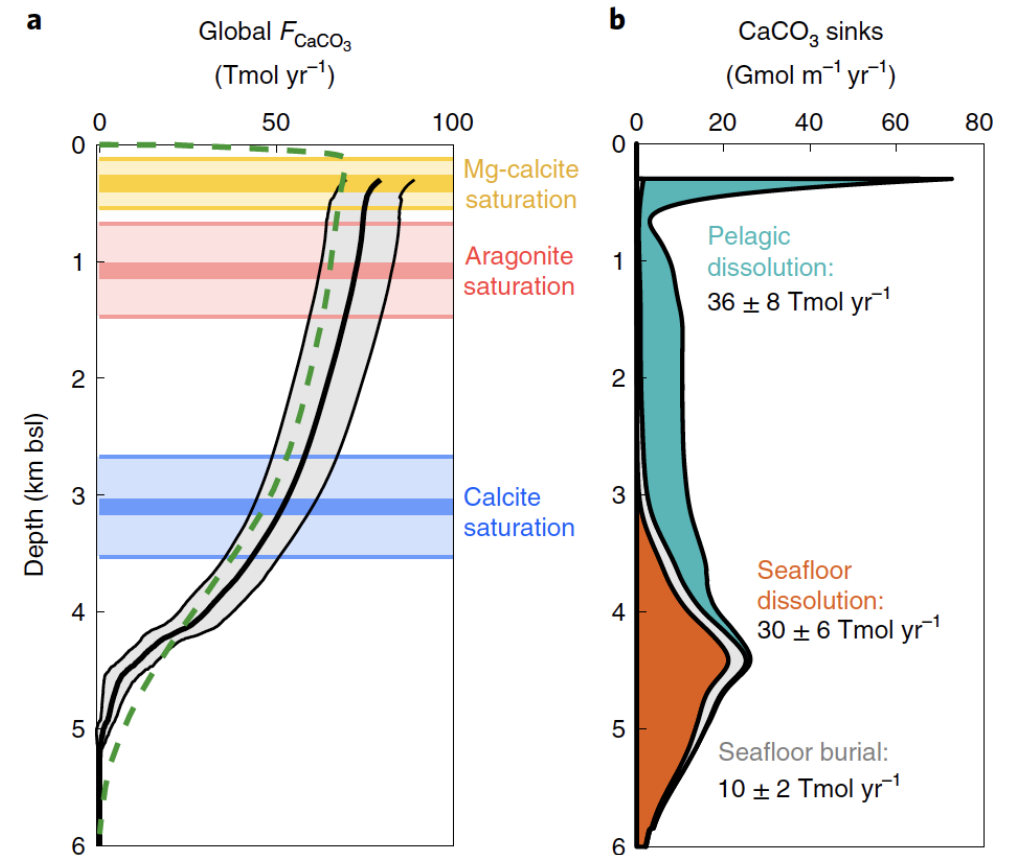
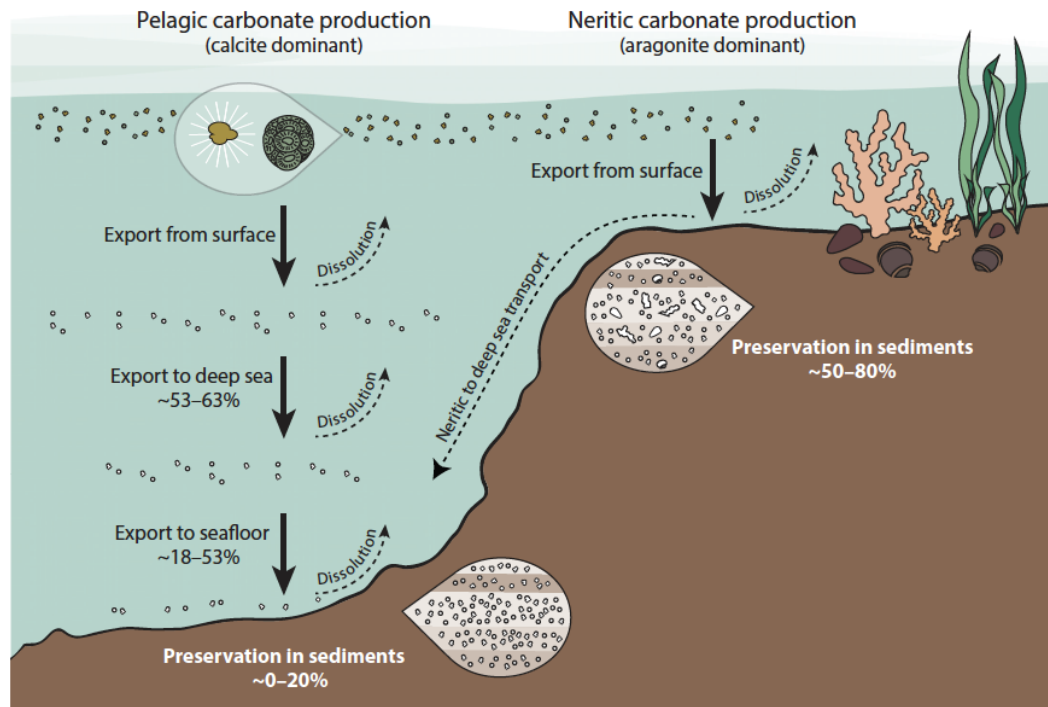
Saturation horizon: set by mineral and carbonate chemistry (S,T, P, ..)

Carbonate compensation depth (CCD): depth where downward flux of carbonate particles is balanced by dissolution rate

Lysocline: zone between saturation horizon and CCD

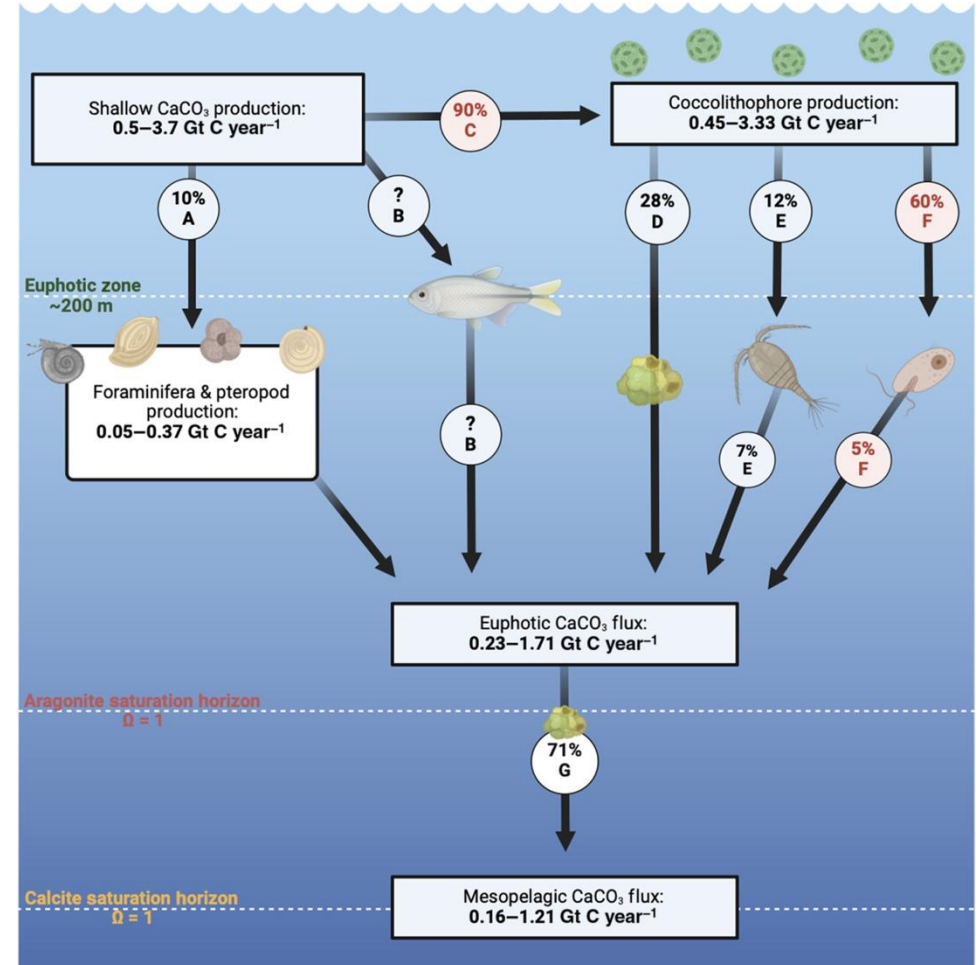
Snow line: depth where no carbonate is found in sediment

Dissolution occurs at all depths

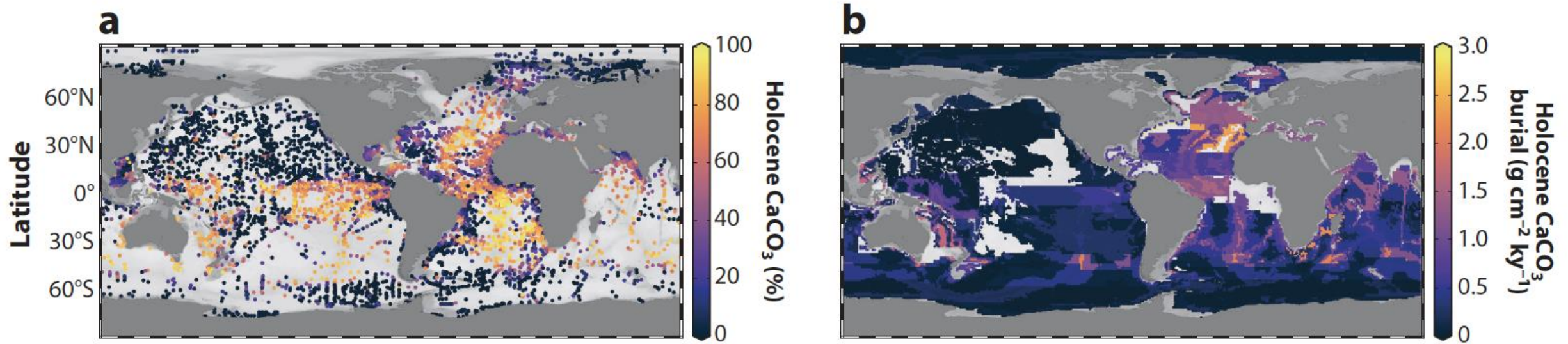


Shallow Dissolution

- High Mg-calcite (e.g. fish) and aragonite undersaturated
- Respiration of organic matter releases carbon dioxide that causes local undersaturation in fecal pellets and sinking particles.
- Dissolution in coastal and ocean margin sediments due to carbon dioxide production (OM) and re-oxidation (strong acids)
- Zooplankton grazing



Carbonates at seafloor



Calcium carbonate in Atlantic Ocean, but little if any in North Pacific Ocean due to **dissolution** (corrosive waters because of accumulated respiration derived CO_2)

The Global Carbon Cycle currency

- 1 Pg (Petagram) = 10^{15} gr
- 1 Pg = 1 Gt (Gigaton)
- 1 Pg = 100,000 swimming pools
(100*50*2 m)

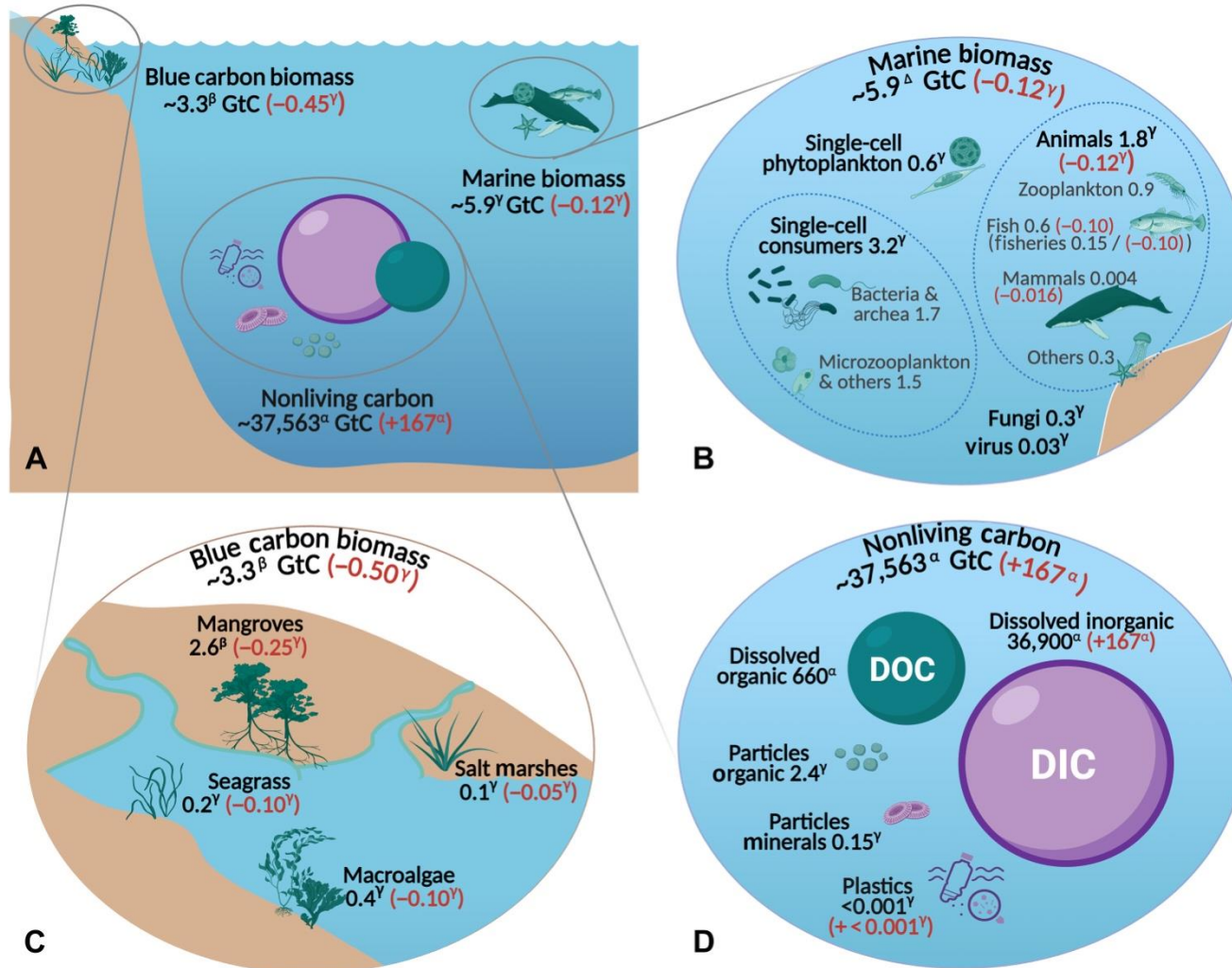
Global Carbon units:

- 2.124 ppm CO₂
- 1 Pg C (10^{15} g)
- 1 Gt C
- 3.66 Pg CO₂
- 83.33 Tmol C

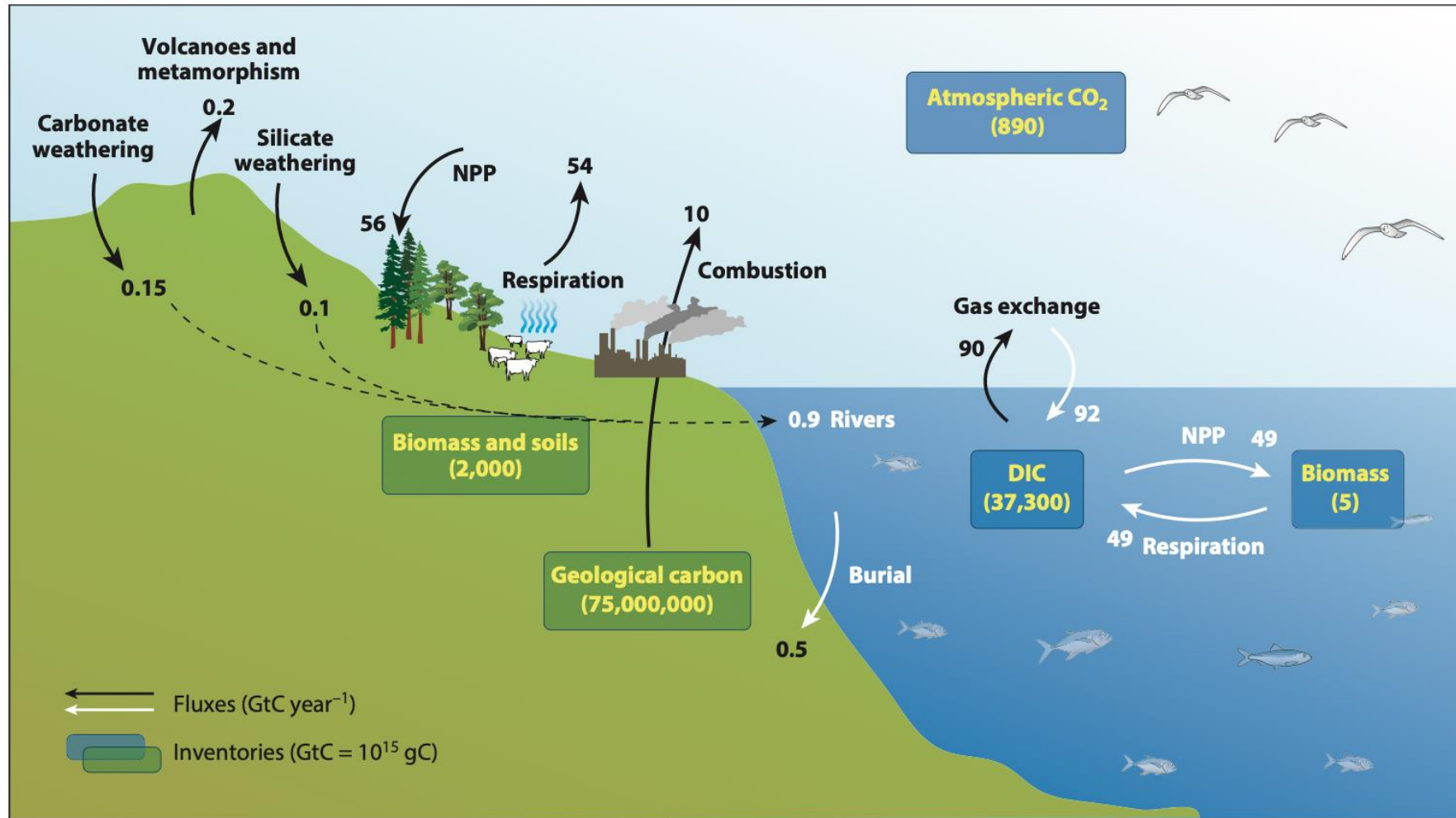
Ocean Carbon Stocks

Ocean carbon stocks (in GtC)

Uncertainty estimates: $\alpha < 25\%$, β 25—100%, $\gamma > 100\%$



Contemporary Carbon Cycle (2020)

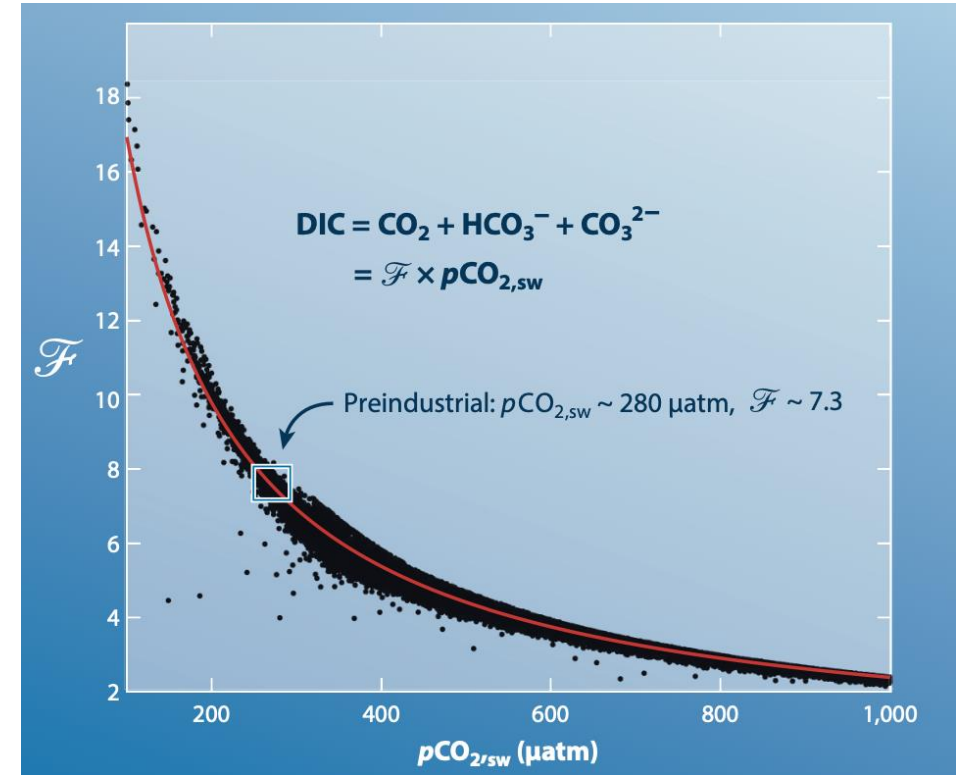


A simple box model for global C cycle

$C_{\text{atm}} = M_{\text{atm}} \times p\text{CO}_{2,\text{air}}$	Atmosphere
$C_{\text{oce}} = M_{\text{oce}} \times \text{DIC}$	Ocean

Table 1 Metrics of the ocean and atmosphere carbon cycle

Quantity (units)	Value		Meaning
	Preindustrial	Contemporary	
M_{oce} (kg)	1.38×10^{21}	1.38×10^{21}	Mass of the ocean ^a
M_{atm} (mol)	1.77×10^{20}	1.77×10^{20}	Molar mass of the atmosphere ^b
$x\text{CO}_{2,\text{air}}$ (ppm)	280	410	Atmospheric CO_2 concentration
DIC_s ($\mu\text{mol kg}^{-1}$)	2,040	2,090	Average surface ^c seawater DIC
DIC_d ($\mu\text{mol kg}^{-1}$)	2,250	2,260	Average deep ocean ^d DIC
$\mathcal{F}$ [$\text{mol air (kg seawater)}^{-1}$]	7.2	– ^e	Ratio of $\text{DIC}/x\text{CO}_2$ at chemical equilibrium
C_{oce} (GtC)	$37,100^f$	$37,300^g$	Total mass of inorganic carbon in the ocean
C_{atm} (GtC)	595	890	Total mass of carbon as CO_2 in the atmosphere



$$F = \text{DIC} / p\text{CO}_{2,\text{sw}}$$

F is the seawater equilibrium carbon capacity

(de Vries, 2022 AnRev)

Distribution C among atmosphere and well-mixed ocean

$$C_{\text{atm}} = M_{\text{atm}} \times p\text{CO}_{2,\text{air}}$$

$$C_{\text{oce}} = M_{\text{oce}} \times \text{DIC}$$

The two-box model is based on three equations:

- For the atmosphere: $C_{\text{atm}} = M_{\text{atm}} \times p\text{CO}_{2,\text{air}}$
- For the ocean: $C_{\text{oce}} = M_{\text{oce}} \times \text{DIC}$
- Equilibrium partitioning: $F = \text{DIC} / p\text{CO}_{2,\text{sw}}$

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These can be re-arranged to:

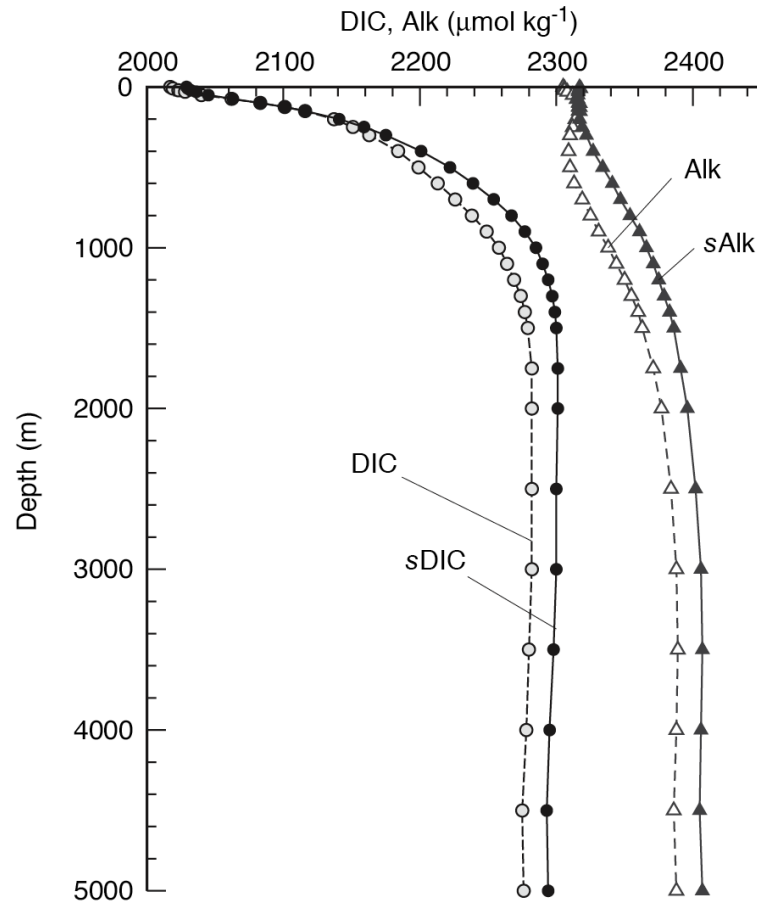
$$\frac{C_{\text{oce}}}{C_{\text{atm}}} = F \frac{M_{\text{oce}}}{M_{\text{atm}}}$$

Where $\frac{M_{\text{oce}}}{M_{\text{atm}}}$ is 7.8 kg seawater/mol air

Pre-industrial Earth with $p\text{CO}_2$ of 280 ppm has an F value of 7.3, hence $\frac{C_{\text{oce}}}{C_{\text{atm}}}$ is predicted to be 57 (7.3×7.8) rather than expected 62.4 ($37100/595$) from observations.

Quite ok, but some space for improvement!

Distribution of inorganic Carbon



Low DIC



High DIC

This gradient in dissolved inorganic carbon is maintained by carbon pumps

- Physical-chemical pump
- Soft-tissue or Biological pump
- Carbonate (counter) pump

Distribution C among atmosphere and two-layer ocean

The equilibrium partitioning equation then becomes:

$$\frac{C_{oce}}{C_{atm}} = F \left(\frac{M_{oce,s}}{M_{atm}} + \frac{DIC_d}{DIC_s} \frac{M_{oce,d}}{M_{atm}} \right)$$

Which can be simplified because $M_{oce,s} \ll M_{oce,d}$

$$\frac{C_{oce}}{C_{atm}} = F \left(\frac{DIC_d}{DIC_s} \frac{M_{oce,d}}{M_{atm}} \right)$$

$C_{atm} = M_{atm} \times pCO_{2,air}$
$C_{oce,s} = M_{oce,s} \times DIC_s$
$C_{oce,d} = M_{oce,d} \times DIC_d$

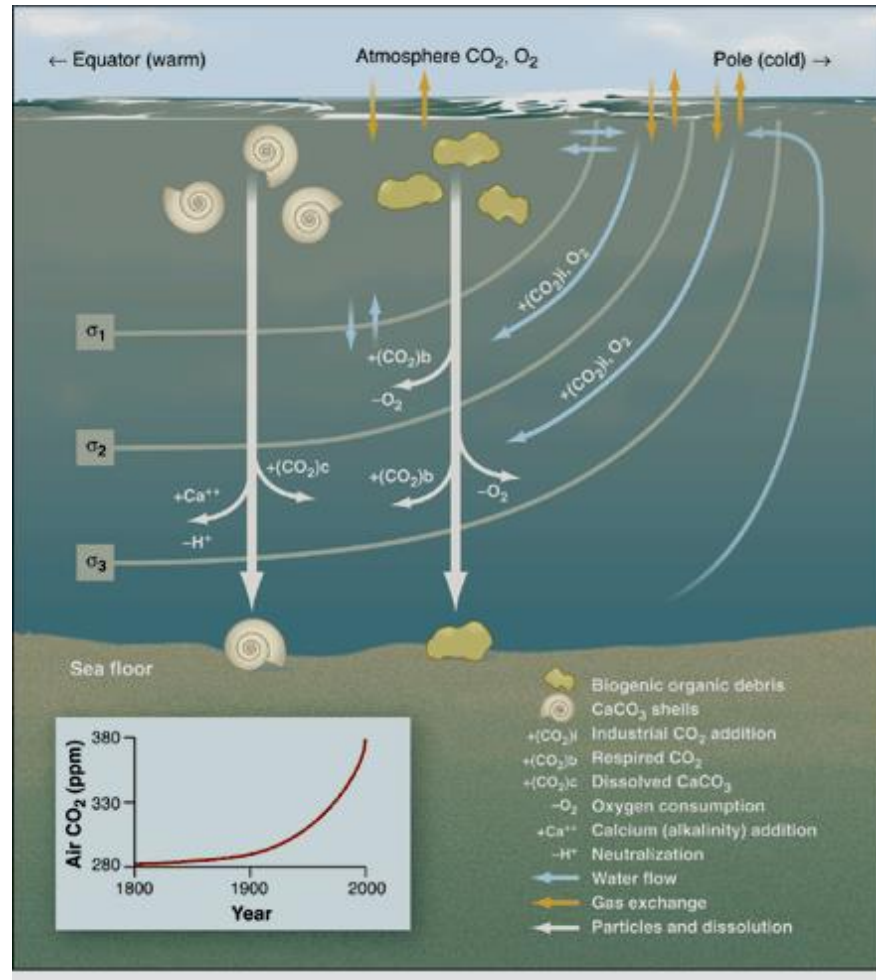
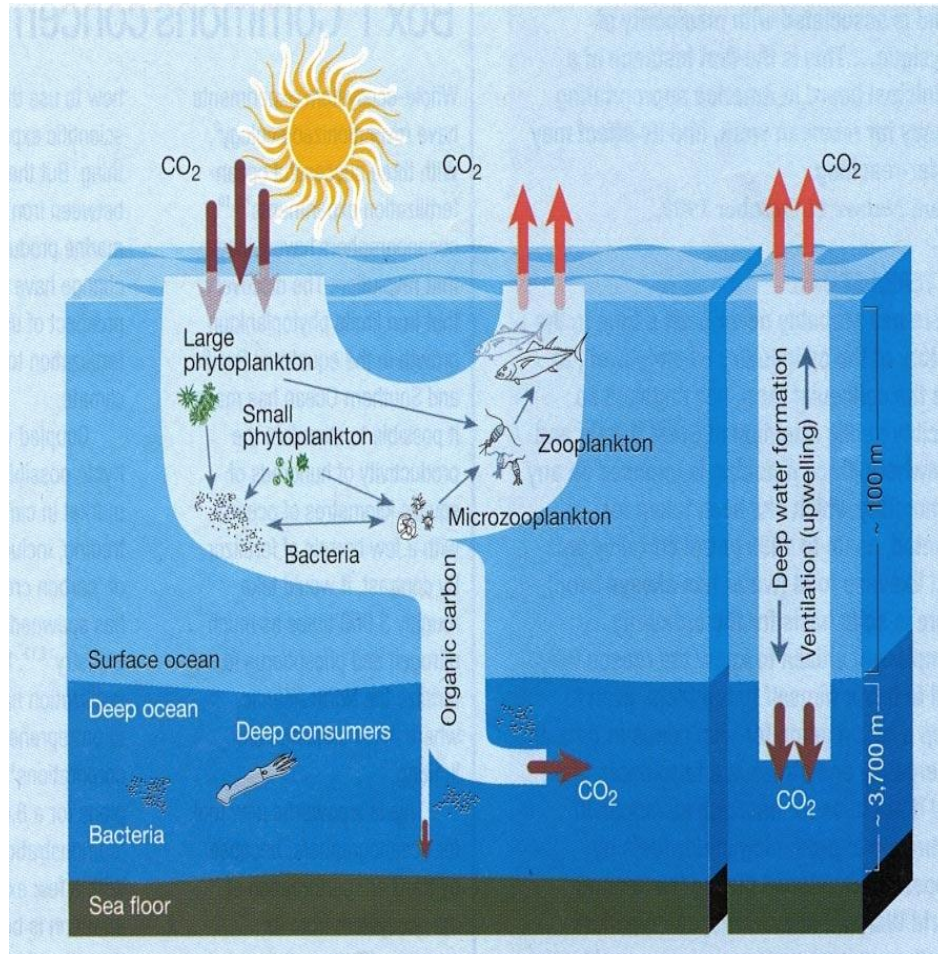
This model is again based on three equations:

- For the atmosphere: $C_{atm} = M_{atm} \times pCO_{2,air}$
- For the ocean: $C_{oce} = M_{oce,s} \times DIC_s + M_{oce,d} \times DIC_d$
- Equilibrium partitioning: $F = DIC/pCO_{2,sw}$

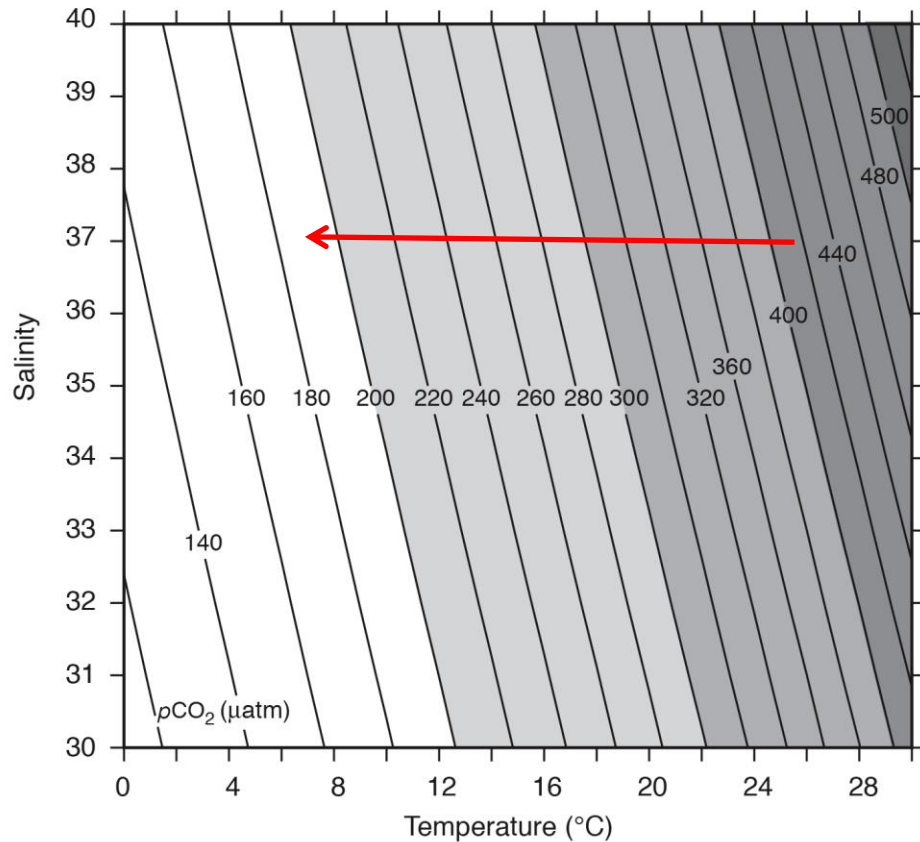
Pre-industrial Earth with pCO_2 of 280 ppm would then have a $\frac{C_{oce}}{C_{atm}}$ ratio of 62, very close to the observed 62.4 (37100/595)

Carbon pumps enhance ocean DIC by 10% and reduce atmospheric CO_2 by a factor of 2.

Biological and Physical Pumps

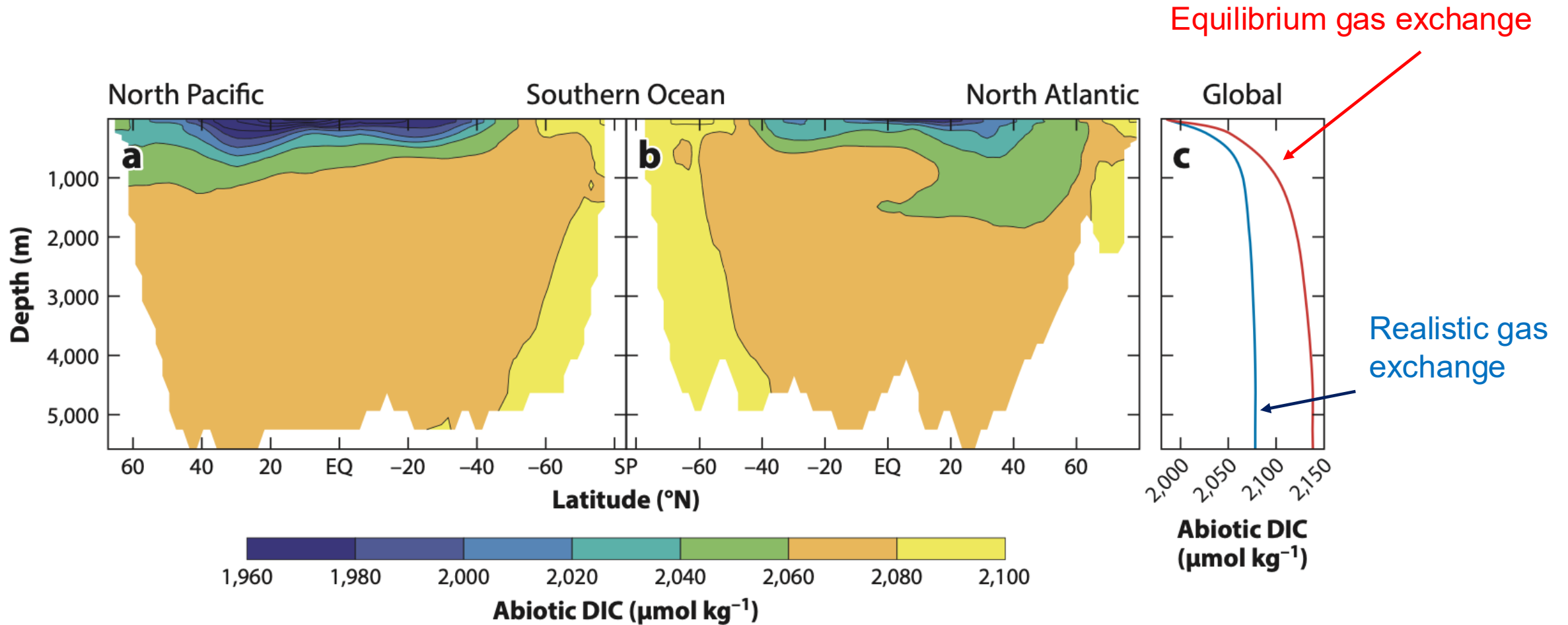


Physical (Chemical) Pump



- Water cooling (e.g. Gulf Stream from low-latitude to high-latitude) results in low $p\text{CO}_2$
- Atmospheric CO_2 can invade seawater
- Underlying mechanisms:
 - CO_2 invades cold water because of chemistry
 - Cold water sinks because of high density (physics)

Physical pump: major role high-latitude ocean



Soft-tissue and Carbonate pumps

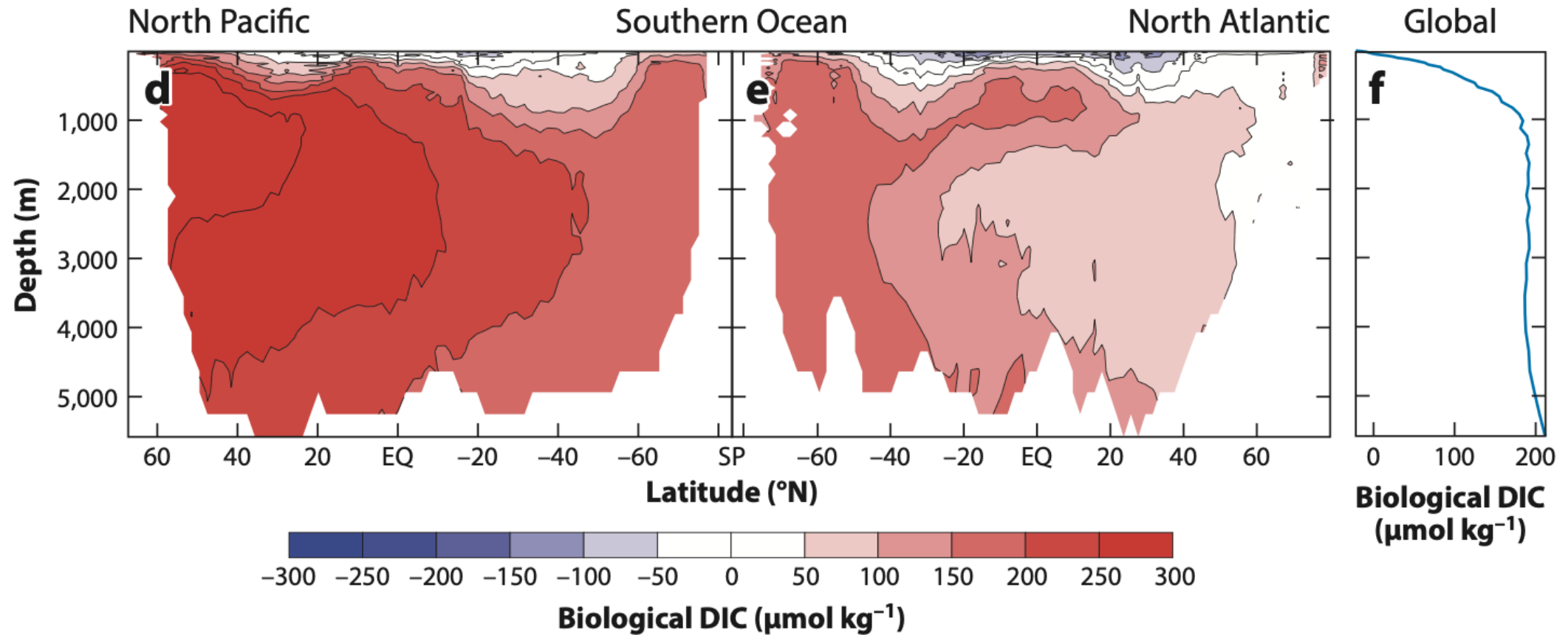
Soft-tissue/Biological pump

- Utilisation of CO_2 by primary production in the biologically active surface waters
- C is transported downward in the form of organics
- The enrichment of CO_2 from deep water through decomposition of detritus originating from biological processes in surface waters

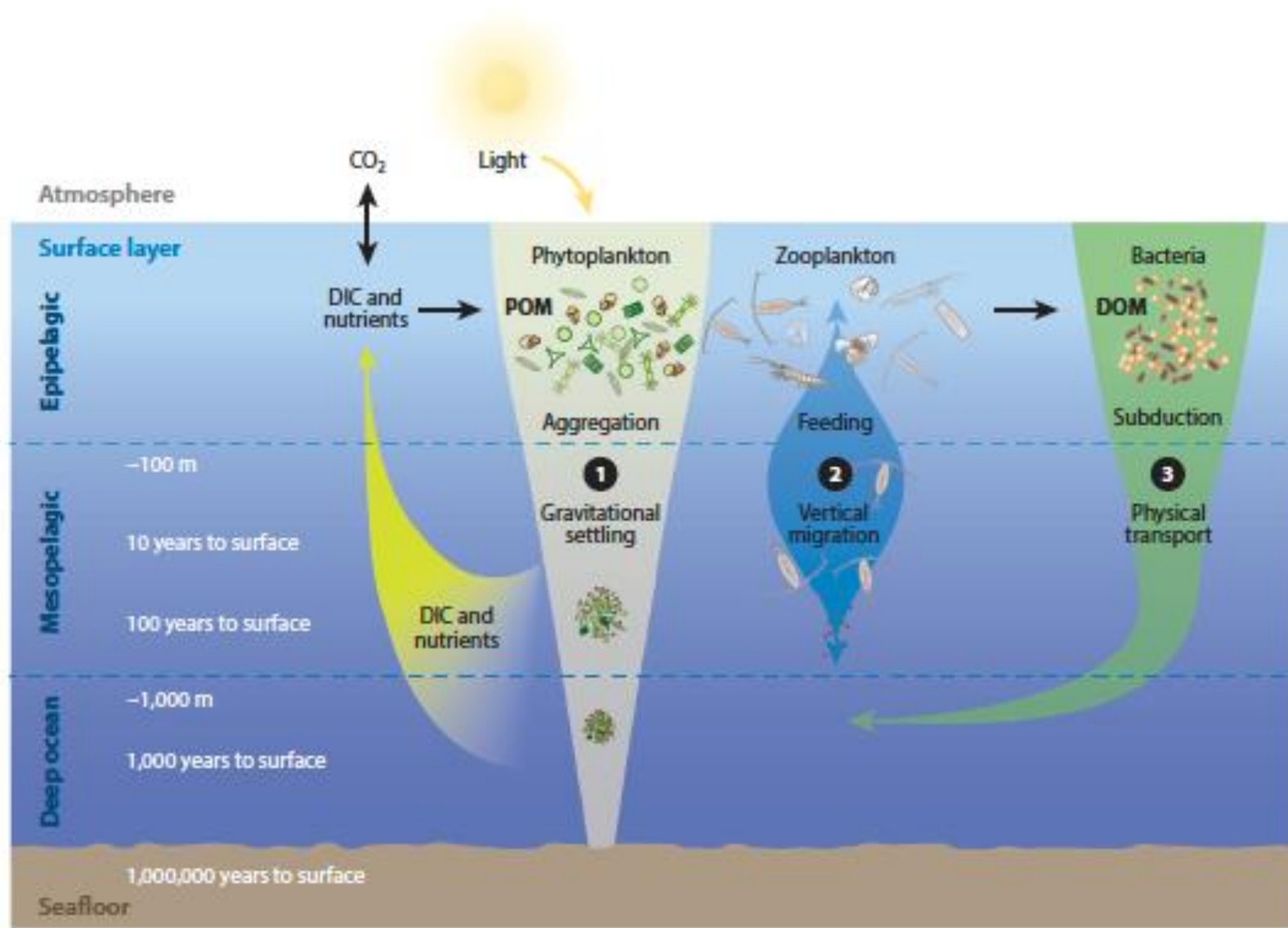
Carbonate pump

- Formation of CaCO_3 skeletons in the biologically active surface waters
- Dissolution of skeletons in undersaturated waters that have settled down from biological processes in surface waters release in DIC and Alkalinity

Biological pump

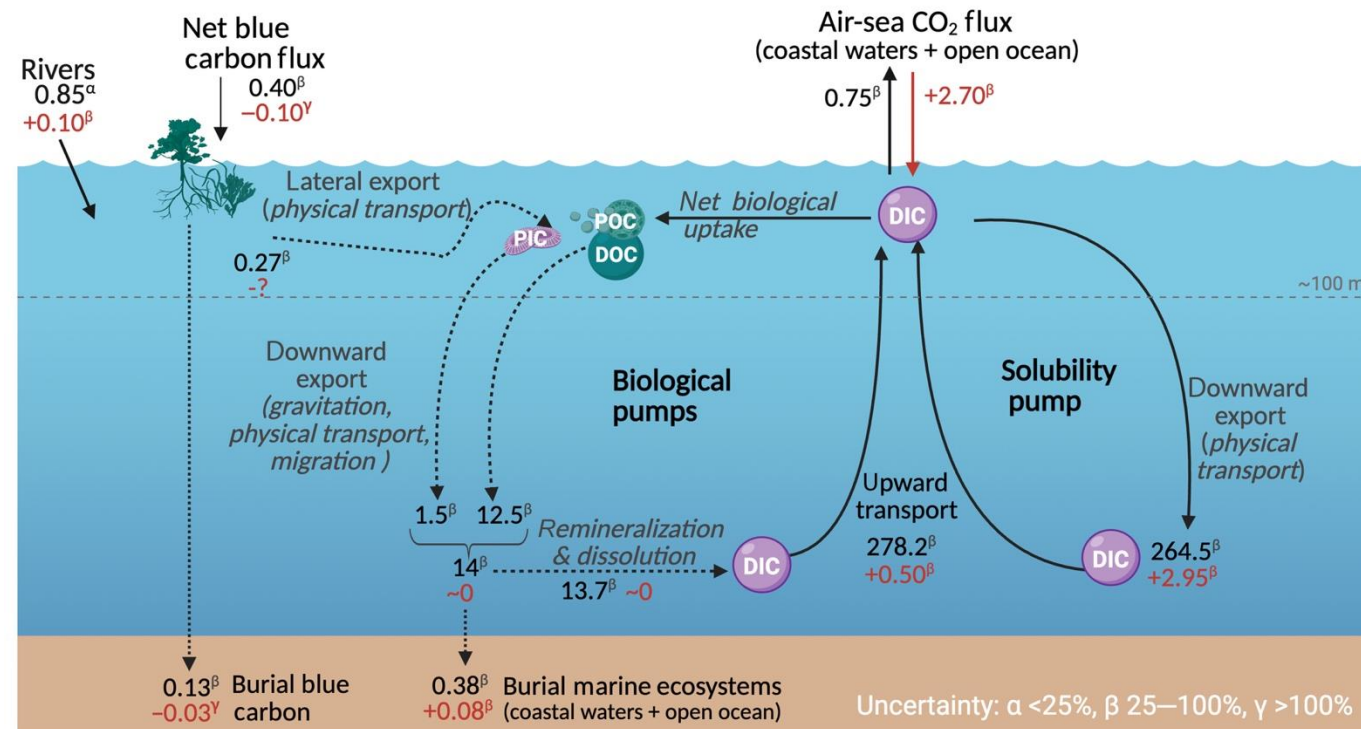


Biological Pump

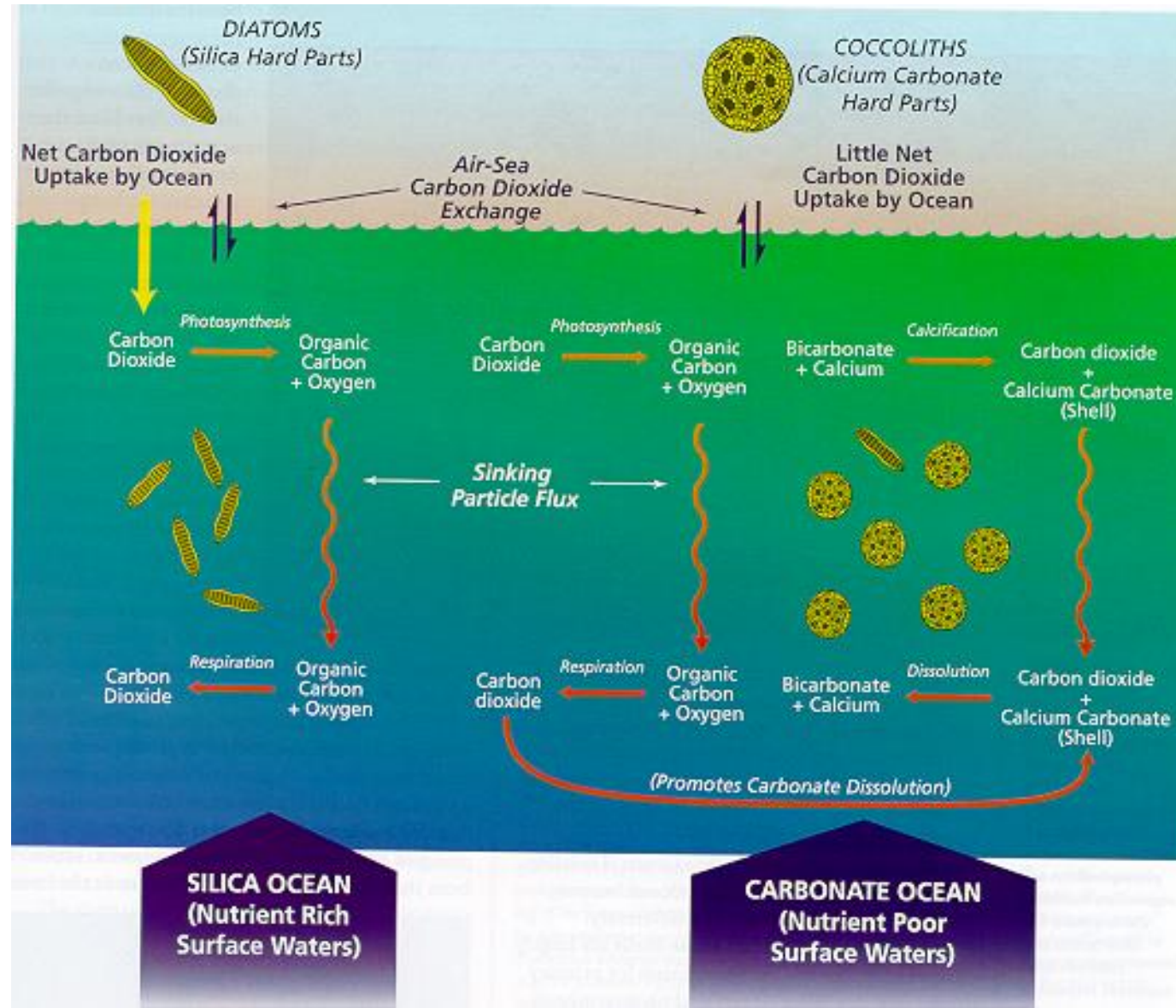


Ocean Carbon Pumps (including coastal)

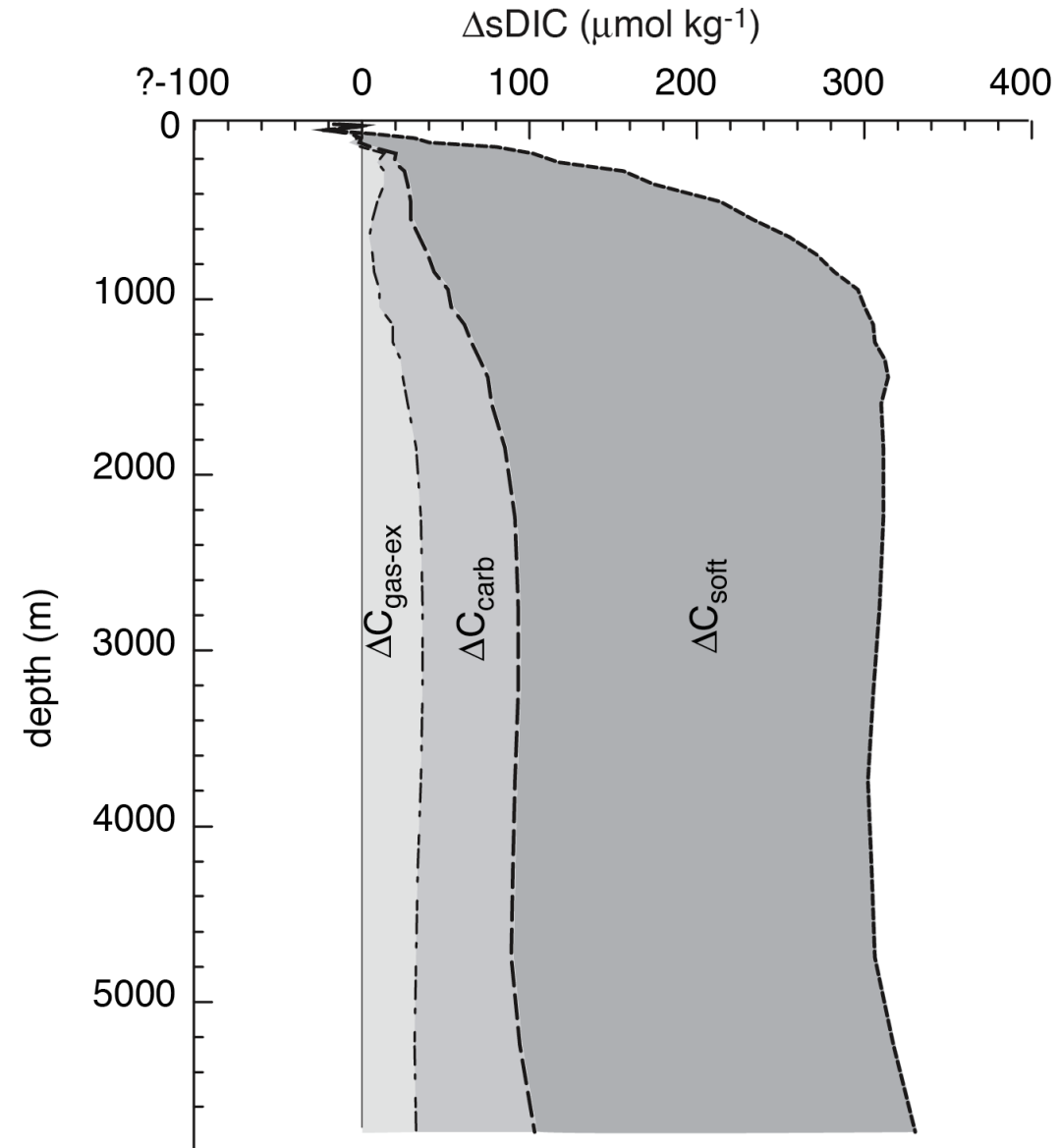
Ocean carbon pumps (in GtC/year)



Carbonate Pump

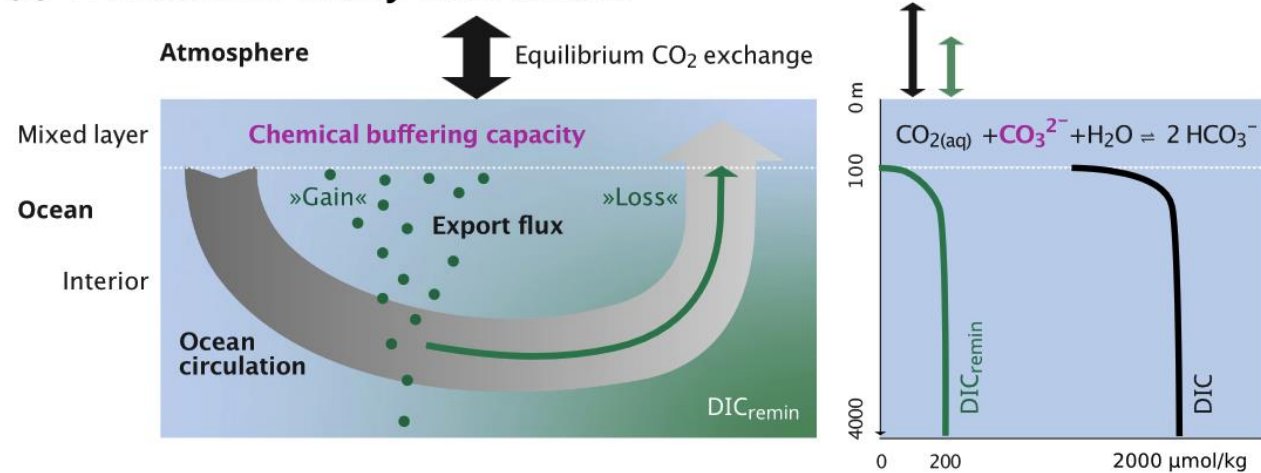


Contribution of the pumps



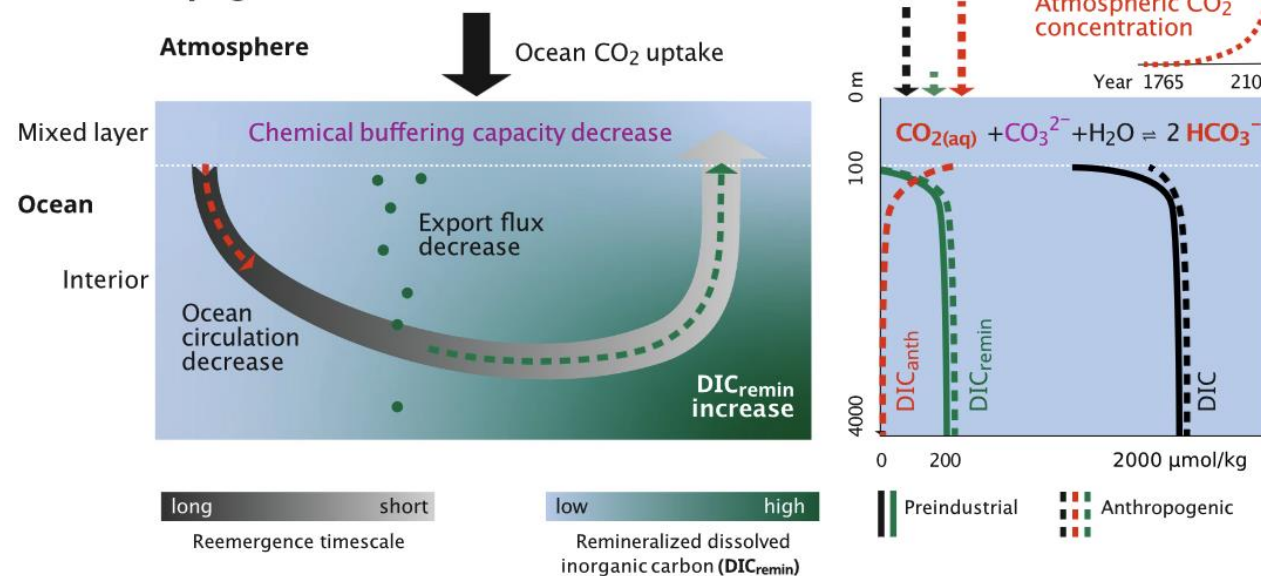
Biological Pump in a Changing Climate

(a) Preindustrial steady-state climate



- At steady-state: Gains and Losses of ocean interior DIC balance
- Climate change induced enhanced stratification lowers export flux.
- A decrease in export flux does not imply a less efficient biological carbon pump because circulations slows down as well. The loss decreases more than the gains so that more DIC accumulates in ocean

(b) Anthropogenic transient climate



Svante Arrhenius: Grandfather of climate change

THE
LONDON, EDINBURGH, AND DUBLIN
PHILOSOPHICAL MAGAZINE
AND
JOURNAL OF SCIENCE.

[FIFTH SERIES.]

APRIL 1896.

XXXI. *On the Influence of Carbonic Acid in the Air upon the Temperature of the Ground.* By Prof. SVANTE ARRHENIUS*.

I. *Introduction: Observations of Langley on Atmospheric Absorption.*

A GREAT deal has been written on the influence of the absorption of the atmosphere upon the climate. Tyndall† in particular has pointed out the enormous importance of this question. To him it was chiefly the diurnal and annual variations of the temperature that were lessened by this circumstance. Another side of the question, that has long attracted the attention of physicists, is this: Is the mean temperature of the ground in any way influenced by the presence of heat-absorbing gases in the atmosphere? Fourier‡ maintained that the atmosphere acts like the glass of a hot-house, because it lets through the light rays of the sun but retains the dark rays from the ground. This idea was elaborated by Pouillet§; and Langley was by some of his researches led to the view, that "the temperature of the earth under direct sunshine, even though our atmosphere were present as now, would probably fall to -200° C., if that atmosphere did not possess the quality of selective

* Extract from a paper presented to the Royal Swedish Academy of Sciences, 11th December, 1895. Communicated by the Author.

† 'Heat & Mode of Motion,' 2nd ed. p. 405 (London, 1865).

‡ *Mém. de l'Ac. R. d. Sci. de l'Inst. de France*, t. vii. 1827.

§ *Comptes rendus*, t. vii. p. 41 (1838).



On the influence of carbonic acid in the air upon the temperature on the ground (Arrhenius, 1896)

Roger Revelle & Hans Suess: Irreversible experiment

Carbon Dioxide Exchange Between Atmosphere and Ocean and the Question of an Increase of Atmospheric CO₂ during the Past Decades

By ROGER REVELLE and HANS E. SUESS, Scripps Institution of Oceanography, University of California, La Jolla, California

(Manuscript received September 4, 1956)

Abstract

From a comparison of C¹⁴/C¹² and C¹³/C¹² ratios in wood and in marine material and from a slight decrease of the C¹⁴ concentration in terrestrial plants over the past 50 years it can be concluded that the average lifetime of a CO₂ molecule in the atmosphere before it is dissolved into the sea is of the order of 10 years. This means that most of the CO₂ released by artificial fuel combustion since the beginning of the industrial revolution must have been absorbed by the oceans. The increase of atmospheric CO₂ from this cause is at present small but may become significant during future decades if industrial fuel combustion continues to rise exponentially.

Present data on the total amount of CO₂ in the atmosphere, on the rates and mechanisms of exchange, and on possible fluctuations in terrestrial and marine organic carbon, are inadequate for accurate measurement of future changes in atmospheric CO₂. An opportunity exists during the International Geophysical Year to obtain much of the necessary information.

Introduction

In the middle of the 19th century appreciable amounts of carbon dioxide began to be added to the atmosphere through the combustion of fossil fuels. The rate of combustion has continually increased so that at the present time the annual increment from this source is nearly 0.4 % of the total atmospheric carbon dioxide. By 1960 the amount added during the past century will be more than 15 %.

CALLNDAR (1938, 1940, 1949) believed that nearly all the carbon dioxide produced by fossil fuel combustion has remained in the atmosphere,

and he suggested that the increase in atmospheric carbon dioxide may account for the observed slight rise of average temperature in northern latitudes during recent decades. He thus revived the hypothesis of T. C. CHAMBERLIN (1899) and S. ARRHENIUS (1903) that climatic changes may be related to fluctuations in the carbon dioxide content of the air. These authors supposed that an increase of carbon dioxide in the upper atmosphere would lower the mean level of back radiation in the infrared and thereby increase the average temperature near the earth's surface.

Subsequently, other authors have questioned Callendar's conclusions on two grounds. First, comparison of measurements made in the 19th century and in recent years do not demonstrate that there has been a significant increase in

Contribution from the Scripps Institution of Oceanography, New Series, No. 900. This paper represents in part results of research carried out by the University of California under contract with the Office of Naval Research.

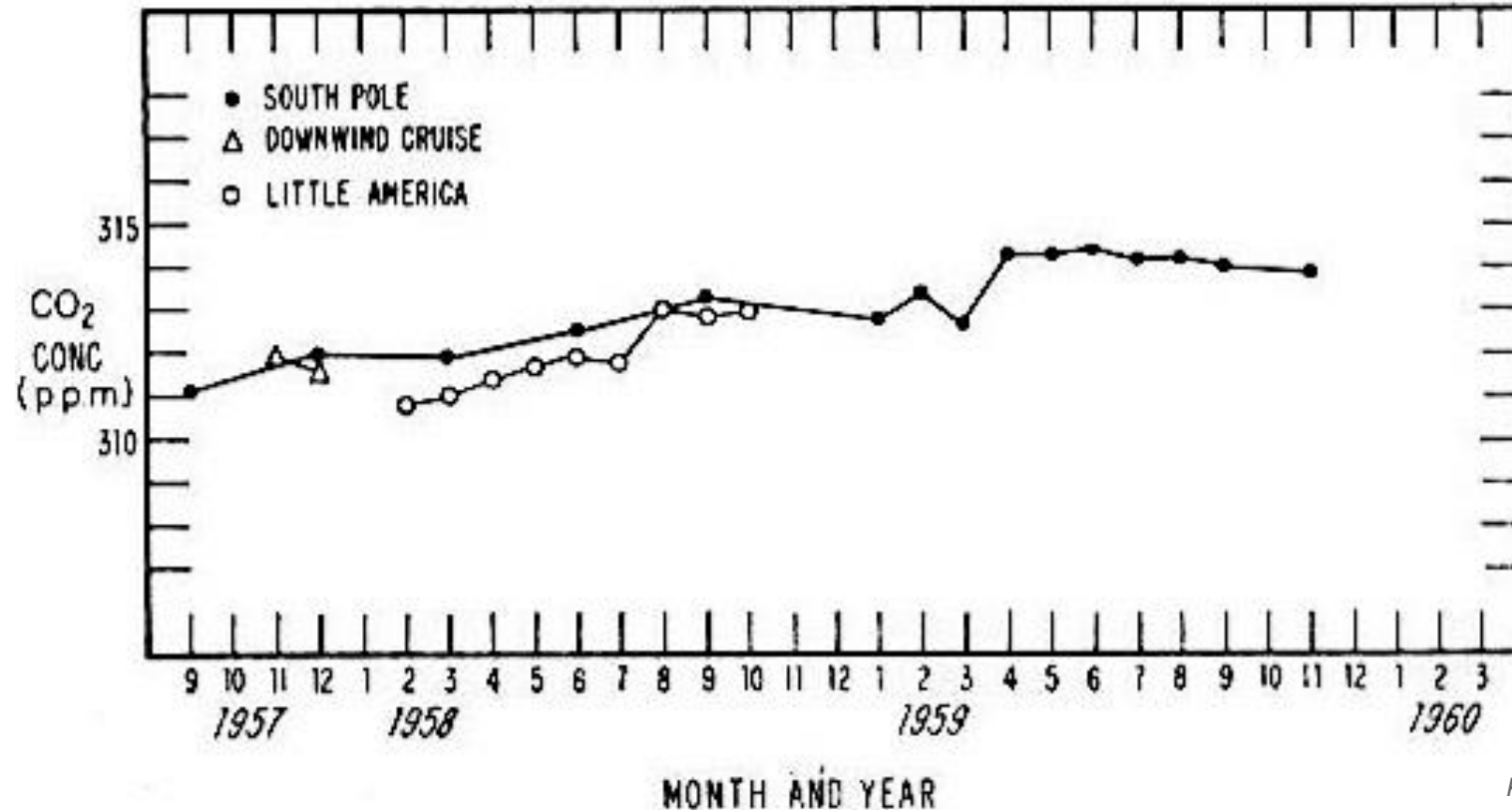
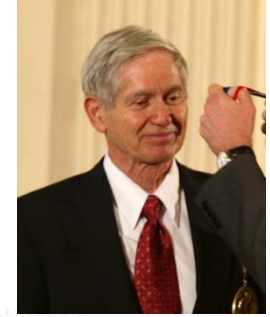
Tellus IX (1957), 1



The increase of atm. CO₂ from this cause is small at present but may become significant during future decades [].

Present data [] are inadequate for accurate measurement of future changes in atm. CO₂. An opportunity exists during the IGY [].

Atmospheric CO₂: 1958-1960



Keeling (1960)

Keeling detected an increase in atmospheric CO₂ within two years

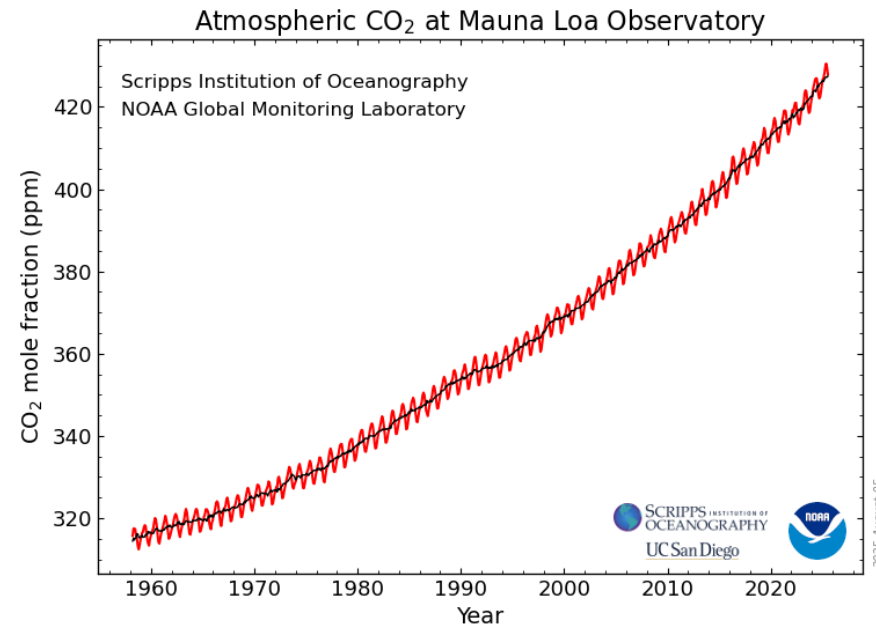
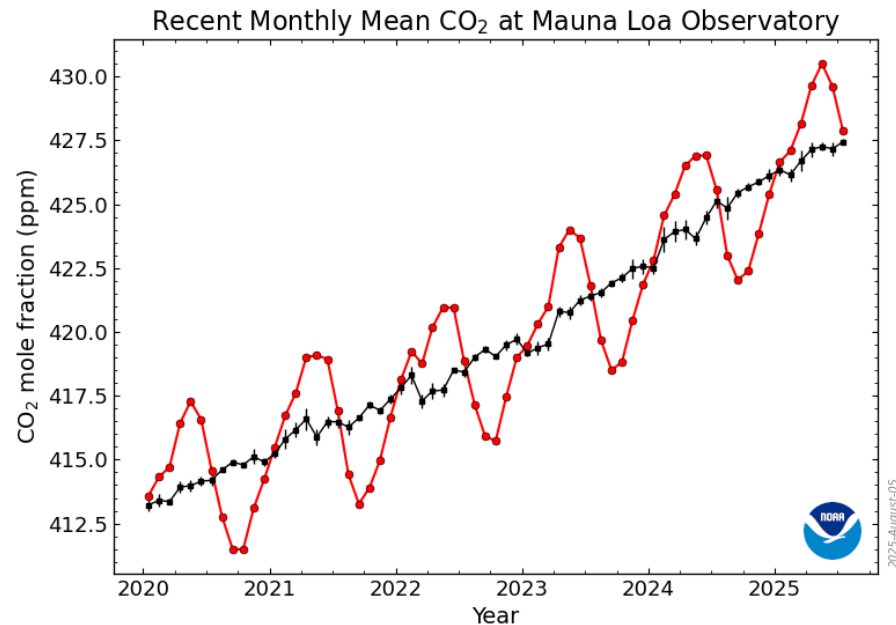
ppm = parts per million; in other words 300 molecules CO₂ out of 1,000,000 molecules

Increase in carbon dioxide from prehistoric **280** to **430** ppm now

July 2025: 427.87 ppm

July 2024: 425.55 ppm

Last updated: Aug 05, 2025



Fate of Anthropogenic CO₂ Emissions (2014-2023)

Friedlingstein et al. 2024

1.1 Pg C y⁻¹ (10%)



+

9.7 Pg C y⁻¹ (90%)



Atmosphere

5.2 GtC/yr

48%



Land

3.2 GtC/yr

29%



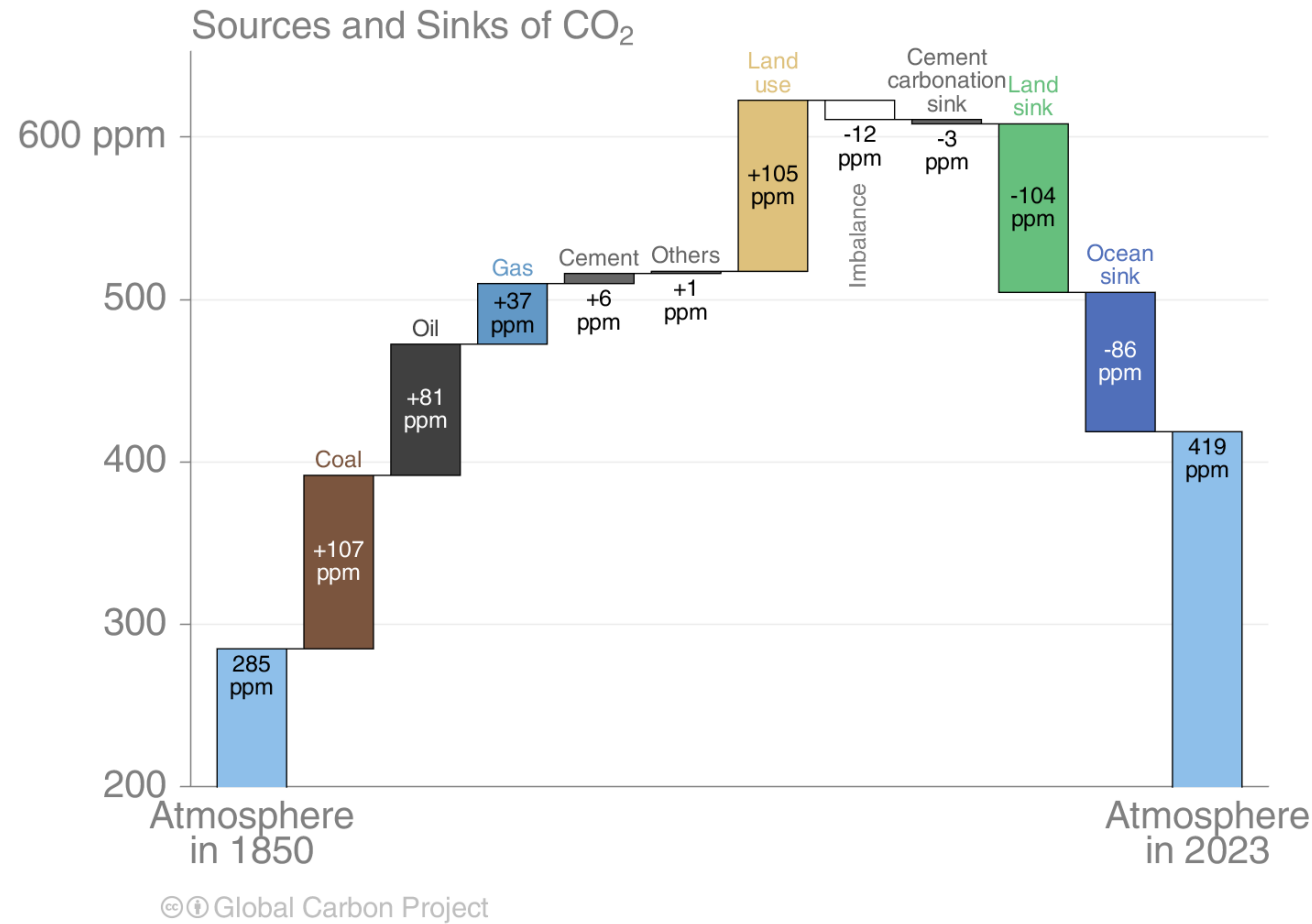
Oceans

3.2 GtC/yr

26%

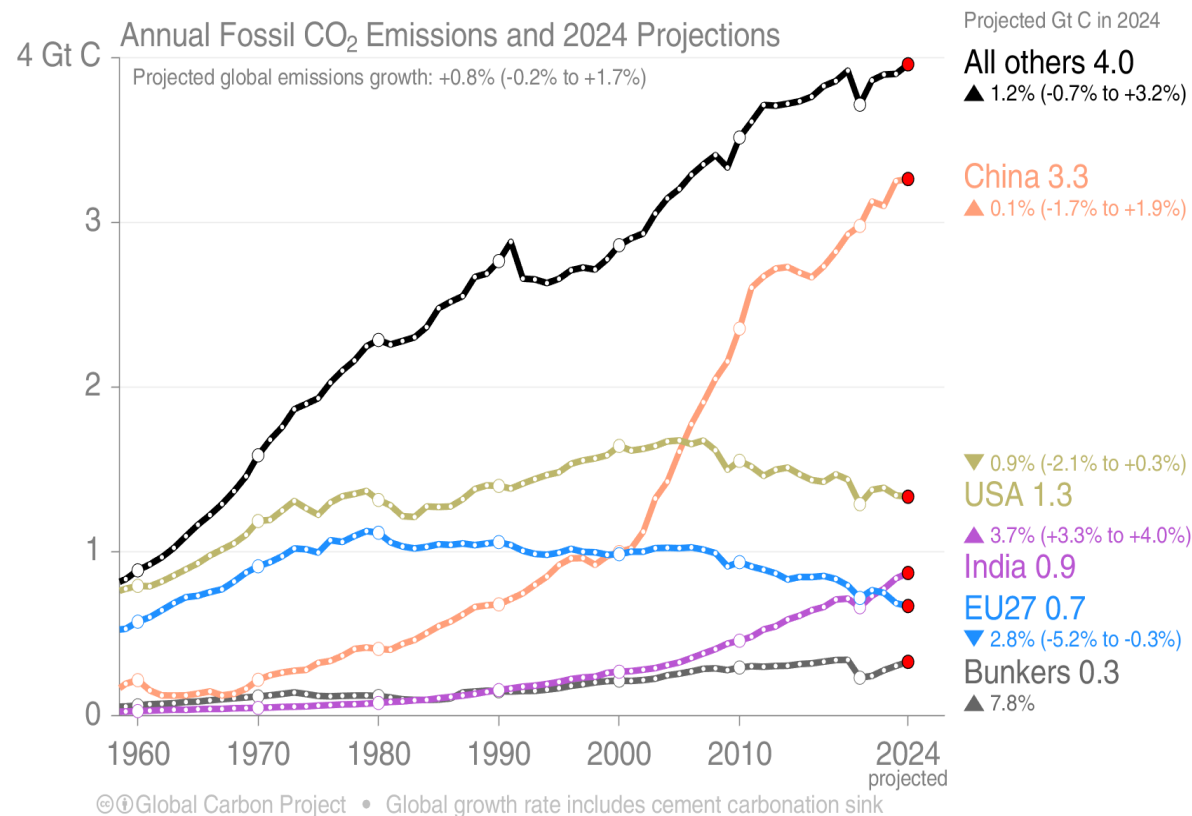
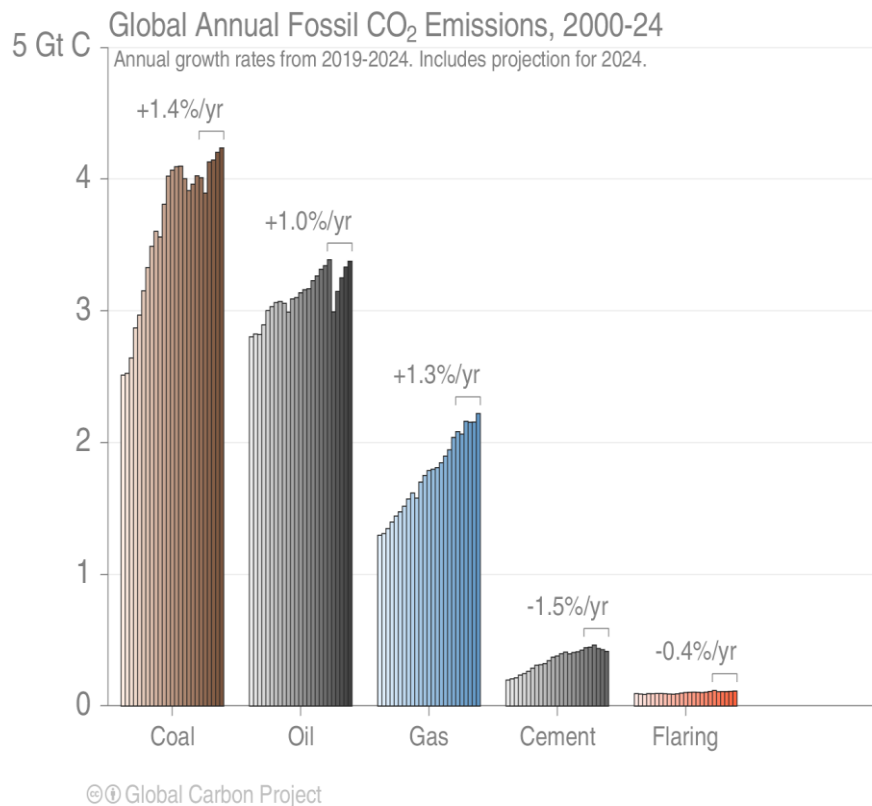


Cumulative budget from 1850 onwards



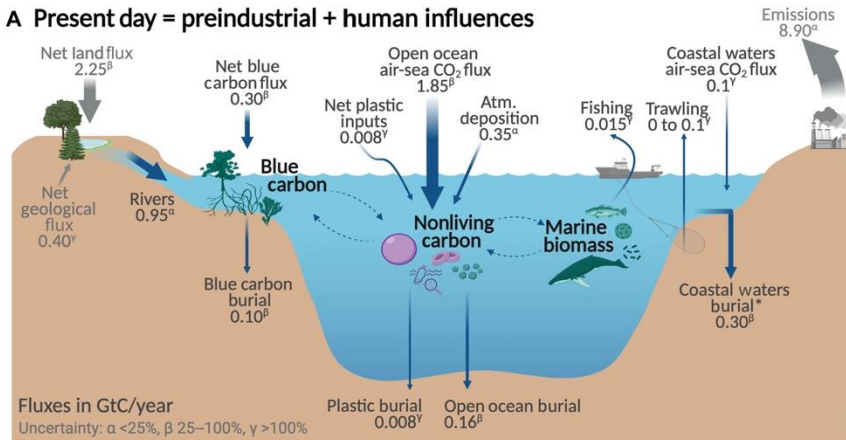
Global Carbon project numbers from Friedlingstein et al. (2024)

Recent Change: Source and Country?

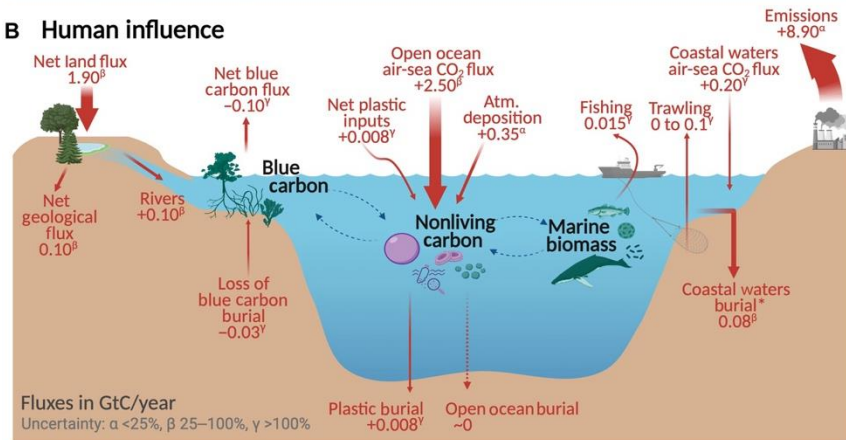


Bunkers = international shipping and aviation

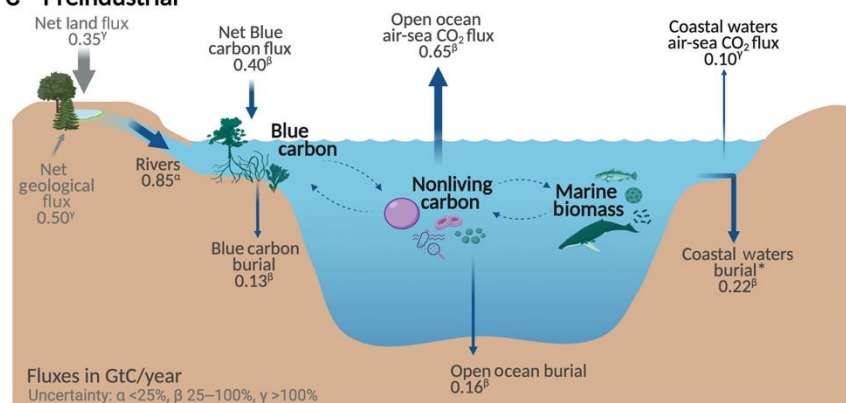
A Present day = preindustrial + human influences



B Human influence



C Preindustrial



Pre-industrial ocean was a **source** of carbon (to compensate for river C input)

Modern ocean is a **sink** (because of atmospheric CO2 increase)

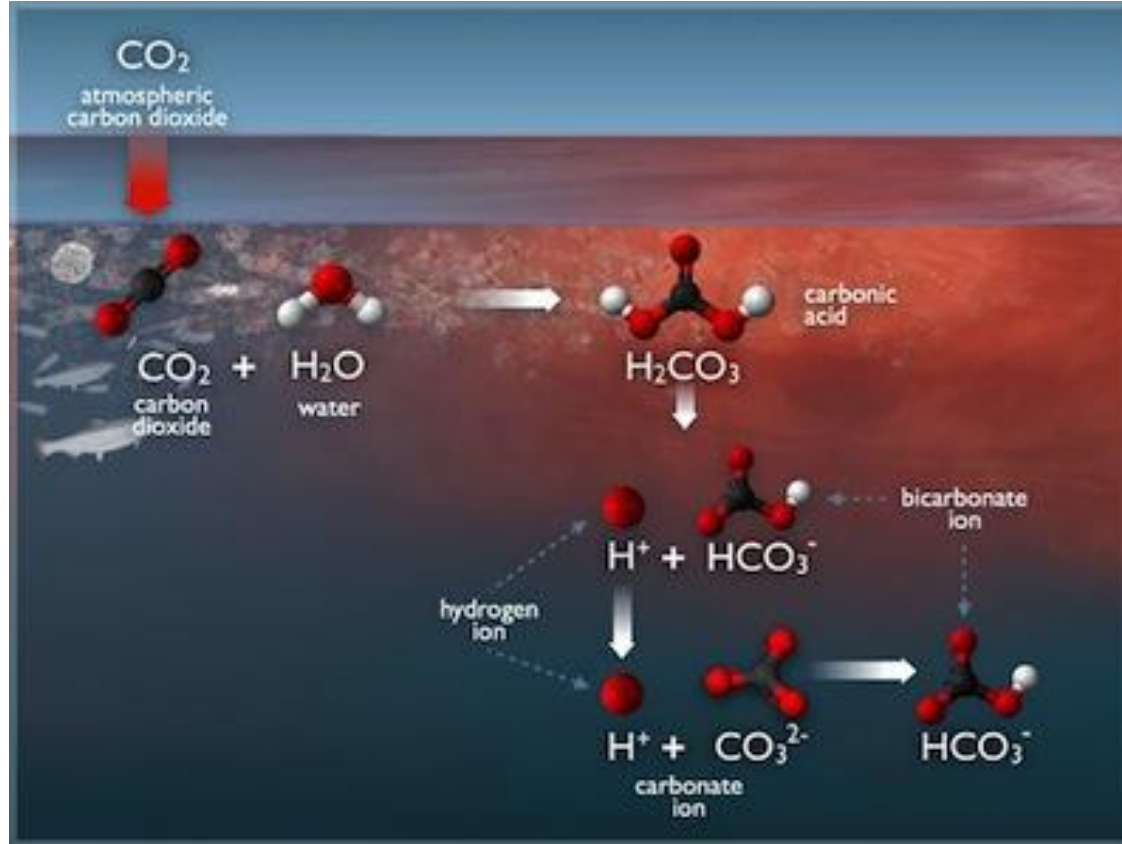
Oceans, CO₂ and climate

- Airborne fraction is 48%: cause of CO₂ problem
- Ocean store about 38000 Pg C vs. 860 Pg C in atmosphere
- Oceans store about 31% of C emitted by humans (fossil fuel and cement)
- This CO₂ storage service of the ocean comes at a price: **Ocean Acidification: the other CO₂ problem**

Why is there so much CO₂ in ocean?

- The ocean has a **large surface area** for exchange (361,900,000 km²)
- And a **large volume** to store (1,335,000,000 km³)
 - Increase in DIC from 1994-2007 is 5-10 μmol kg⁻¹ against background of about 2000 μmol kg⁻¹ (< 0.5%).
- **CO₂ reacts** with H₂O to form carbonic acid (H₂CO₃), a weak acid

When CO₂ dissolves...it becomes an ion that cannot exchange



Carbonic Acid (H₂CO₃)

+

Bicarbonate (HCO₃⁻)

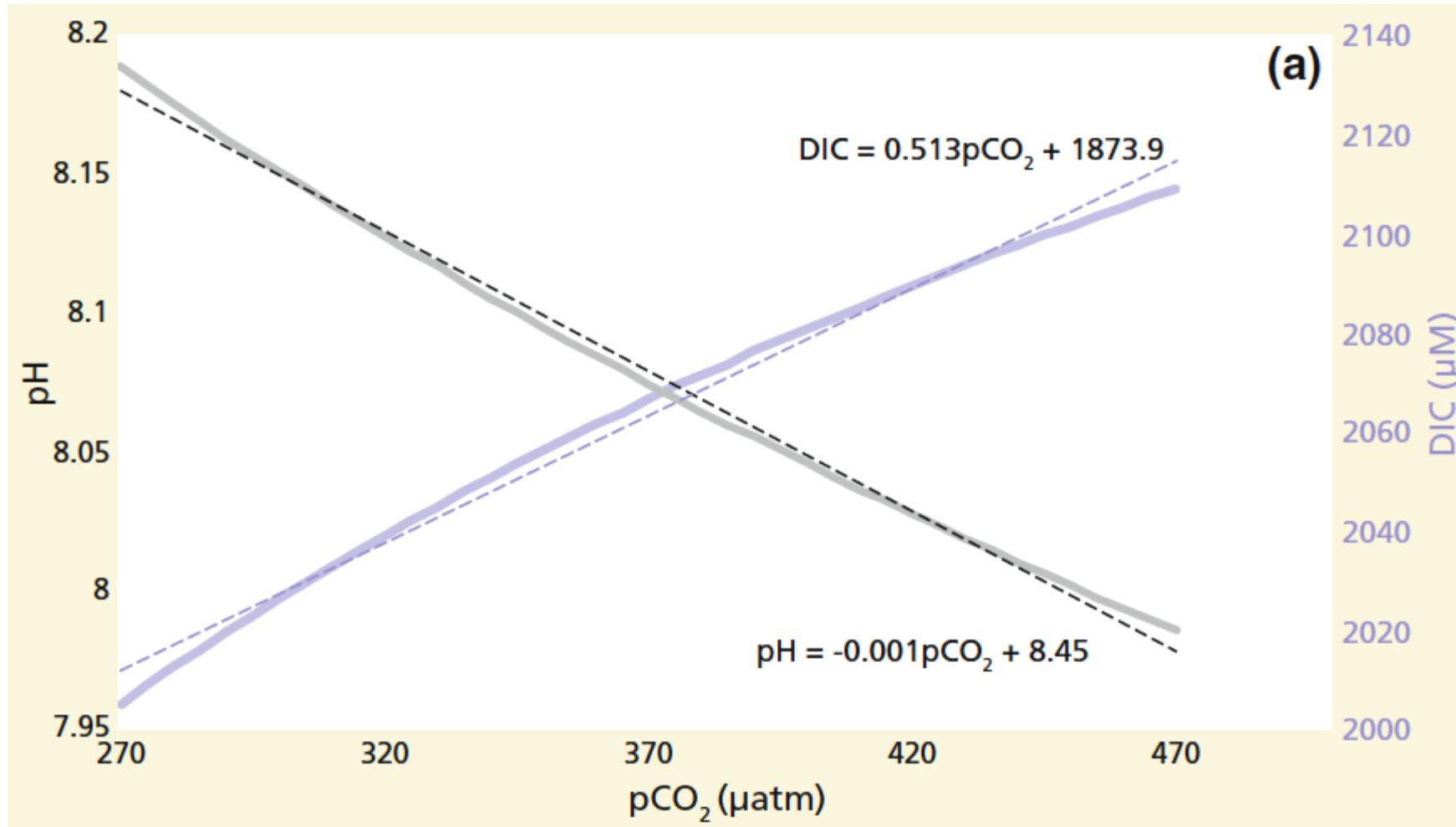
+

Carbonate (CO₃²⁻)

=

Dissolved Inorganic Carbon (DIC or C_T)

pH and DIC as function pCO₂



- Doubling of atmospheric CO₂ from 280 to 560 would give $0.51 \times 280 = 143 \mu\text{mol kg}^{-1}$ increase in DIC
- Relative to background of $2100 \mu\text{mol kg}^{-1}$ this corresponds to 6.8 % increase in DIC.

Revelle Sensitivity Factor

- Given $[\text{HCO}_3^-] \gg [\text{CO}_3^{2-}] \gg [\text{H}_2\text{CO}_3^*]$, the majority of the CO_2 taken up from the atmosphere will not stay as $[\text{H}_2\text{CO}_3^*]$, but react with $[\text{CO}_3^{2-}]$:

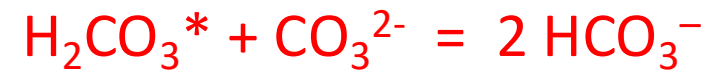


- The (in)efficiency with which this occurs is termed the Revelle factor, i.e.

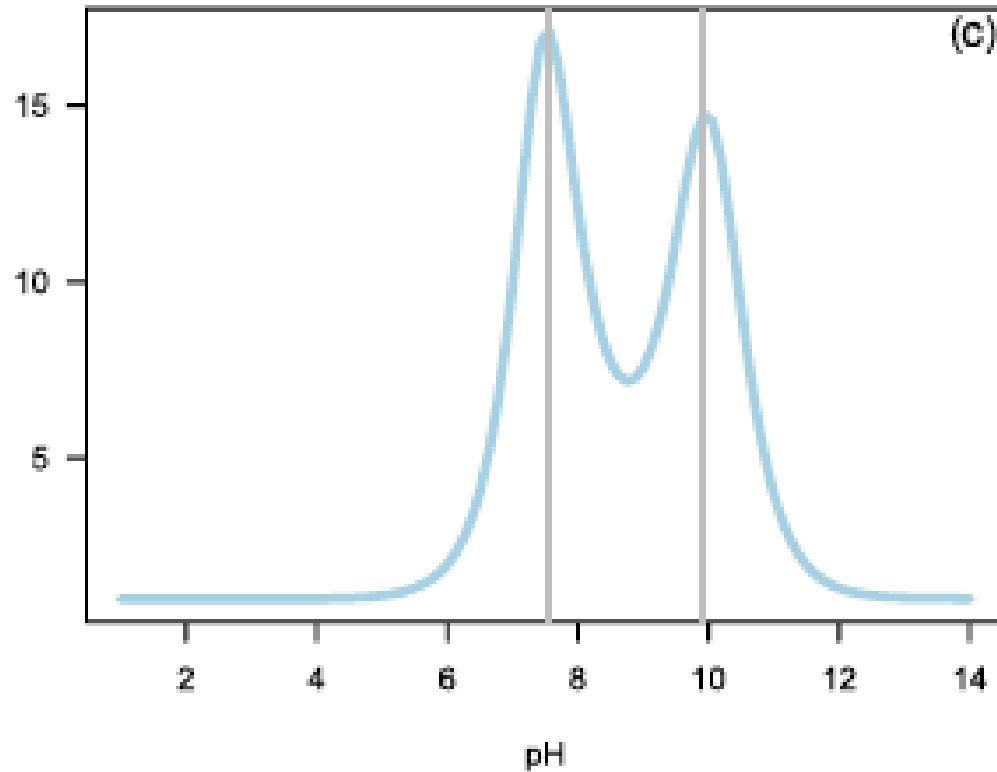
$$R = \frac{\partial \ln p\text{CO}_2}{\partial \ln \text{DIC}} = \frac{\text{DIC}}{p\text{CO}_2} \left(\frac{\partial p\text{CO}_2}{\partial \text{DIC}} \right)$$

- R is high then low efficiency; low R high efficiency

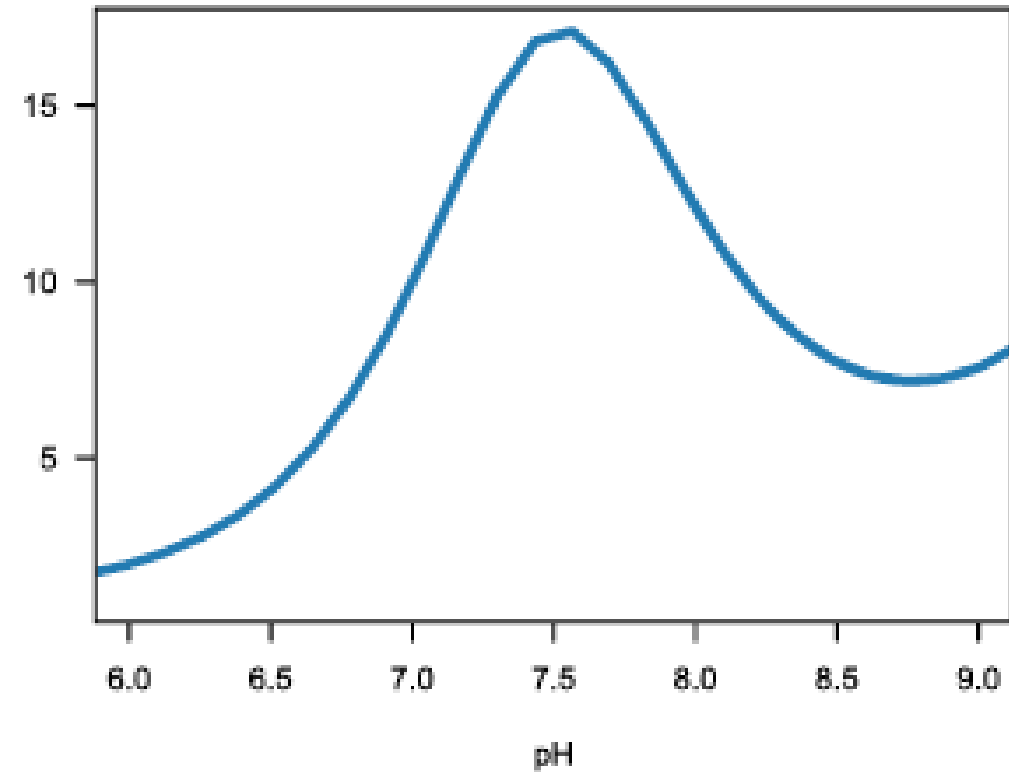
Revelle Sensitivity Factor



Revelle sensitivity factor



Revelle sensitivity factor



Efficiency/Buffering

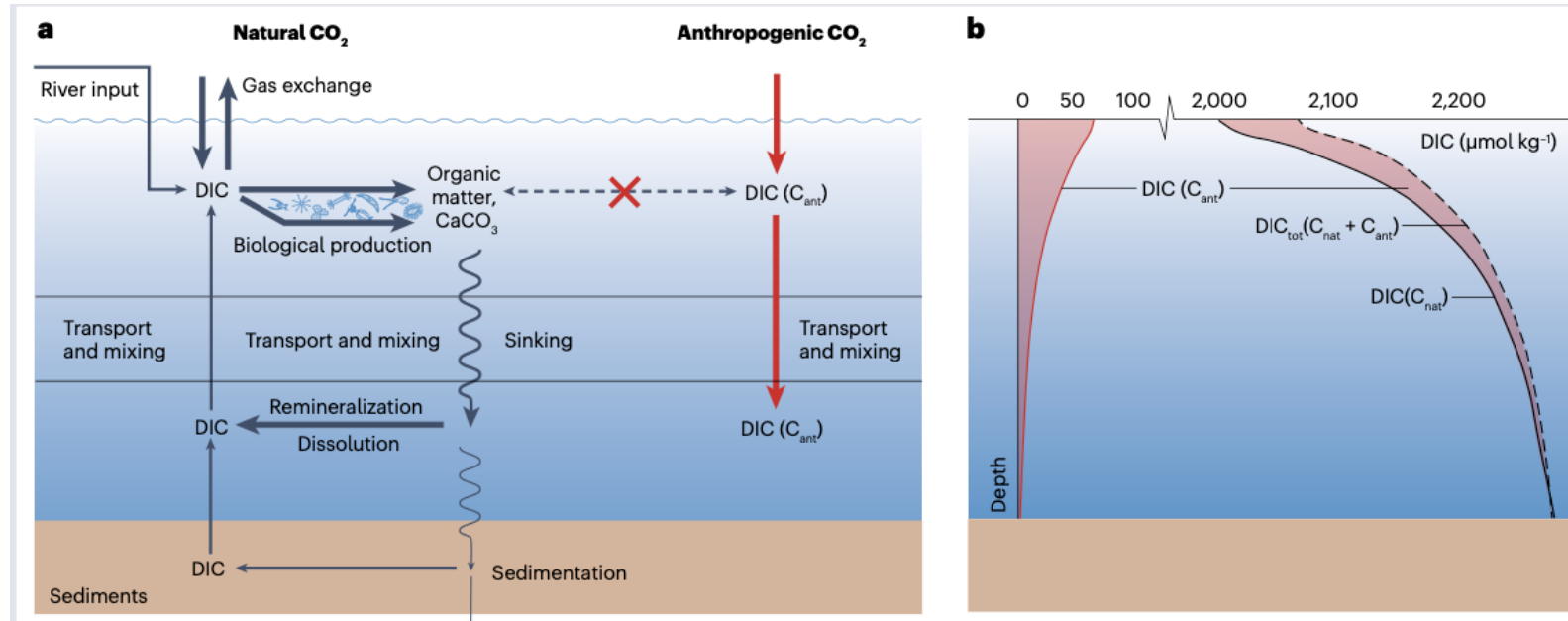
Revelle and increasing CO₂

- R = 15 or 10

$$R = \frac{\partial \ln p\text{CO}_2}{\partial \ln \text{DIC}} = \frac{\text{DIC}}{p\text{CO}_2} \left(\frac{\partial p\text{CO}_2}{\partial \text{DIC}} \right)$$

- Doubling of CO₂ ($\partial p\text{CO}_2 / p\text{CO}_2$) = 1; e.g. (560-280)/(280)
- $\partial \text{DIC} / \text{DIC} = 1/15$ or $1/10 = 0.066$ or 0.1
- Doubling pCO₂ leads to 6.6 or 10 % increase in DIC
(same result as our simple two box model and linear approximation)

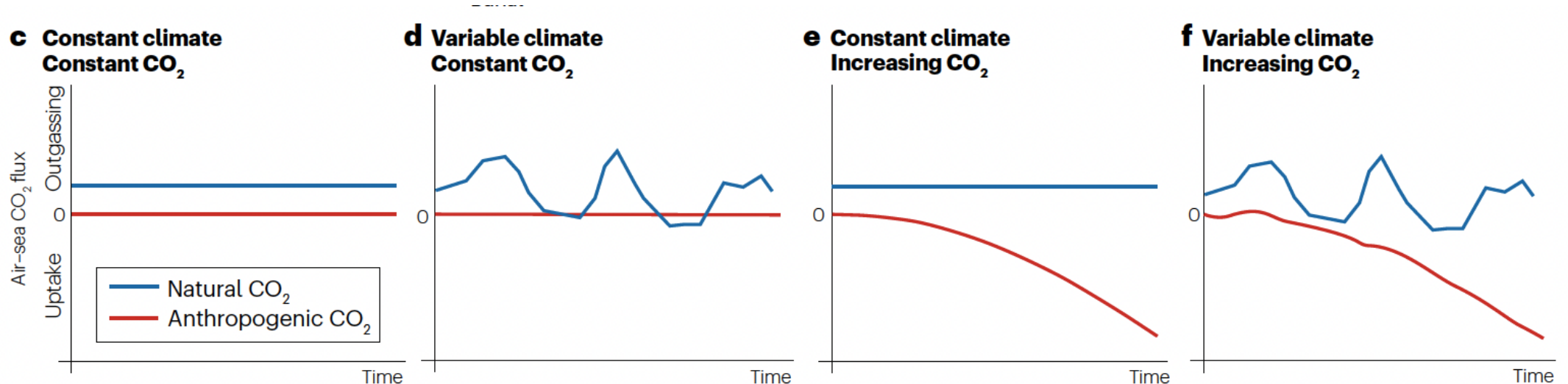
Anthropogenic carbon in the ocean



- Carbon in the ocean participates in multiple processes (biological, chemical, organic/inorganic).
- Anthropogenic carbon is not a directly measurable quantity. It can only be estimated indirectly and is due to physical-chemistry and increasing atmospheric carbon dioxide.
- The anthropogenic signal in the ocean is only a few percent of the (unknown) natural background of dissolved inorganic carbon (DIC).

Anthropogenic carbon in the ocean

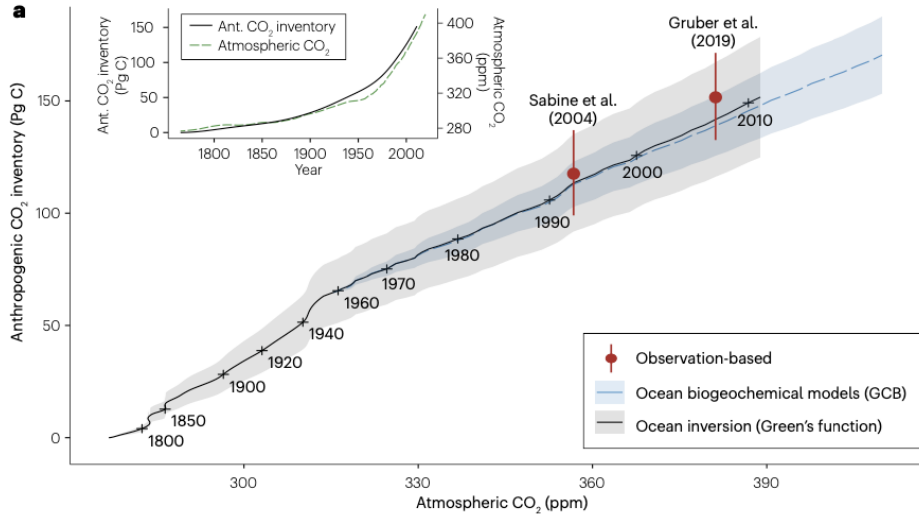
- 90% of DIC in ocean is chemical equilibrium ocean/atmosphere
- 10% of DIC is due to biological pump
- Anthropogenic DIC (C_{ant}) is very small addition due to increase in atmospheric carbon dioxide.
- Important to distinguish between (1) DIC natural, (2) climate change induced DIC changes and (3) anthropogenic DIC.



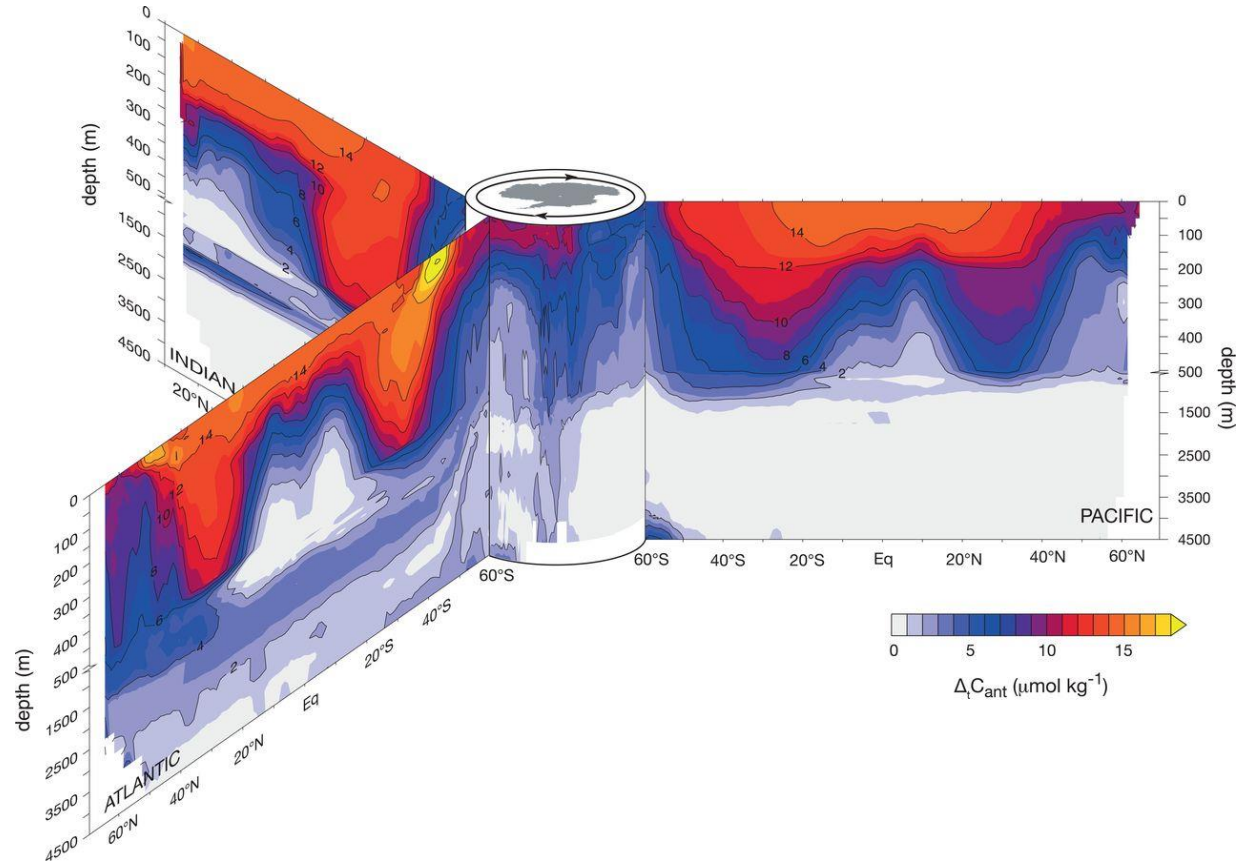
How to quantify C_{ant}

- Cant method:
 - $C_{ant} = DIC - 117 \times PO_4^{3-} - 0.5 (TA + 16 \times PO_4^{3-}) - DIC_{pre-ind.}^0$
 - This removes changes in DIC due to organic matter production/consumption and calcite production/dissolution
 - This term represents the DIC of subducting water in pre-industrial times and is calculated from T, S, TA,..
- Time series of C^*
 - $C^* = DIC - 117 \times PO_4^{3-} - 0.5 (TA + 16 \times PO_4^{3-})$

Anthropogenic C accumulation in Ocean

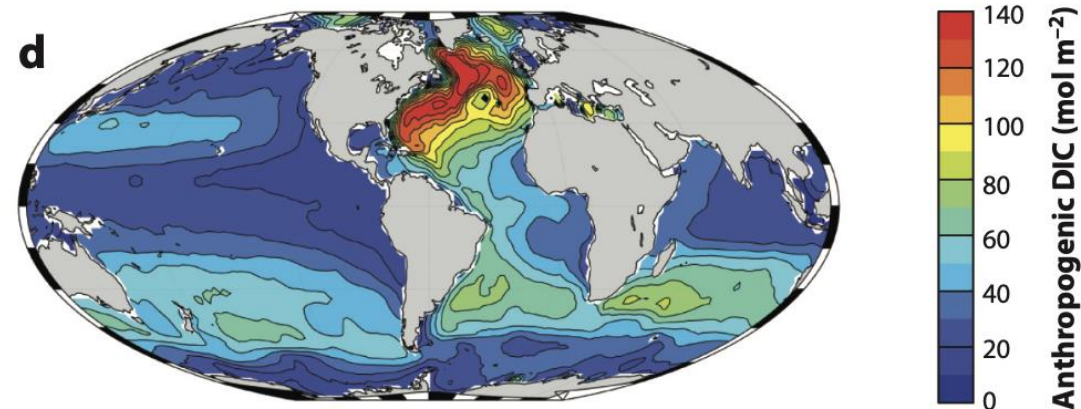
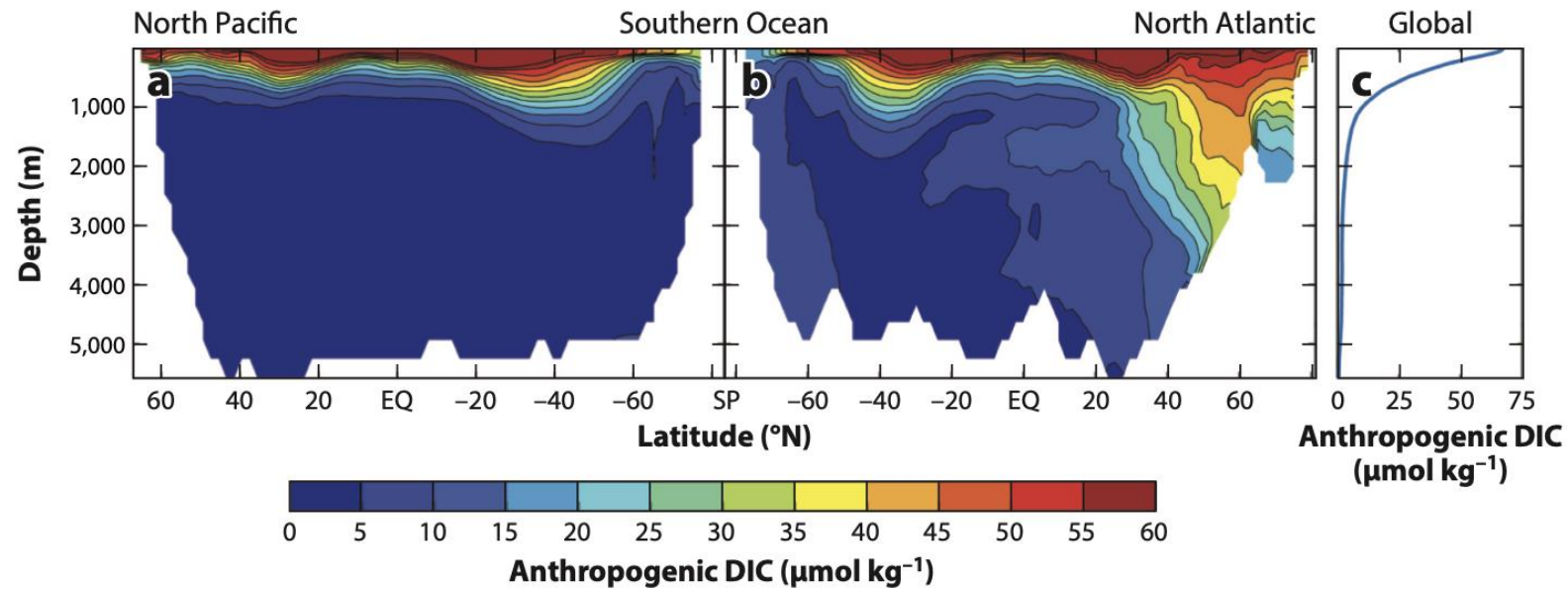


Gruber et al. (2023, Nature Reviews E&E)



Gruber et al. (2019, Science)

Anthropogenic C accumulation in Ocean



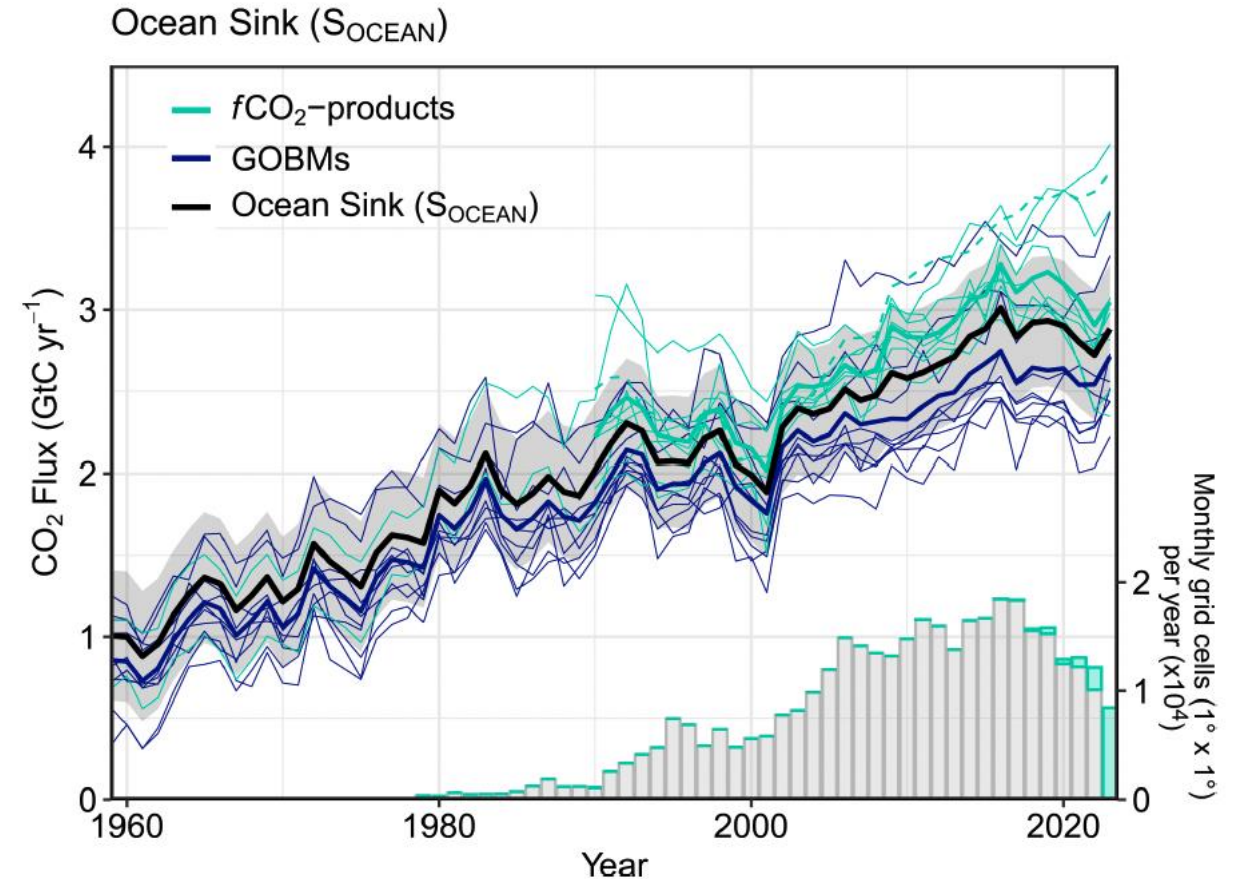
Ocean's Anthropogenic Carbon Uptake

Global Ocean Biogeochemical Models

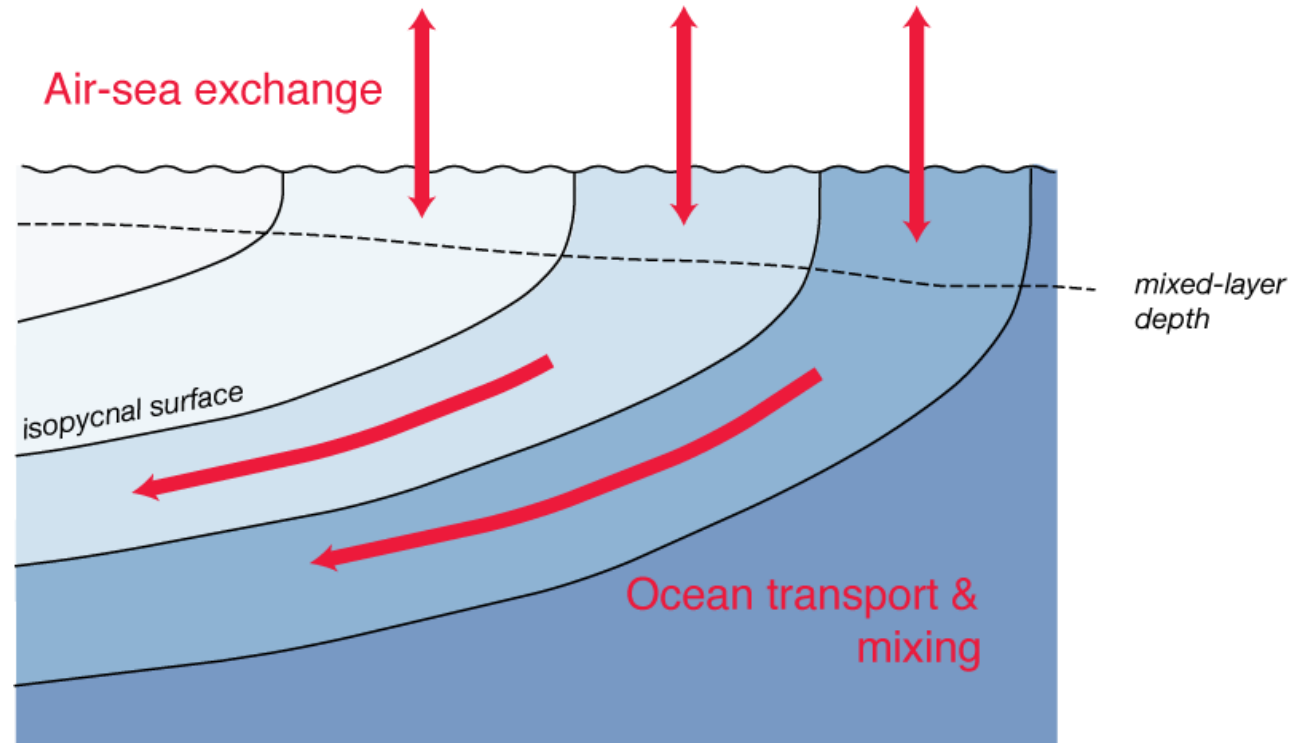
- Physical model to simulate ocean circulation, heat and moisture exchange
- Biogeochemical model to simulate cycles of carbon (organic/inorganic) and a few nutrients
- Changing atmospheric conditions (Carbon Dioxide)

Air-Sea Exchange

- Global observations on CO_2 exchange over ocean surface ($f\text{CO}_2$ -products)
- Corrections needed for river input of carbon minus sediment carbon burial ($0.65 \text{ Gt C yr}^{-1}$).



What is the rate limiting step?



This requires some understanding of the timescales associated with these two processes....

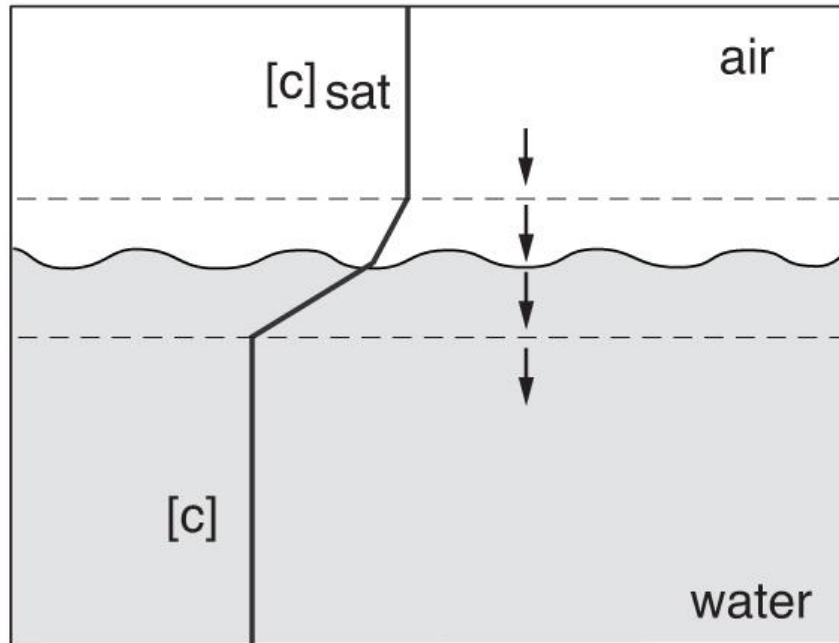
Gas Exchange

Gas exchange rate = FluxDiffusion + BubbleFlux

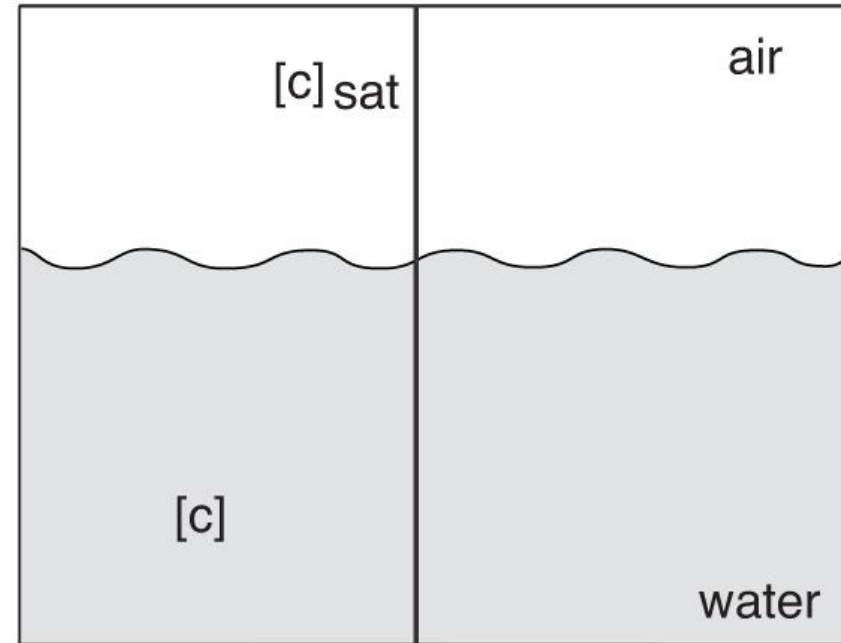
$\text{FluxDiffusion} = k (p_{\text{GASwat}} - p_{\text{GASair}})$

where $k = f(\text{wind speed, turbulence})$

(a) contrast in gas concentration

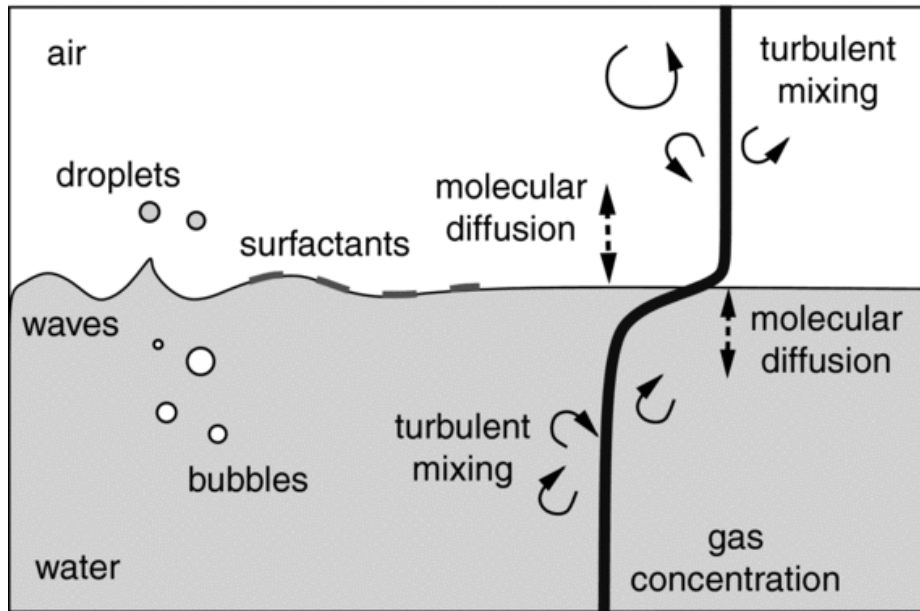


(b) equilibrium state

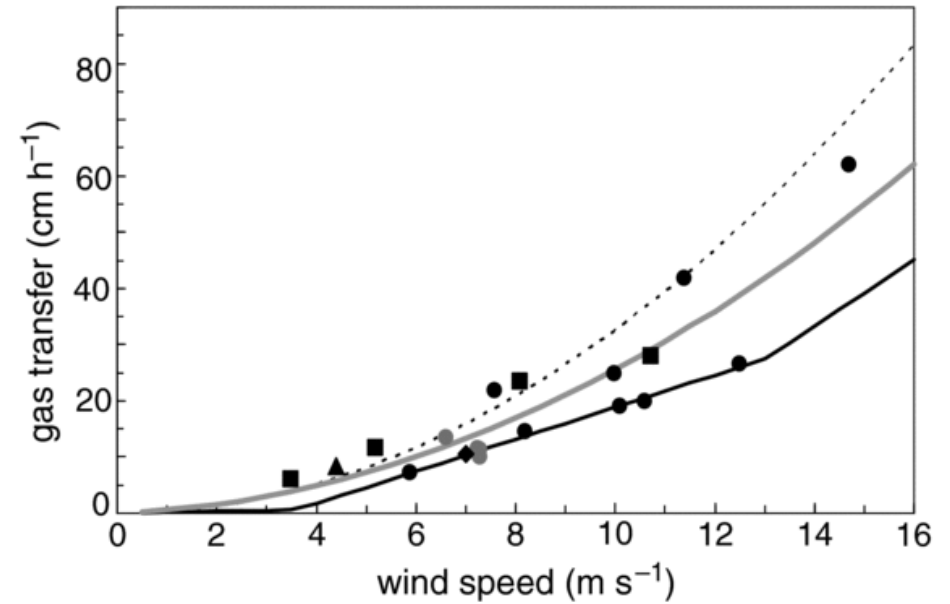


Gas Exchange

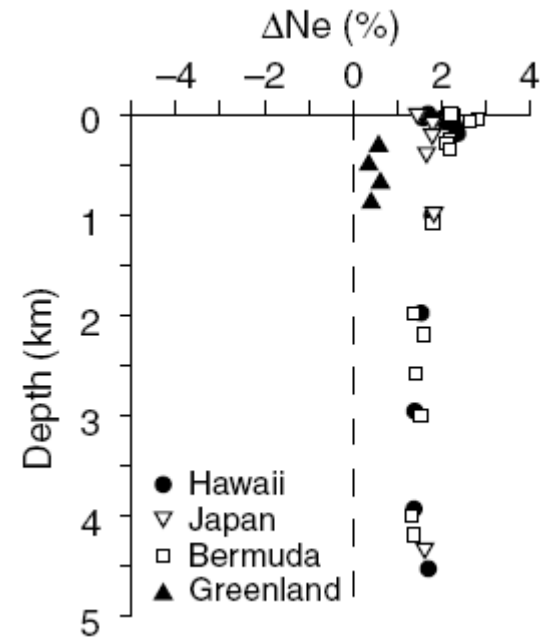
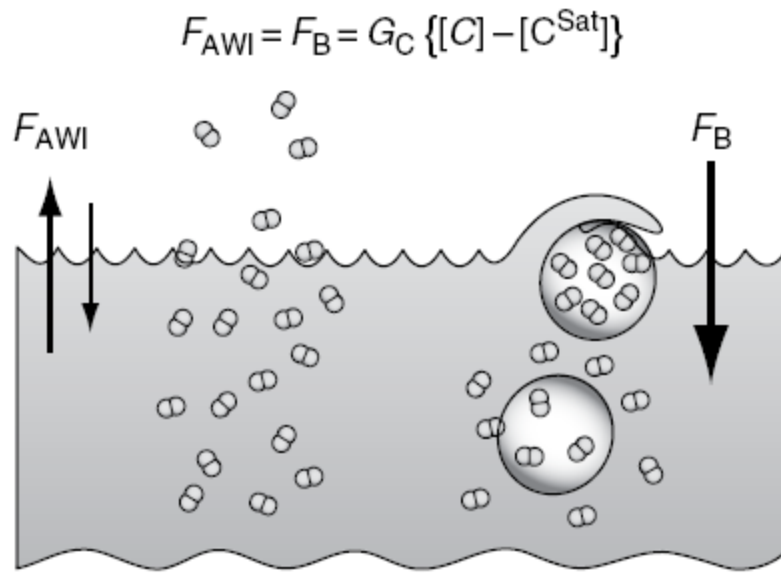
(a) physical processes affecting air–sea transfer



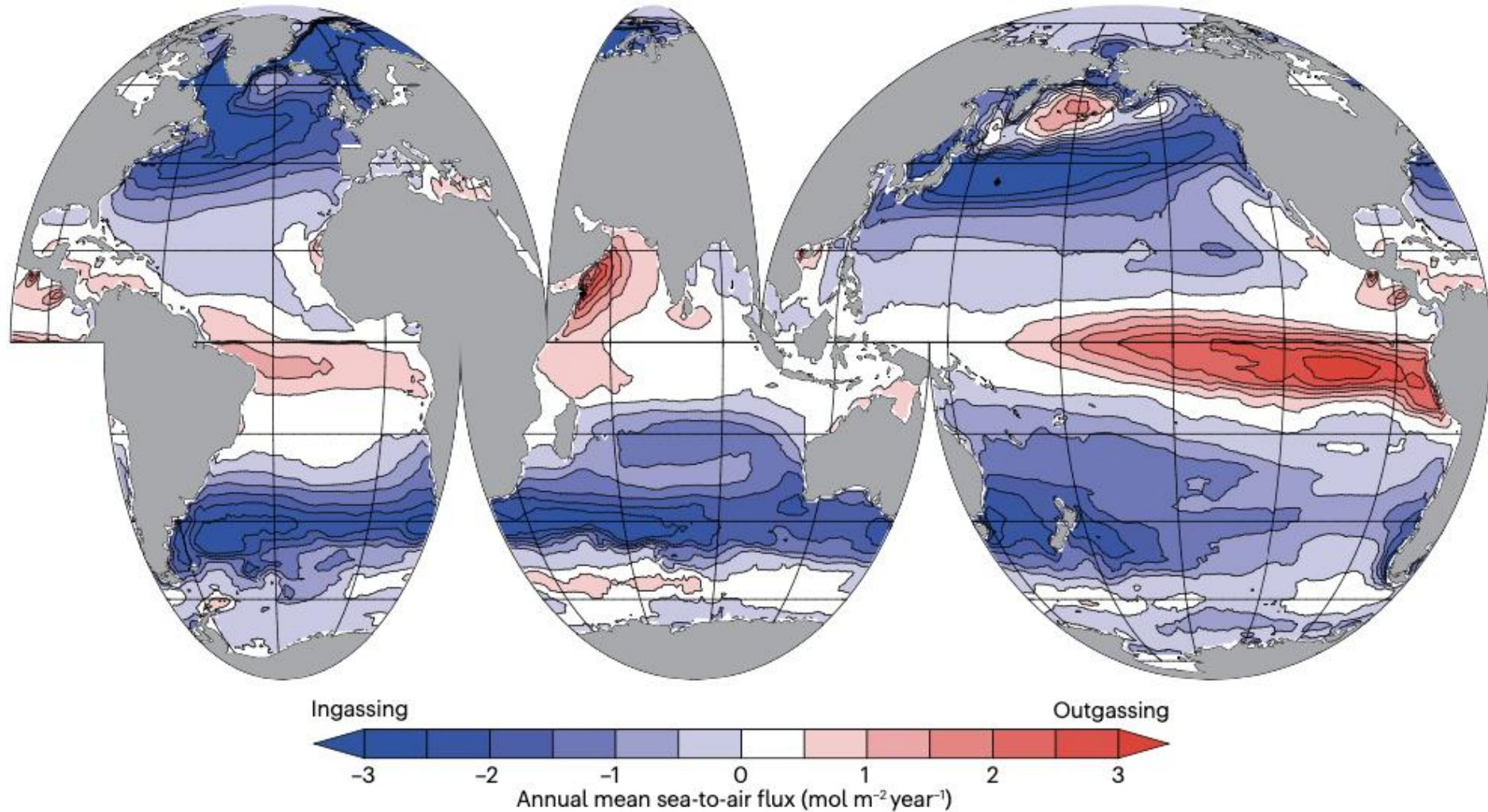
(b) air–sea gas transfer versus wind speed



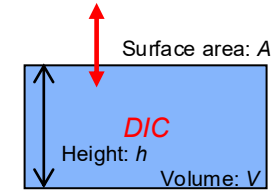
Bubble injection



Carbon dioxide exchange



Gas-exchange: How fast is equilibrium established?



Characteristic time-scale for CO₂

$$\tau_{\text{CO}_2} = \frac{\partial \text{DIC}}{\partial [\text{H}_2\text{CO}_3^*]} \cdot \frac{h}{k_w} \approx 20 \cdot \tau_a$$

buffering

h : depth of surface mixed layer

k_w : gas exchange velocity

τ_a : exchange time-scale of a
“normal” gas

$$\tau_{\text{CO}_2} \approx 20 \cdot \frac{h}{k_w} \approx 20 \cdot \frac{50 \text{ m}}{0.2 \text{ m h}^{-1}} \approx 20 \cdot 250 \text{ h} \approx 210 \text{ days}$$

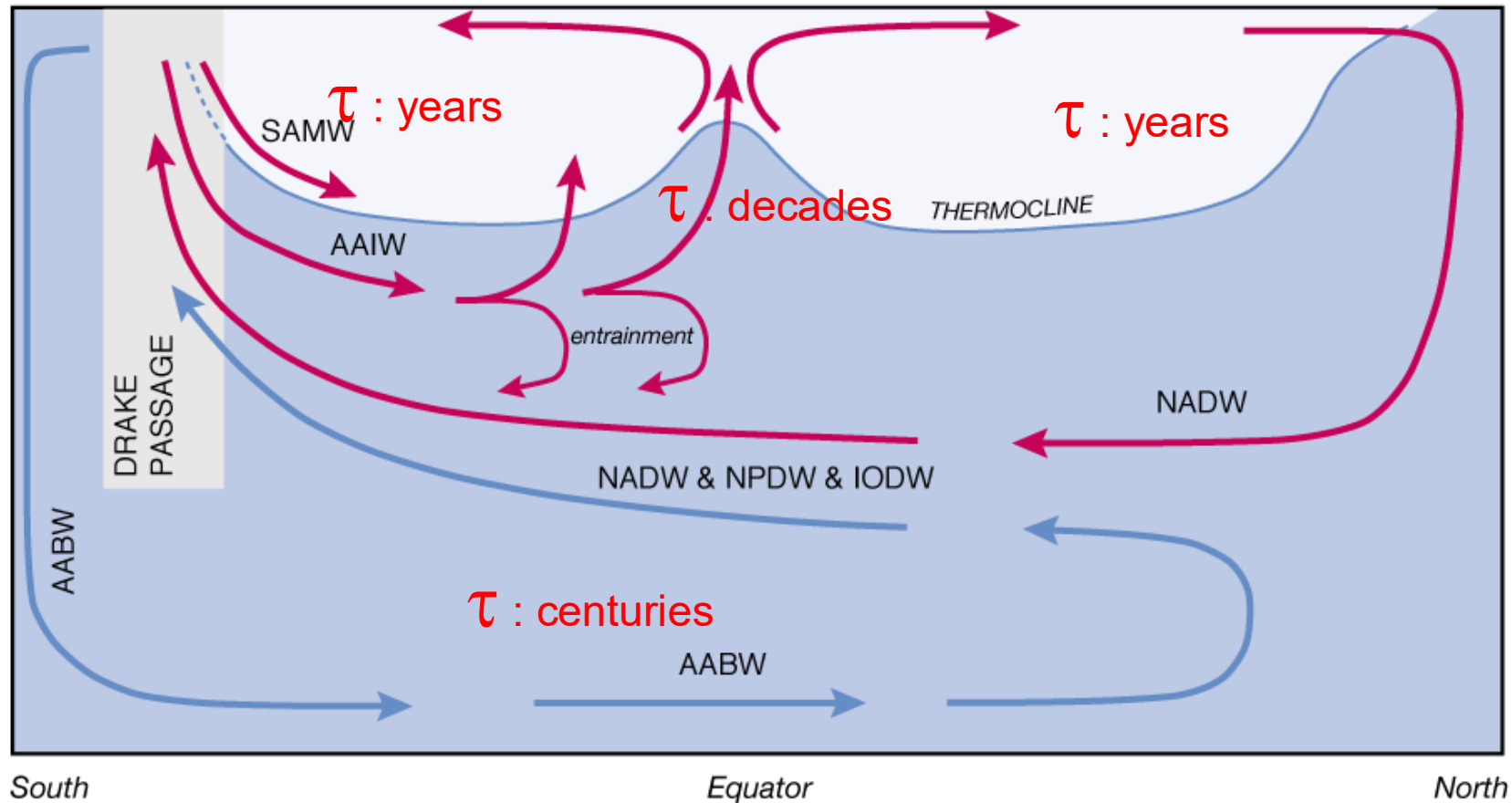
Characteristic time-scale for other gases

$$\tau_a = \frac{h}{k_w} \approx 250 \text{ h} \approx 10 \text{ days}$$

210 days is short relative to ant. perturbation:

gas-exchange is generally NOT limiting ant. CO₂ uptake

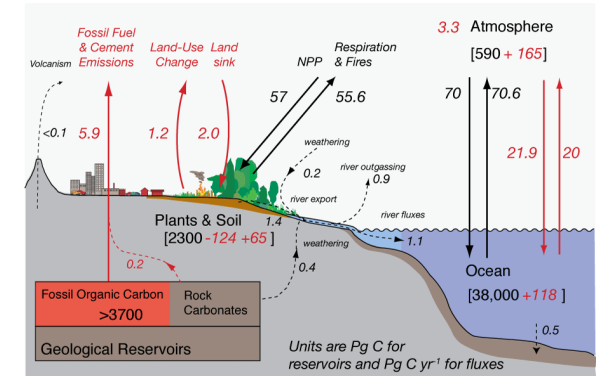
Ocean transport and mixing: time-scales associated with large-scale overturning



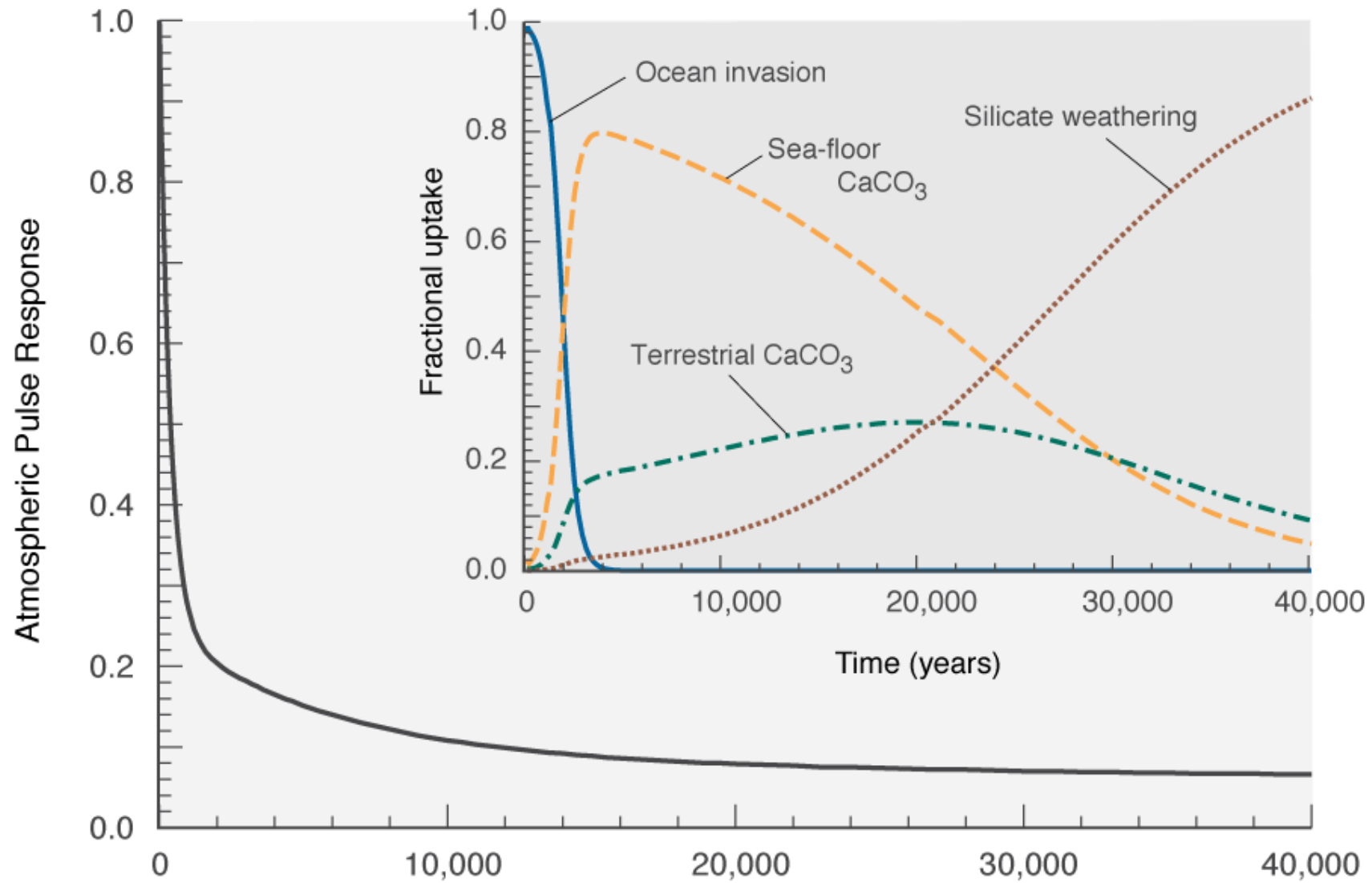
Large-scale ocean interior circulation is occurring on timescales that are long relative to the atmospheric perturbation, i.e. ocean transport is the rate limiting step!

How long does CO₂ stay in atmosphere

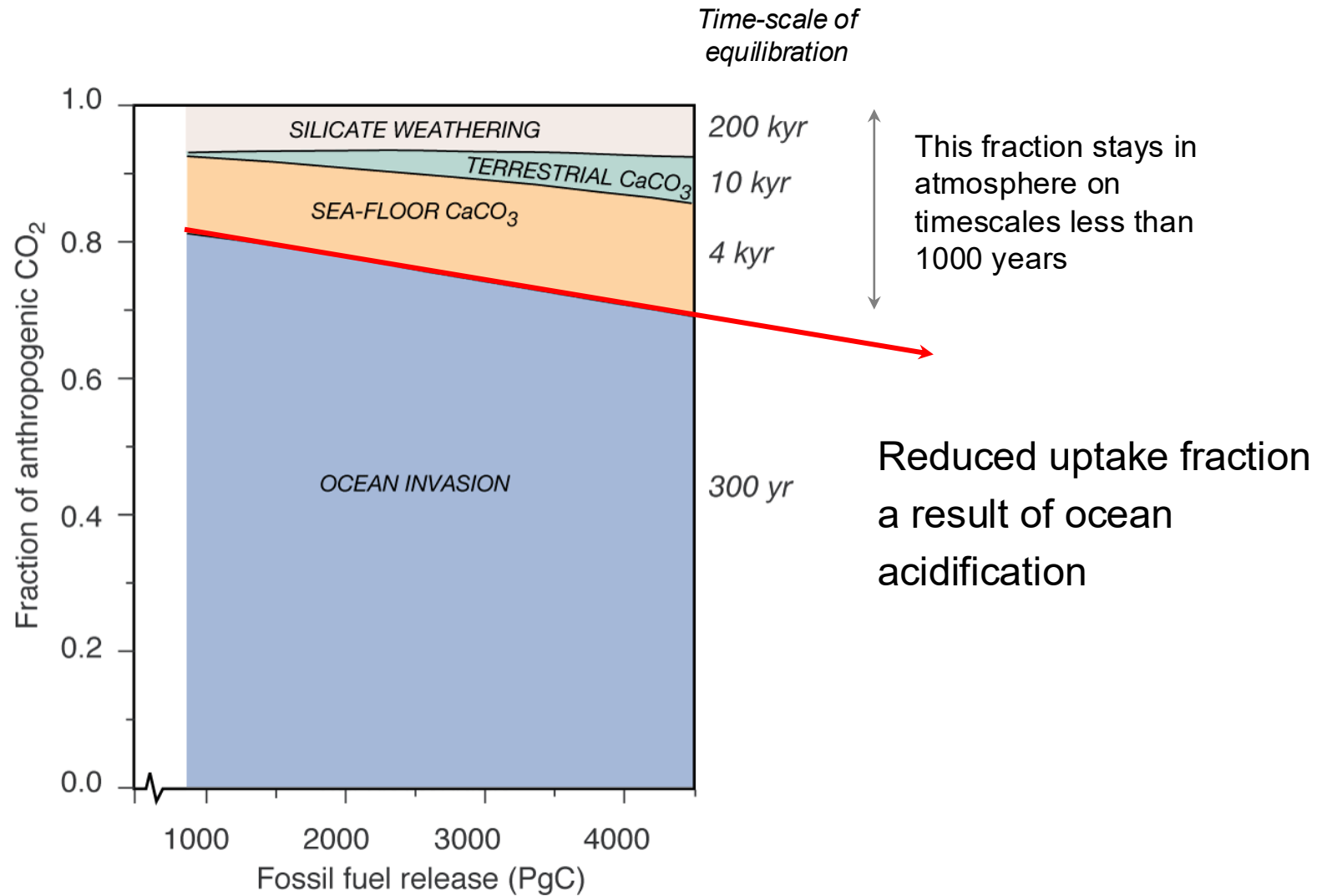
- Residence time of natural C (steady state)
 - Stock/flux = 600/190 (ocean+land) = 3.1 year
- Residence time of anthropogenic C
 - Stock/flux = (160 to 200)/(3 to 4) = 40 to 70 year
- However, this does not apply to anthropogenic carbon dioxide:
a small fraction is left for years to come
 - Geological Record and Models show that steady-state will be reached only after 10,000 to 100,000 years



The ultimate fate of anthropogenic CO₂

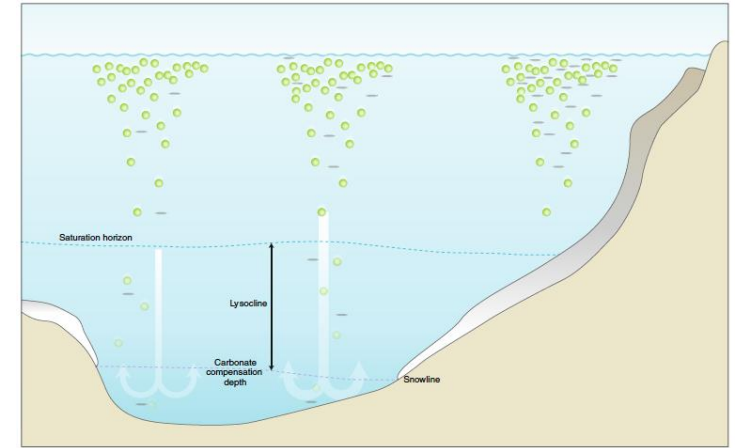


The ultimate fate of anthropogenic CO₂

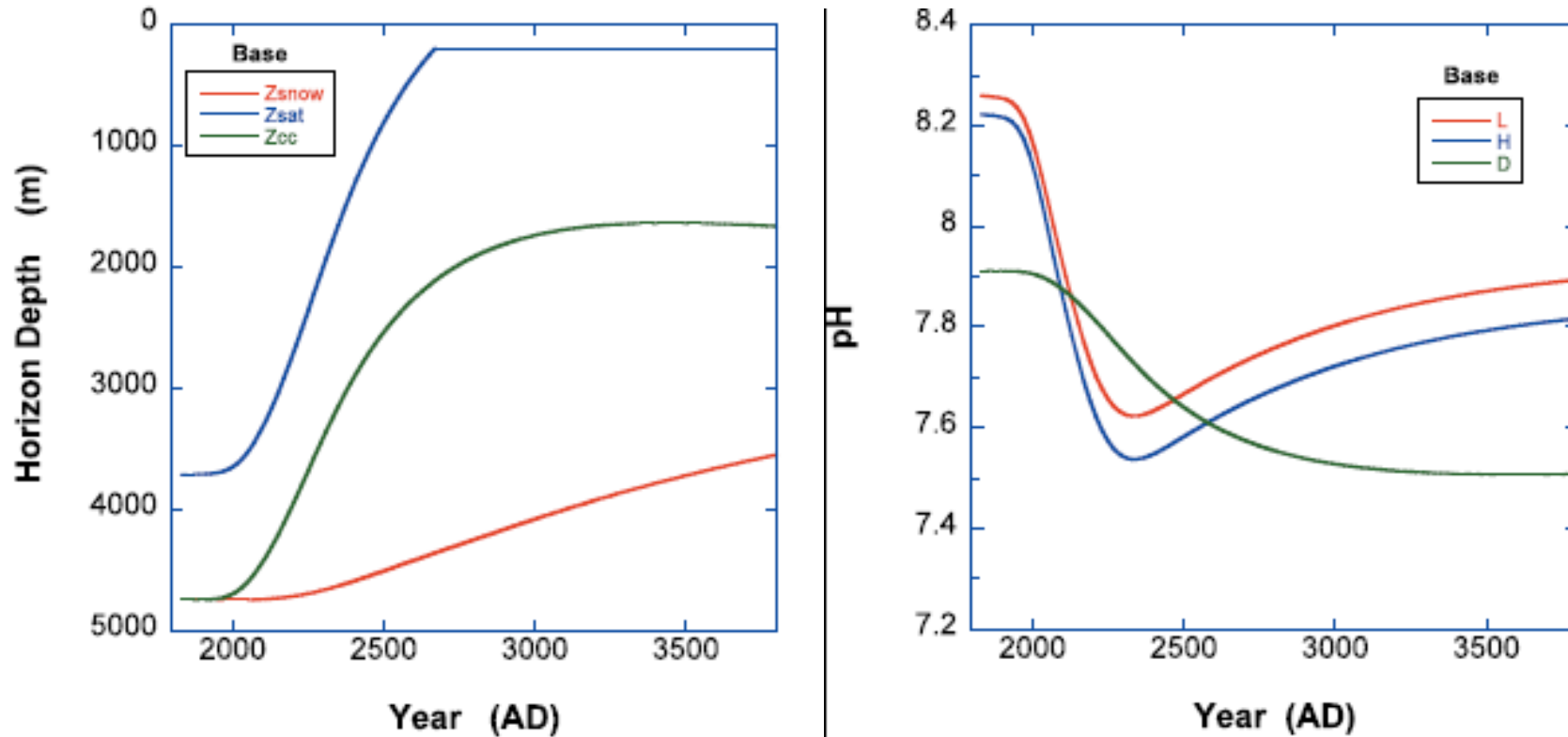


Carbonate Compensation Dynamics

- At steady-state:
 - CCD equals snow-line
 - Saturation depth is ~1 km shallower than CCD
- During Acidification:
 - Saturation depth and CCD diverge and shallow rapidly
 - Snowline rises slowly is uncoupled from CCD
- During transition from high to low CO₂ world:
 - Saturation depth and CCD converge but remain different
 - CCD approaches snowline



Ongoing Transient in Carbonate Compensation



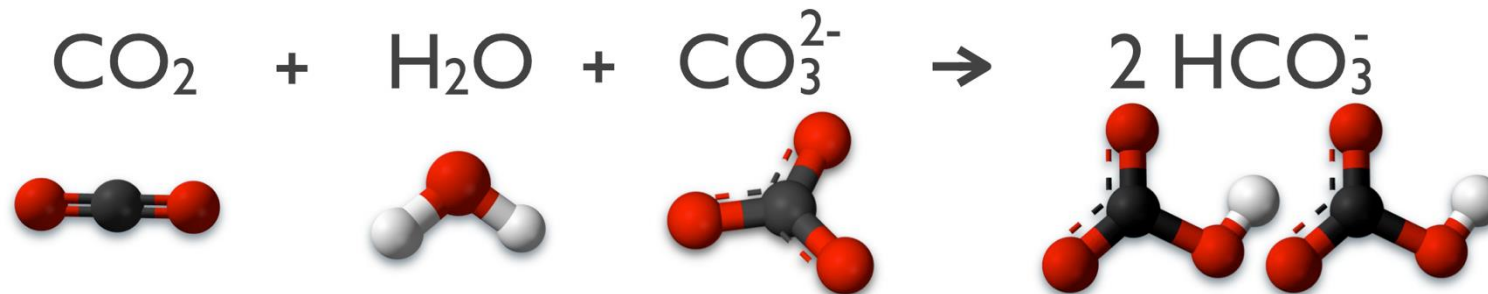
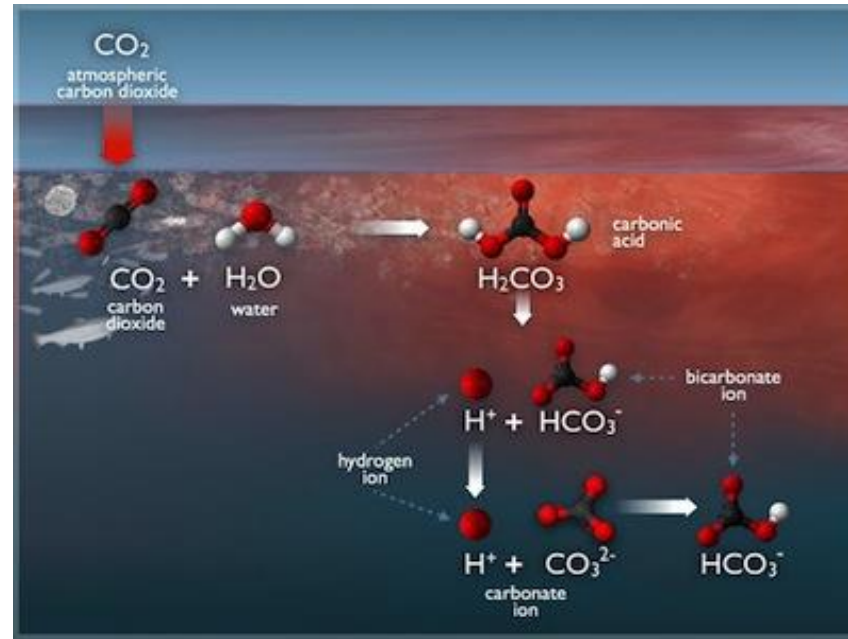
Carbonate Compensation

- Chemical or canonical compensation focusses on dissolution (at the seafloor)
- Biological carbonate compensation takes into account response of calcifying organisms, because of ocean acidification.
- Dissolution increases TA, while precipitation decreases TA, hence calcification matters for TA and thus buffering ocean.

See box 5.2

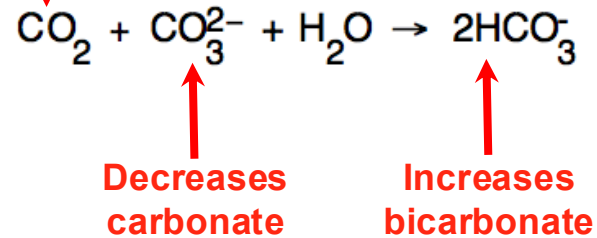
Boudreau et al. (2018)

What happens with the CO₂ in the ocean in near future



Ocean Acidification

Increase in
dissolved CO₂



Carbonation:
increase of dissolved inorganic carbon

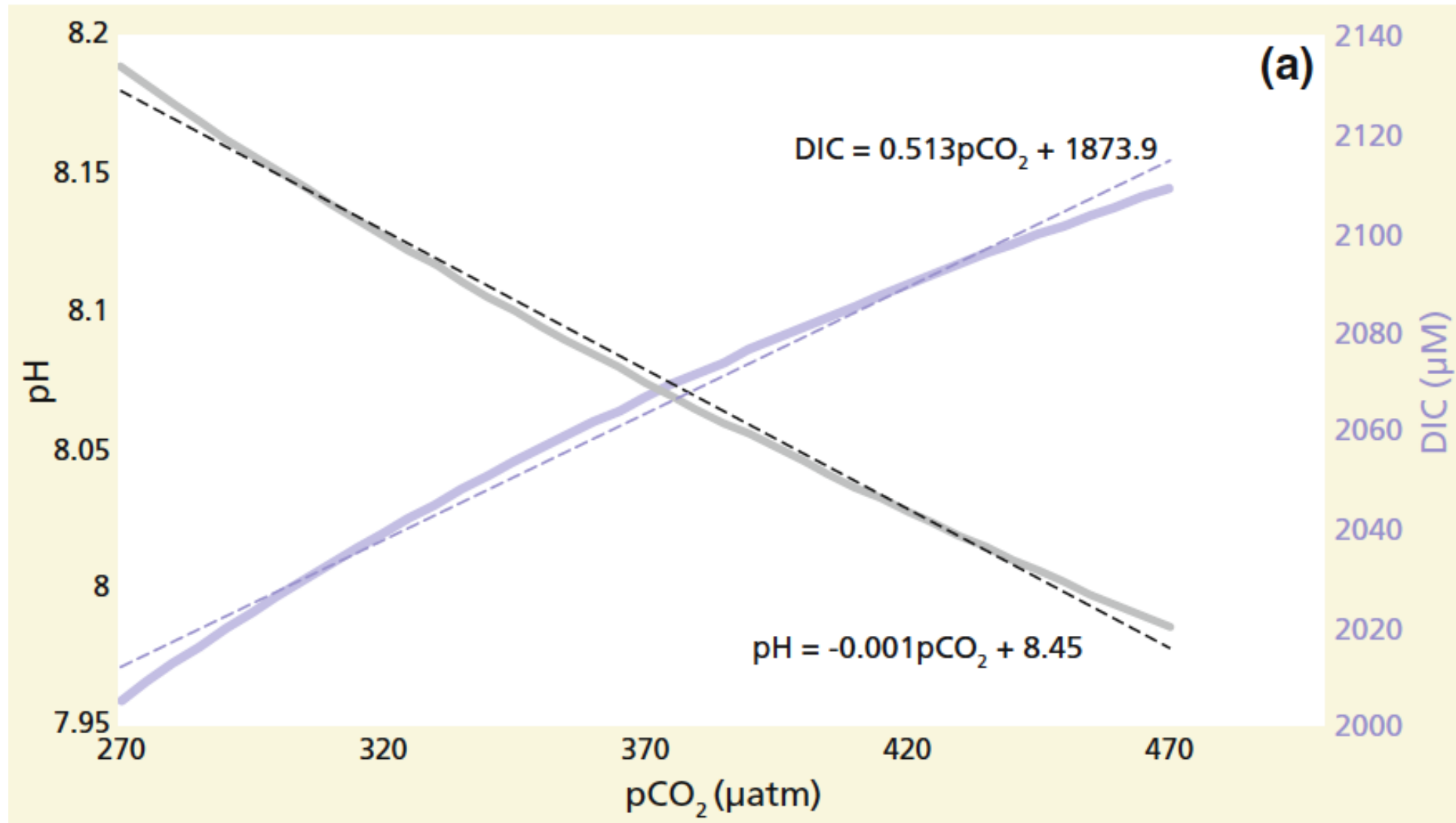
$$[\text{H}^+] = K_2^* \frac{[\text{HCO}_3^-]}{[\text{CO}_3^{2-}]}$$

Hydrogen ion
concentration
increases and pH
decreases

Acidification:
increase of hydrogen ions

Acidity increases:
“*ocean acidification*”

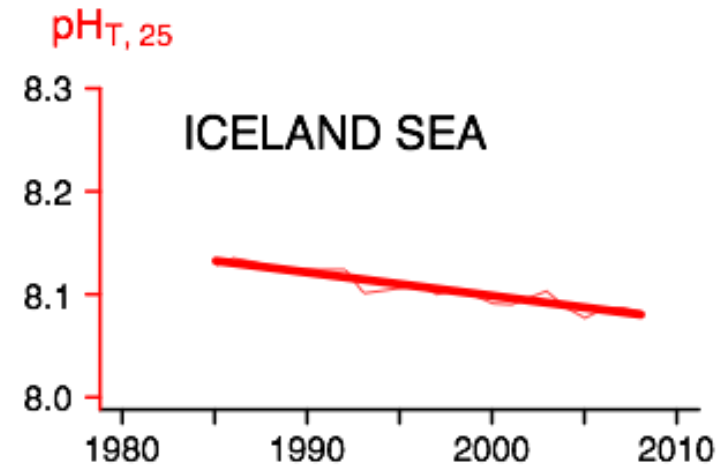
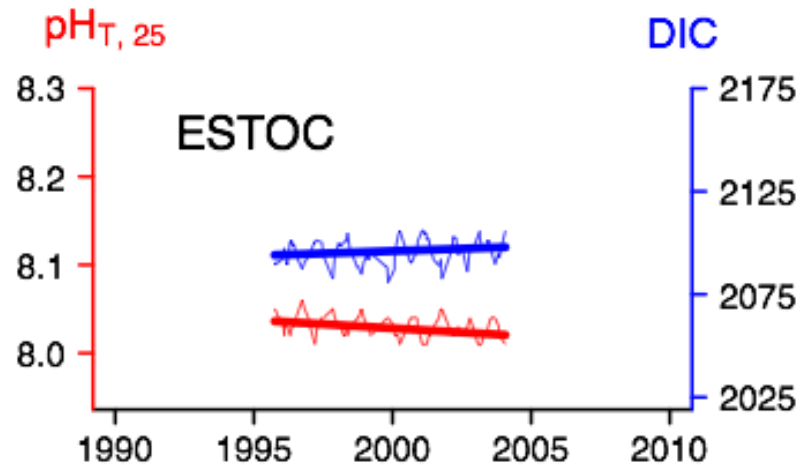
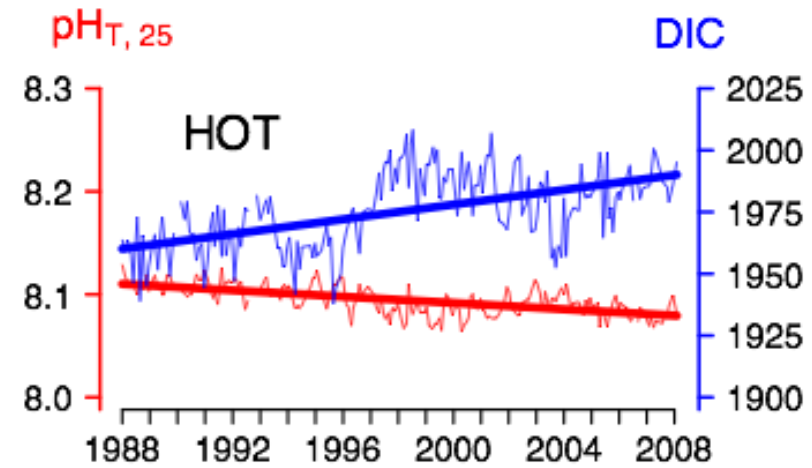
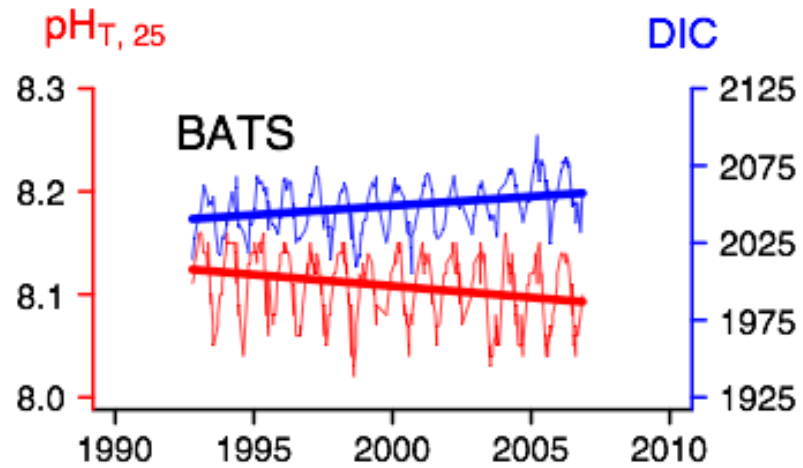
pH and DIC as function pCO₂



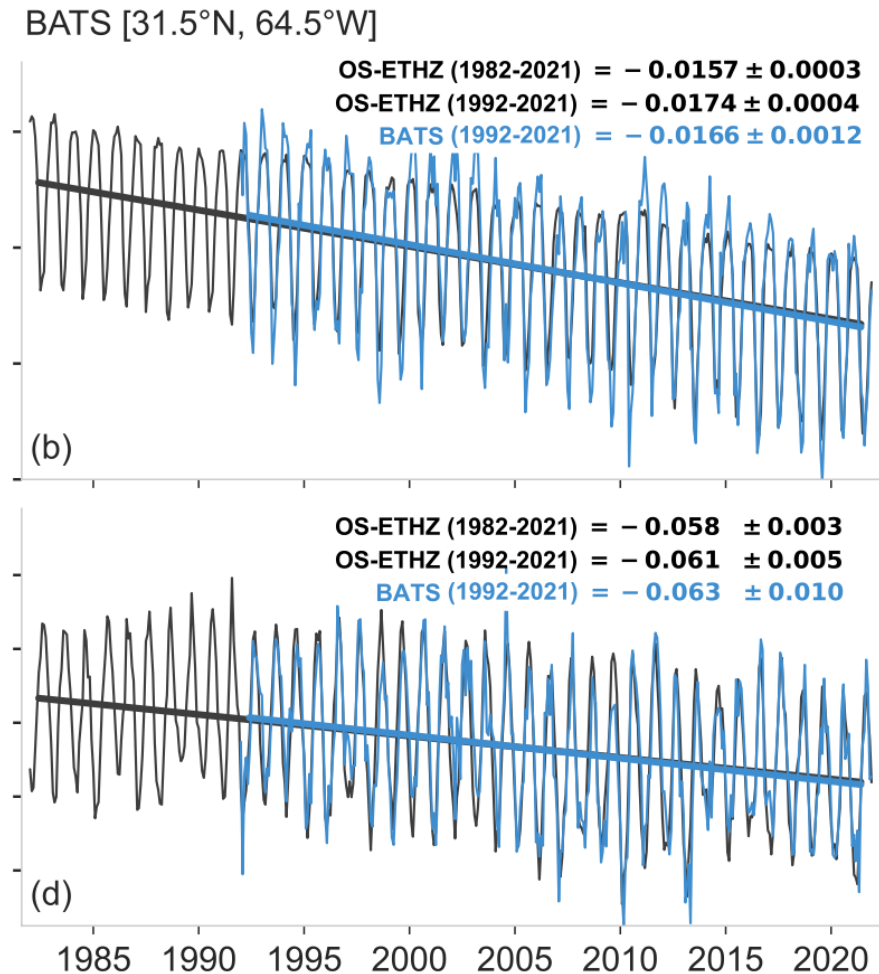
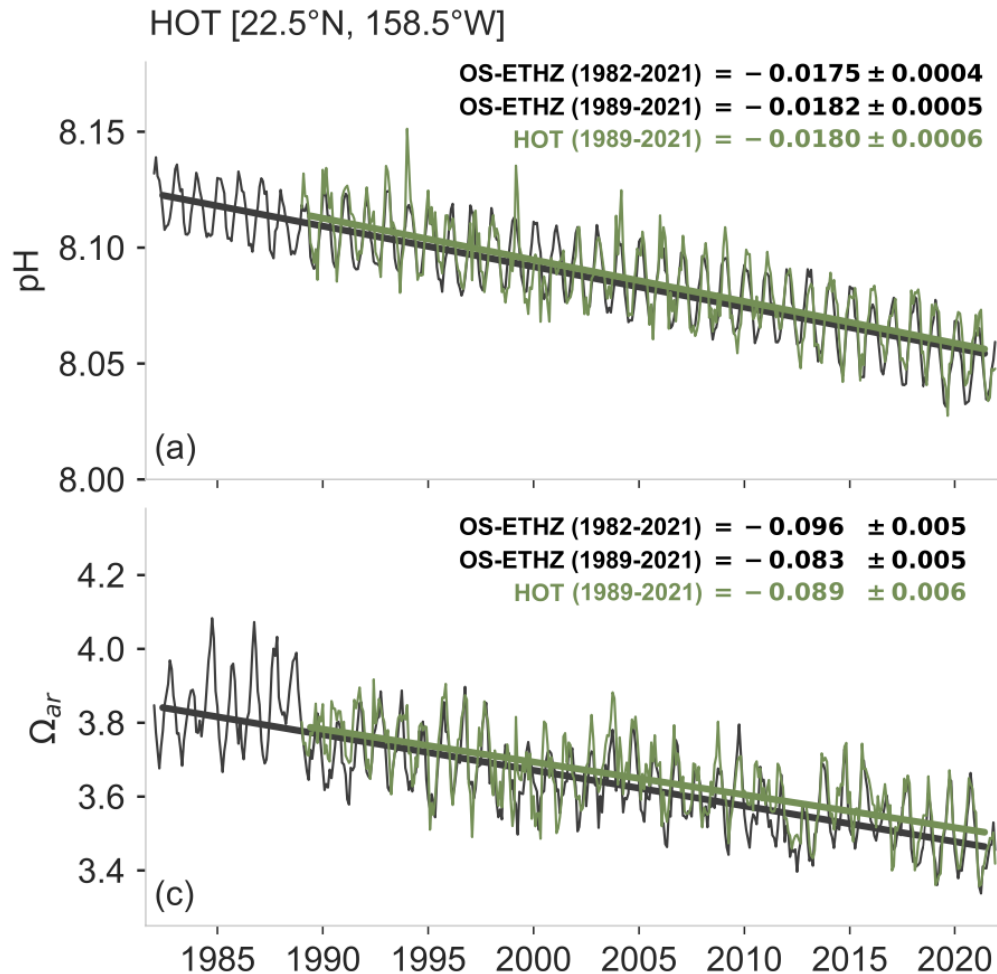
A simple calculation

- Annual CO₂ increase is about 2.8 ppm or μatm
- Regression of pH with $p\text{CO}_2$:
 - $\text{pH} = -0.001 p\text{CO}_2$
 - Annual pH **decline of 0.0028** unit
- Regression of DIC with $p\text{CO}_2$:
 - $\text{DIC} = 0.513 p\text{CO}_2$
 - Annual DIC increase of $1.4 \mu\text{mol kg}^{-1}$

Observed DIC increases and pH decreases as calculated using thermodynamics



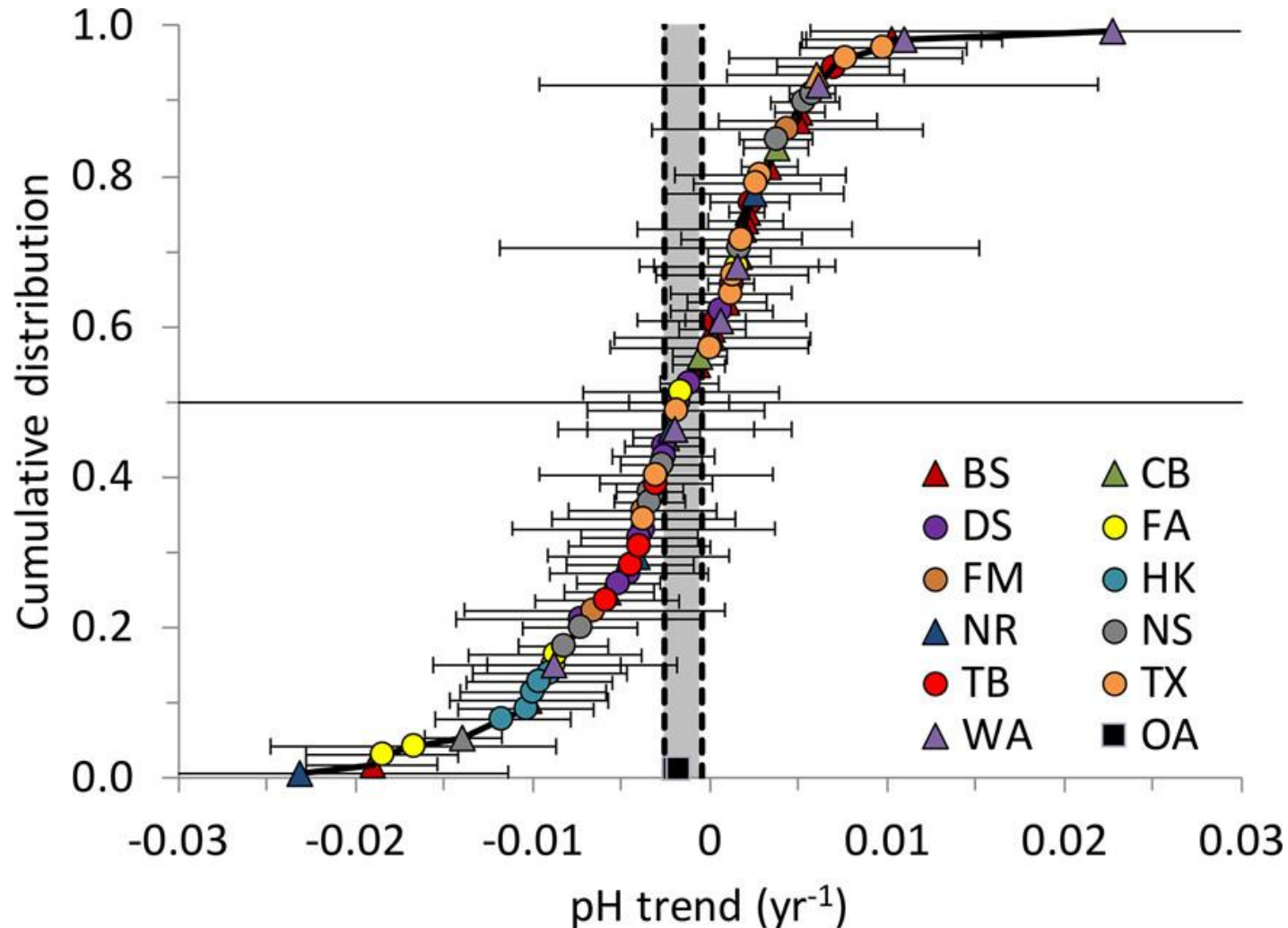
40 year long records



Ocean Acidification

- Observations in **open ocean** fully consistent with thermodynamics and CO_2 increase in atmosphere
- pH dynamics in **coastal systems** poorly understood:
 - much larger decreases, sometimes increase
 - other factors: low oxygen, eutrophication, atmospheric deposition,

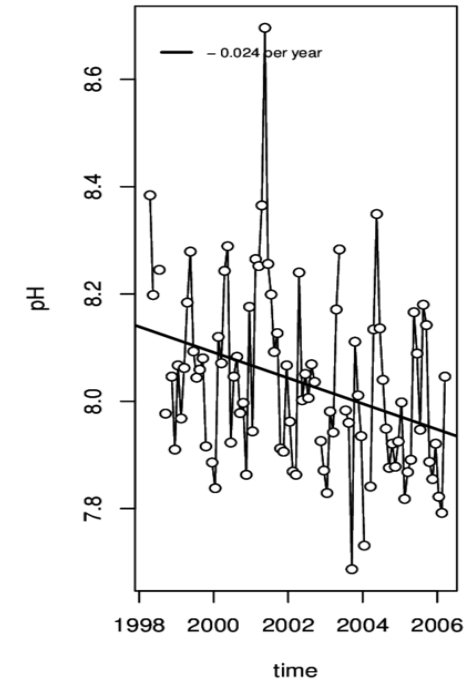
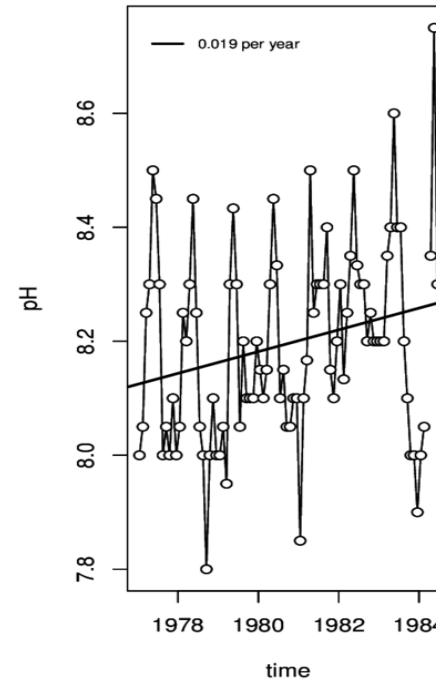
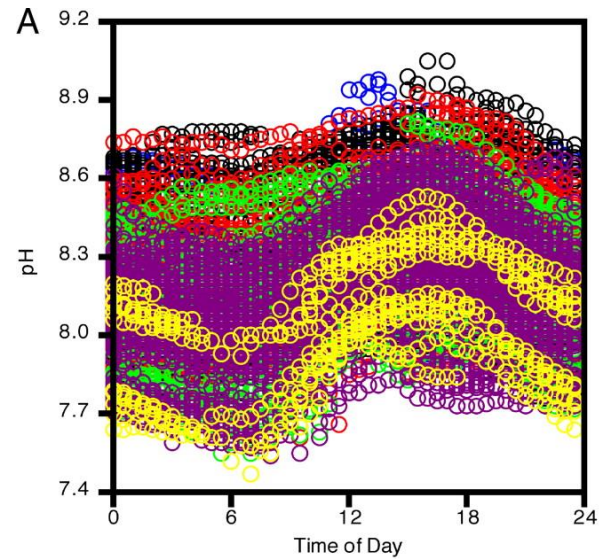
Coastal Acidification



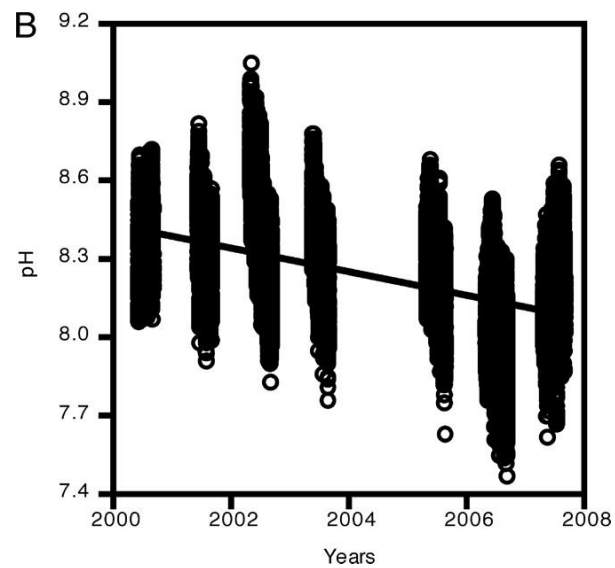
BS = Baltic Sea, CB = Chesapeake Bay, DS = Danish Straits, FA = French Atlantic coast, FM = French Mediterranean coast, HK = Hong Kong coast, NR = Neuse River Estuary, NS = North Sea, TB = Tampa Bay, TX = Texas estuaries, and WA = Western Australia estuaries.

Carstensen & Duarte (2019)

Coastal Acidification



North Sea

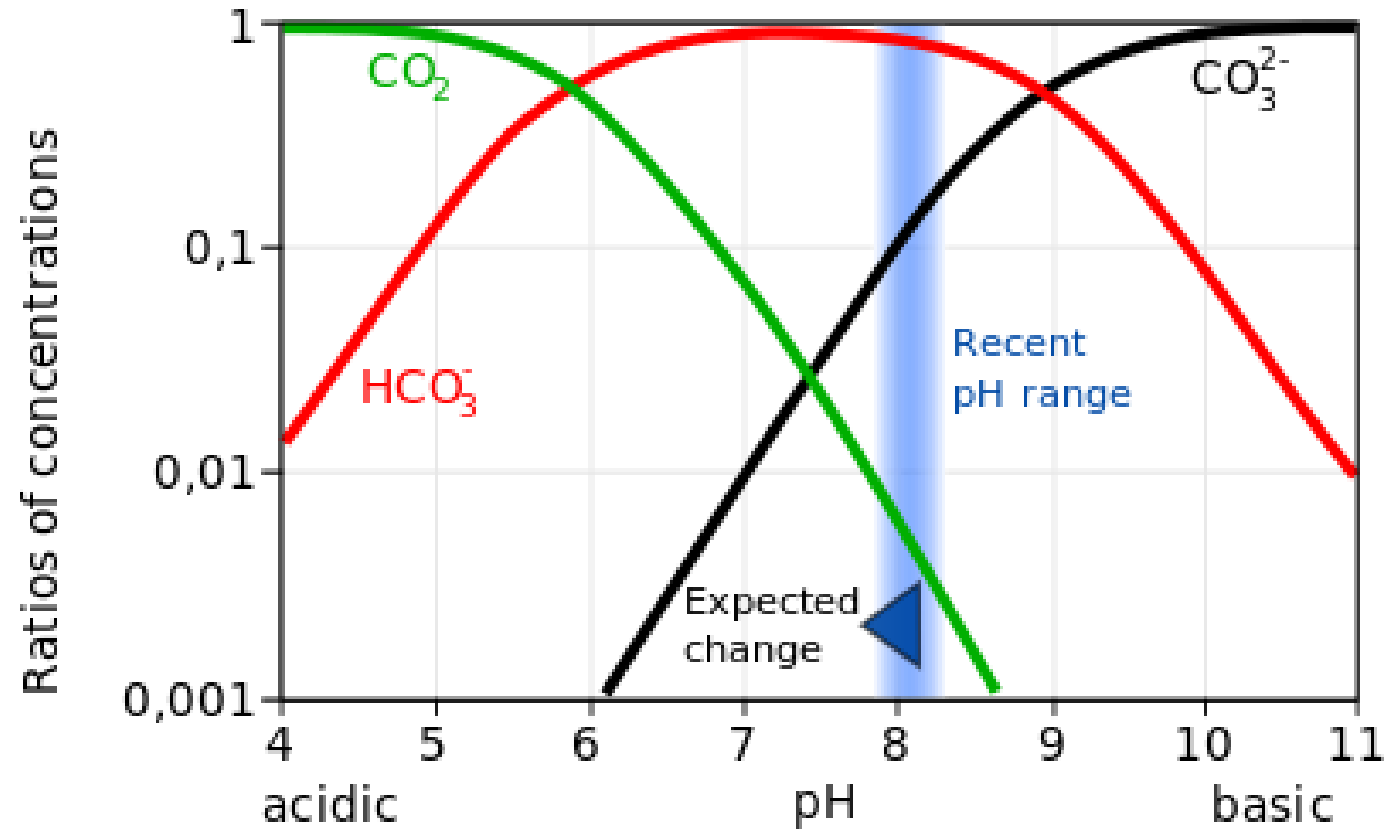


Washington Coast, USA

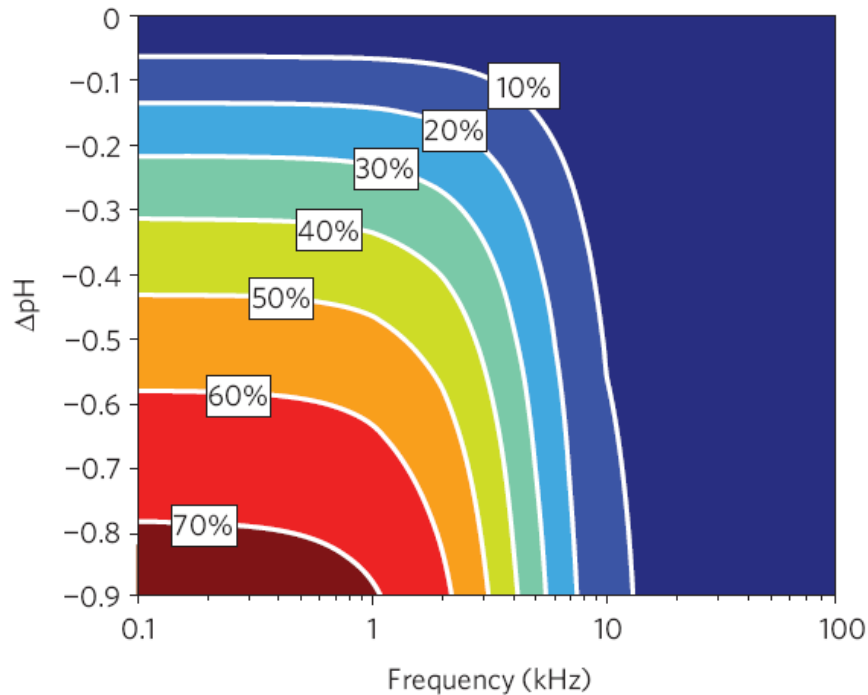
So what, why should we be interested or worry?

1. Consequences for ocean pH:
 - 101 chemistry: master variable
 - influence on sound penetration
 - speciation of trace elements
2. Buffer capacity and uptake of anthropogenic carbon by ocean decline (positive feedback)
3. Carbonate minerals
 - Dissolution
 - Slower calcification (including shell fish)
4. Impact on organisms

Consequences Carbonate System



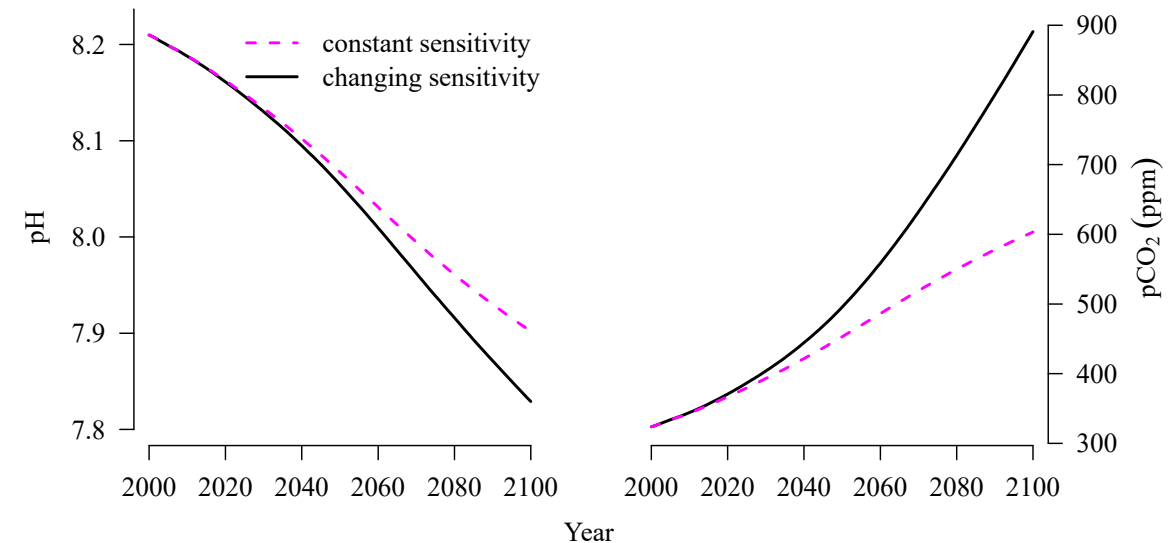
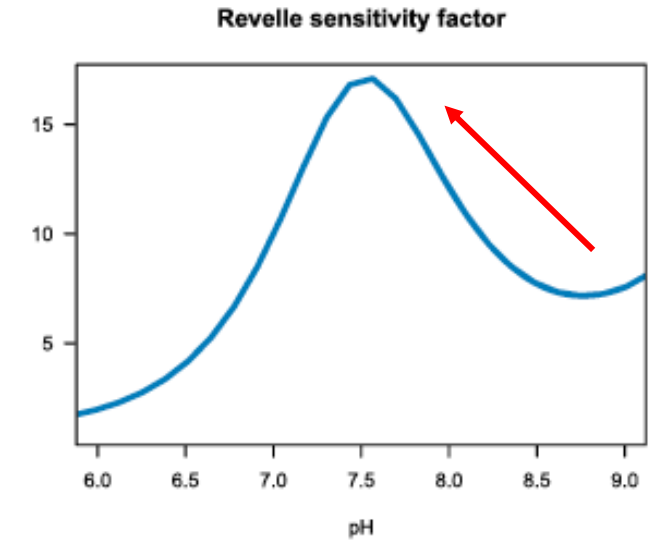
Sound in future ocean



- Attenuation of sound depends on pH (MgSO_4 , borate complexes)
- 0.3 pH decrease, implies 40 % decrease in attenuation low frequencies
- Consequences for underwater noise and life

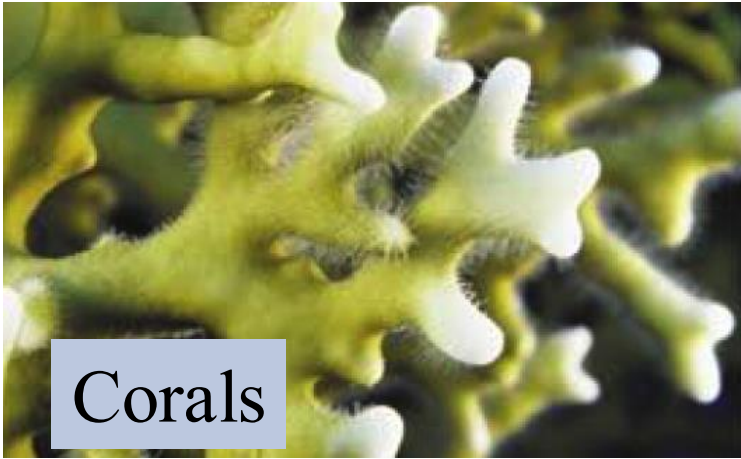
Positive feedback

- CO₂ penetration into ocean, lowers pH
- Lowers buffering capacity (makes ocean more sensitive)
- pH declines faster
- Ocean absorbs (relatively) less CO₂
- More CO₂ remains in atmosphere (900 vs. 600 in 2100)

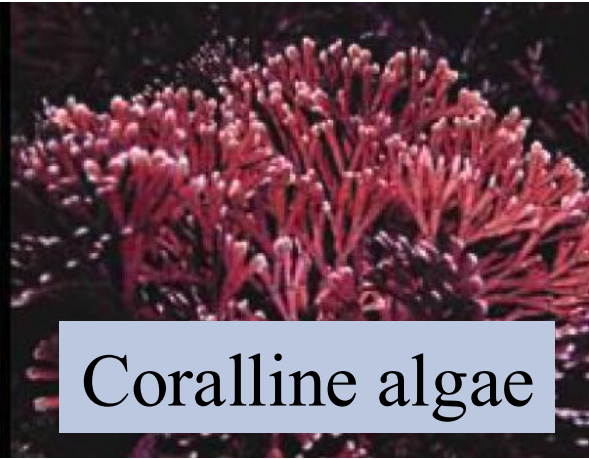


Middelburg et al. (2020, EOS)

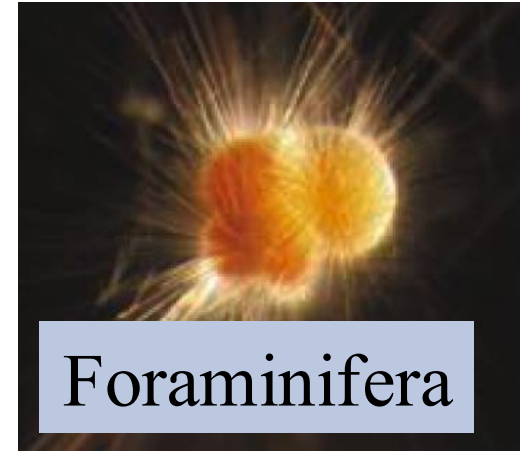
Marine calcifiers at risk



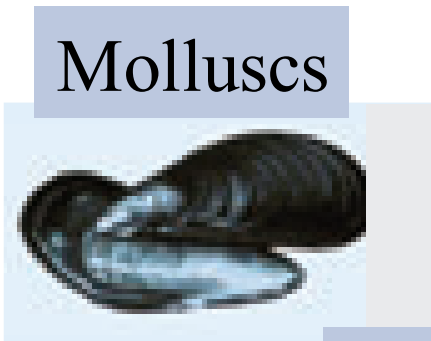
Corals



Coralline algae



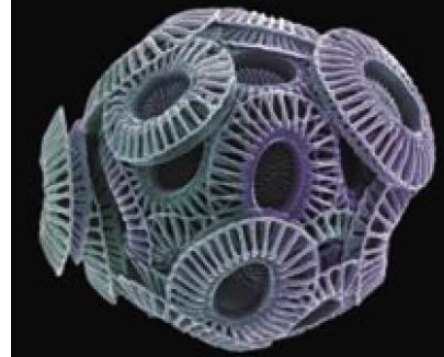
Foraminifera



Molluscs



Echinoderms



Coccolithophorids

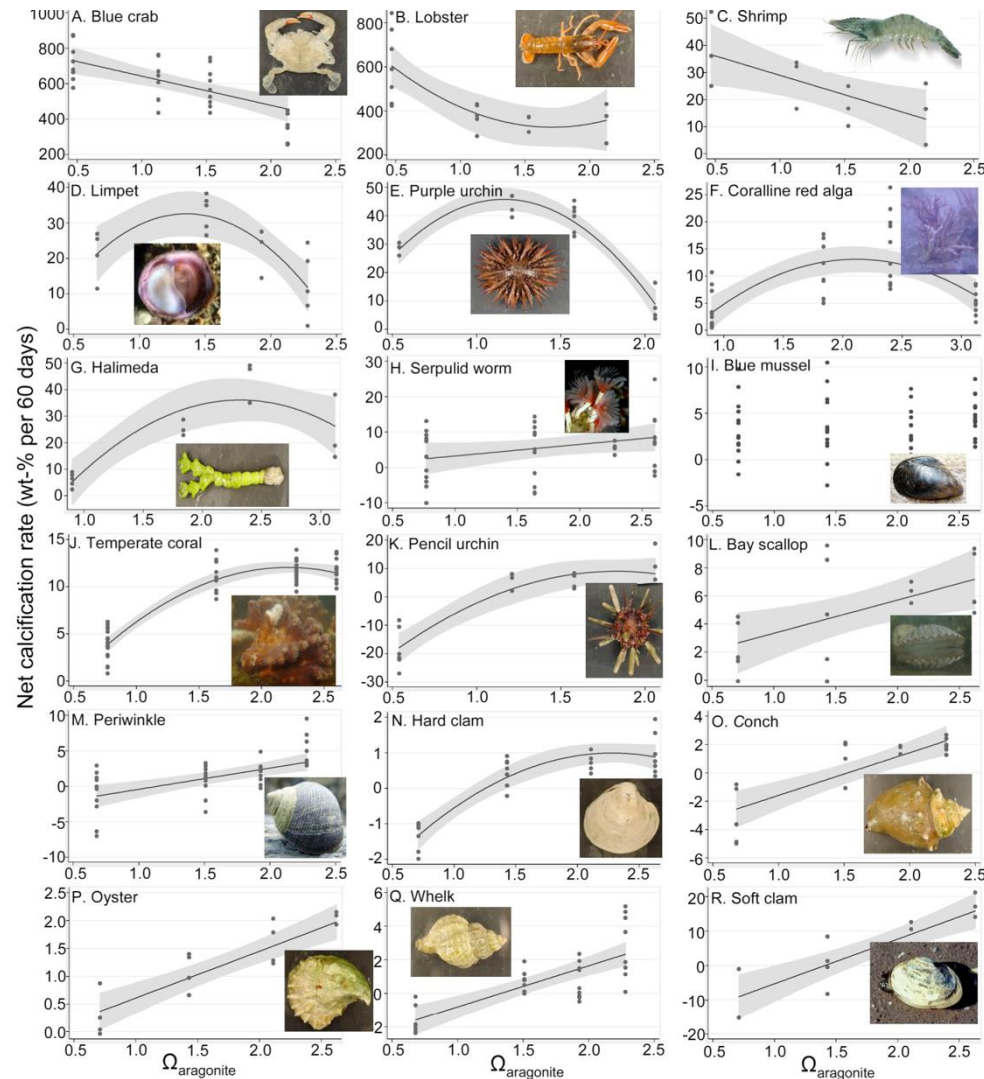


Pteropods

**Calcification (the formation of CaCO_3 from Ca^{2+} and CO_3^{2-})
is strongly affected by a decline in pH**

Calcification and ocean acidification

Calcification

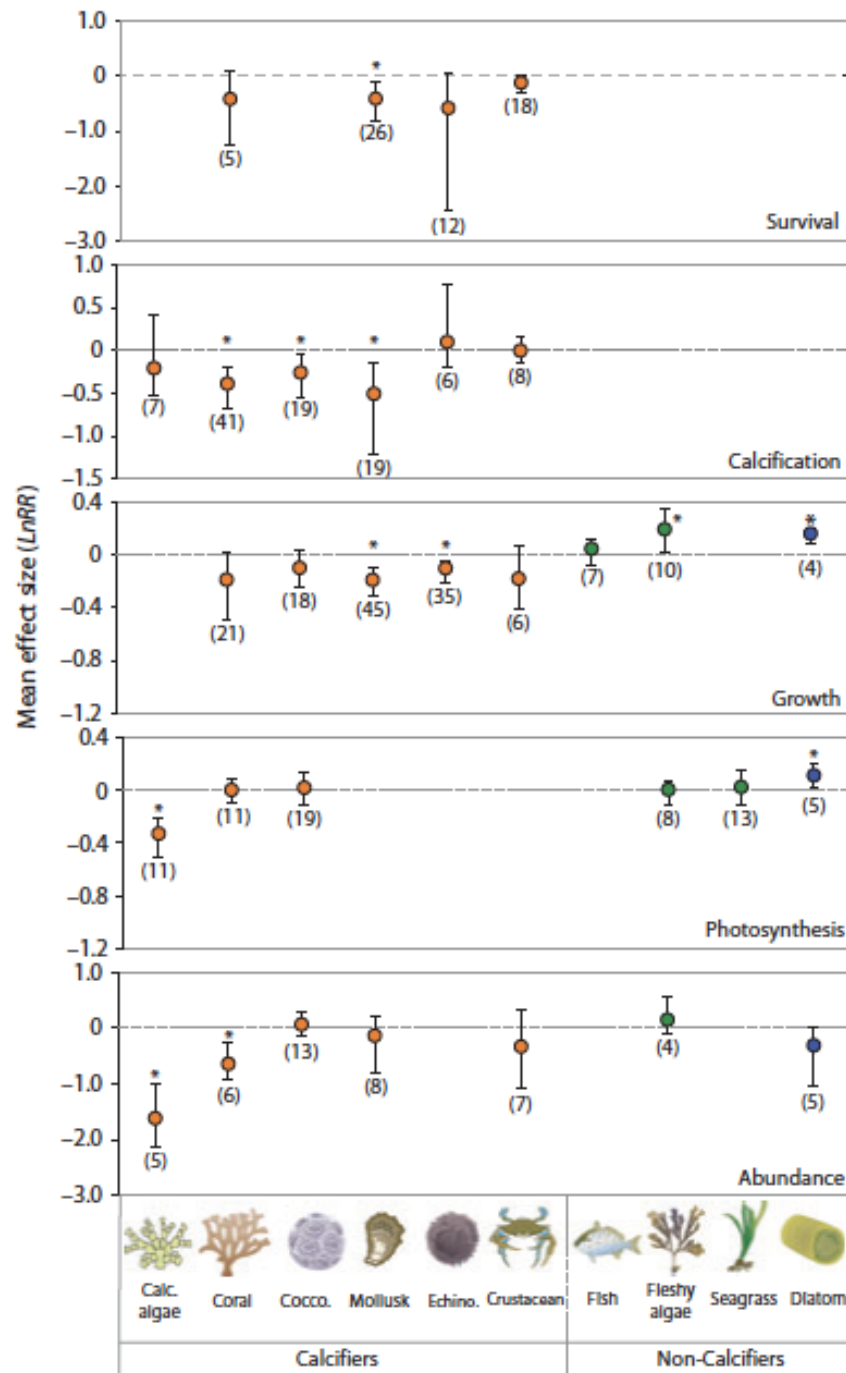


Saturation

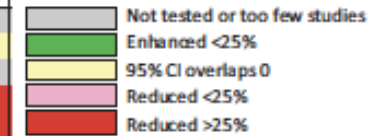
Ries et al., 2009

Biological Consequences

- **Increase H^+**
 - Availability of trace elements for organisms
 - Noise attenuation
- **Decrease CO_3^{2-}**
 - Calcification
- **Increase CO_2 & HCO_3^-**
 - Substrate for primary producers (algae, seagrass, etc.)



Taxa	Response	Mean Effect
 Calcifying algae	Survival	
	Calcification	
	Growth	
	Photosynthesis	-28%
	Abundance	-80%
 Corals	Survival	
	Calcification	-32%
	Growth	
	Photosynthesis	
	Abundance	-47%
 Coccolithophores	Survival	
	Calcification	-23%
	Growth	
	Photosynthesis	
	Abundance	
 Mollusks	Survival	-34%
	Calcification	-40%
	Growth	-17%
	Development	-25%
	Abundance	



Calcification:
Decrease in
 CO_3^{2-} and
saturation degree

 Fleshy algae	Survival	
	Calcification	
	Growth	+22%
	Photosynthesis	
	Abundance	
 Seagrasses	Survival	
	Calcification	
	Growth	
	Photosynthesis	
	Abundance	
 Diatoms	Survival	
	Calcification	
	Growth	+17%
	Photosynthesis	+12%
	Abundance	

Photosynthesis &
Growth:
More CO_2 substrate
available for primary
producers

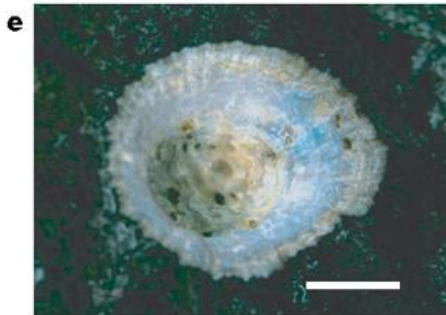
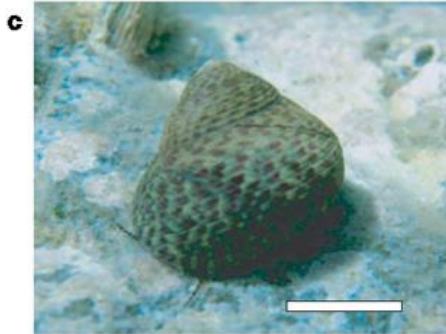
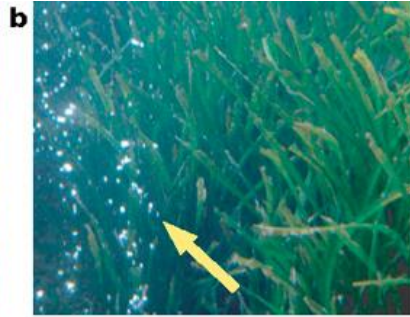
Short-term vs. long-term

- Results shown were based on one single organism, short duration experiments
 - Adaptation
 - Evolution
- Complementary information:
 - Geological record
 - Natural CO₂ seeps
 - Mesocosm experiments (whole ecosystem)

Natural CO₂ seeps

Normal, pH = 8.2

pH = 7.3-7.6



- Seagrass loses calcite precipitates
- No juvenile snails
- Corrosion snail in CO₂ rich water
- 30% less biodiversity

Natural CO₂ seeps



pH

8.1

- Loss of biodiversity
 - Species
 - Habitats



7.9

- 30% less calcification
- Seagrass profits



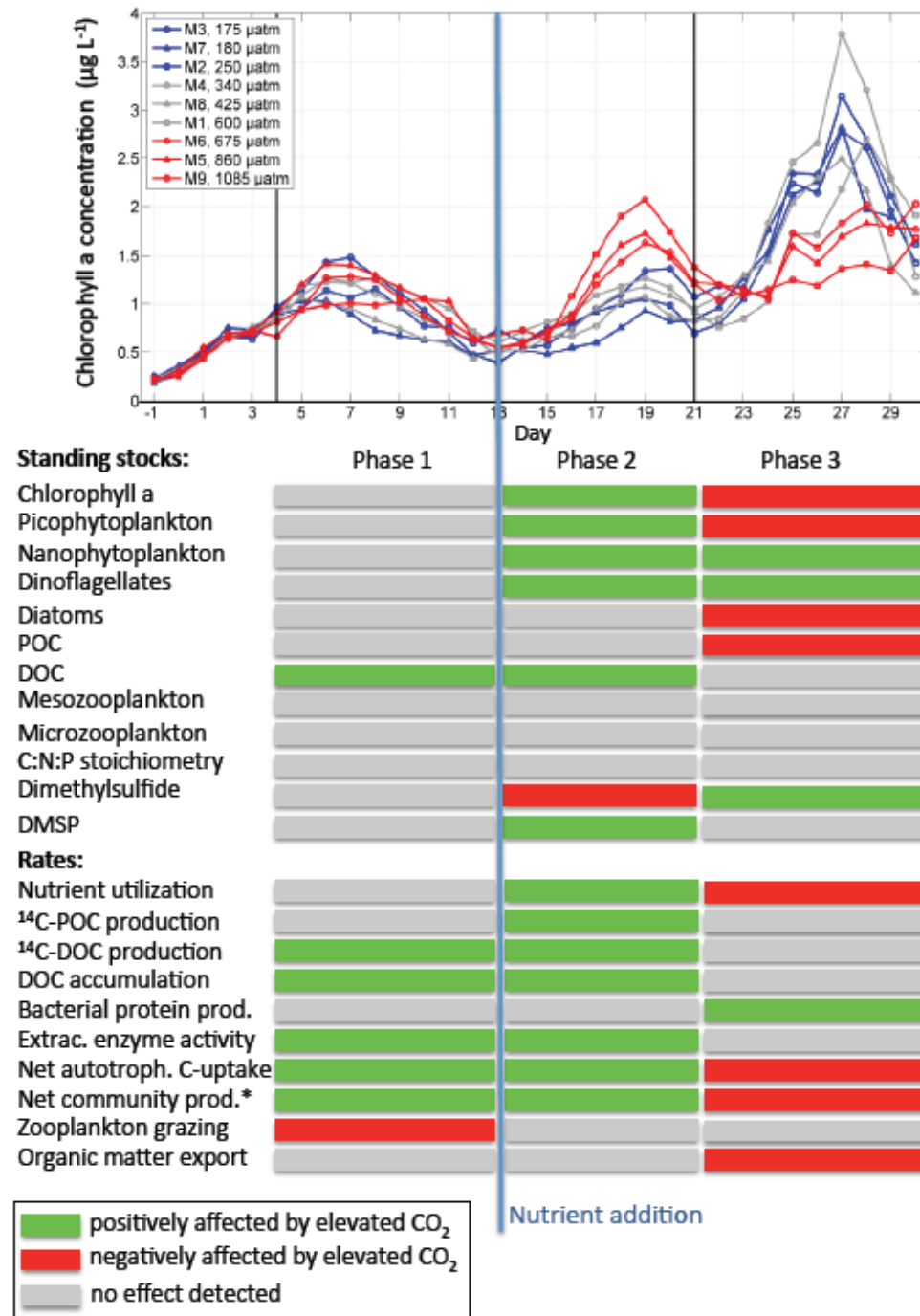
<7.7

Svalbard Experiment

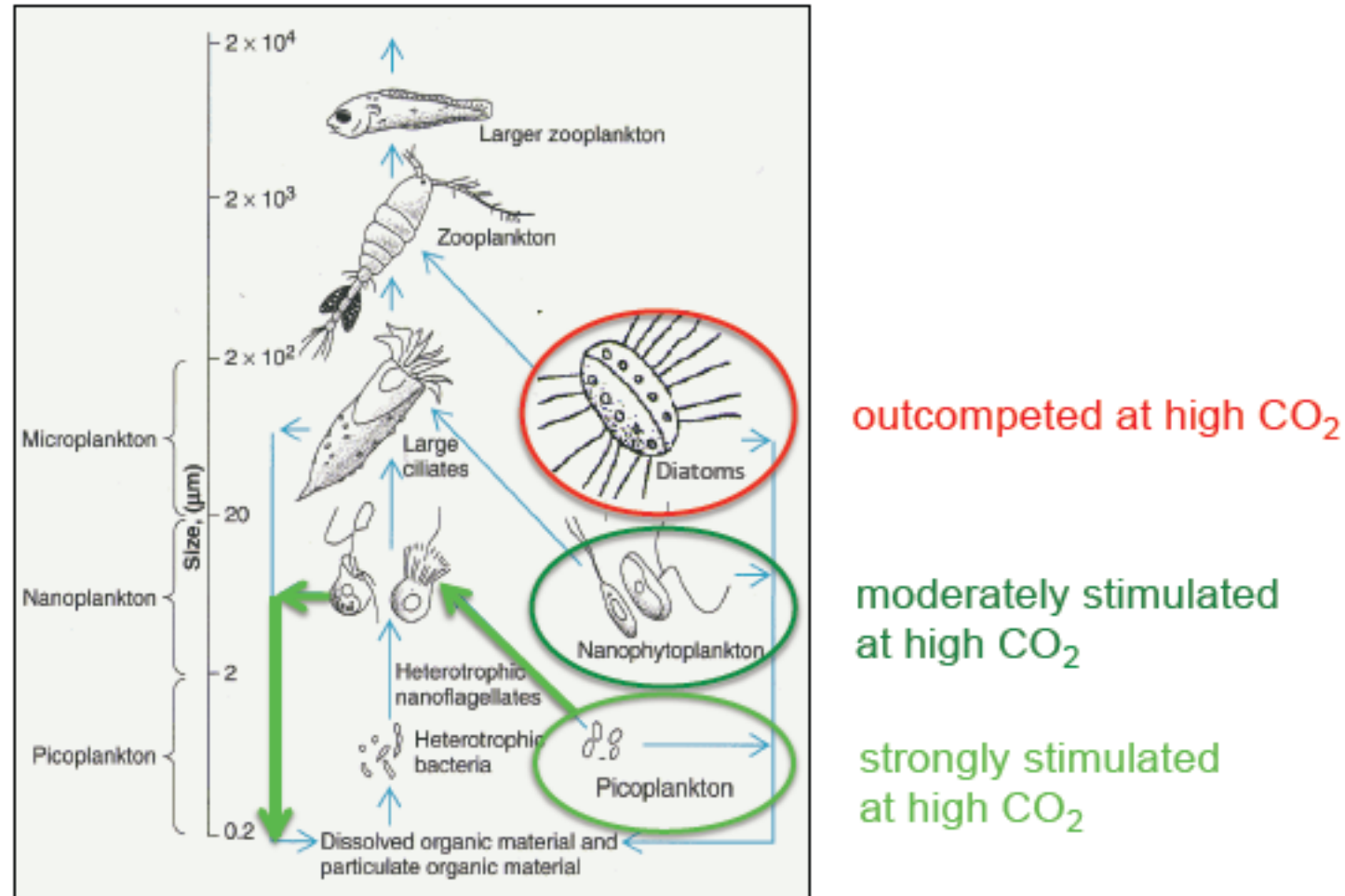




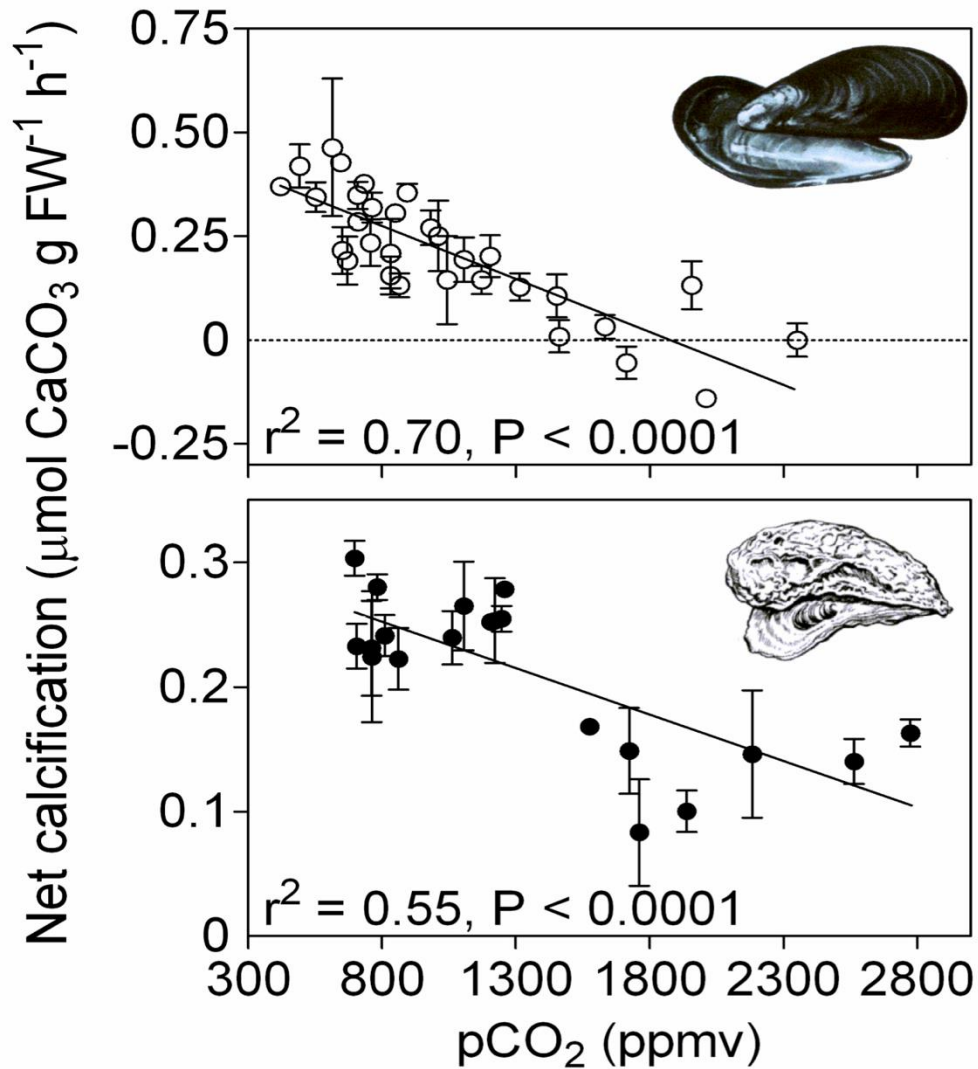
Svalbard Experiment



OA impact on food web



Economical consequences?

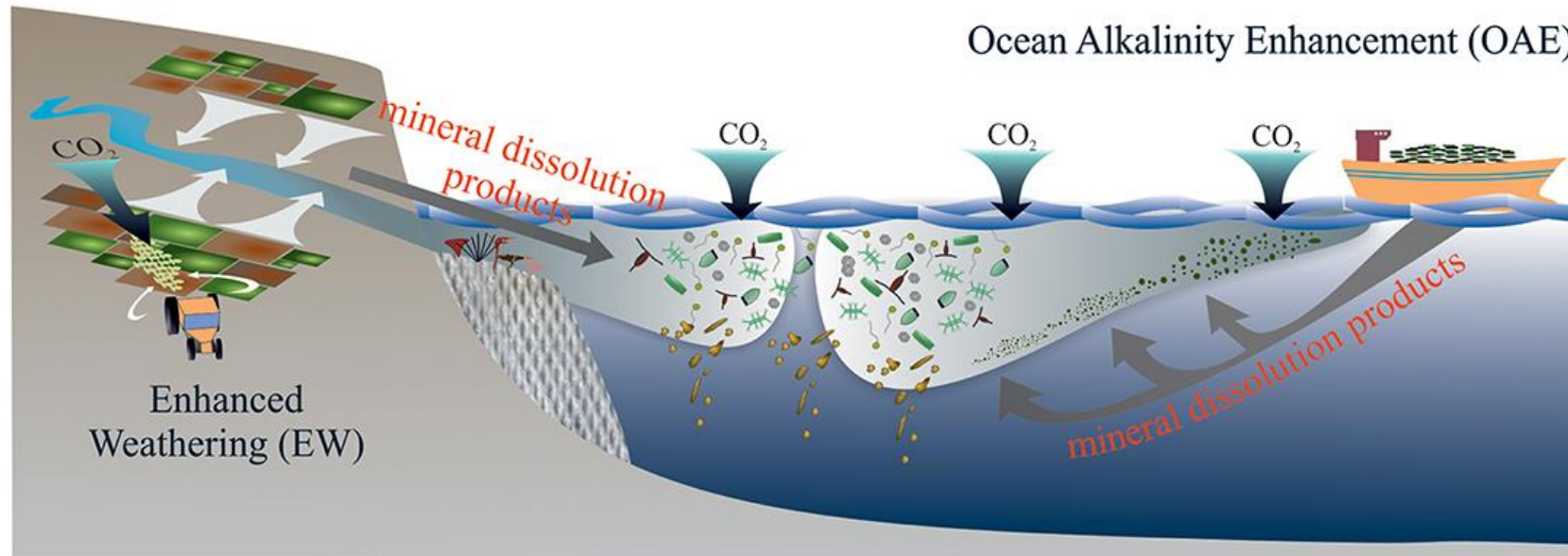


At 740 ppmv CO_2 :

- 25 % less mussel shells
 - 10 % less oyster shell
- 10 billion € revenues worldwide

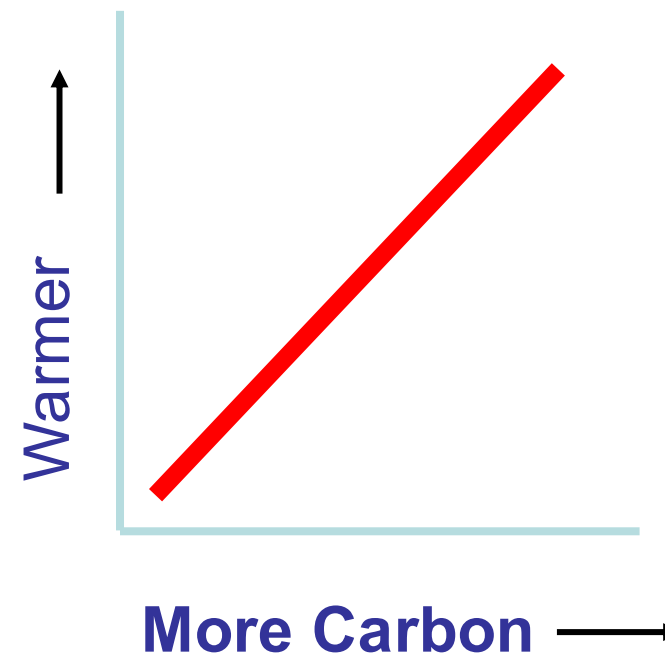
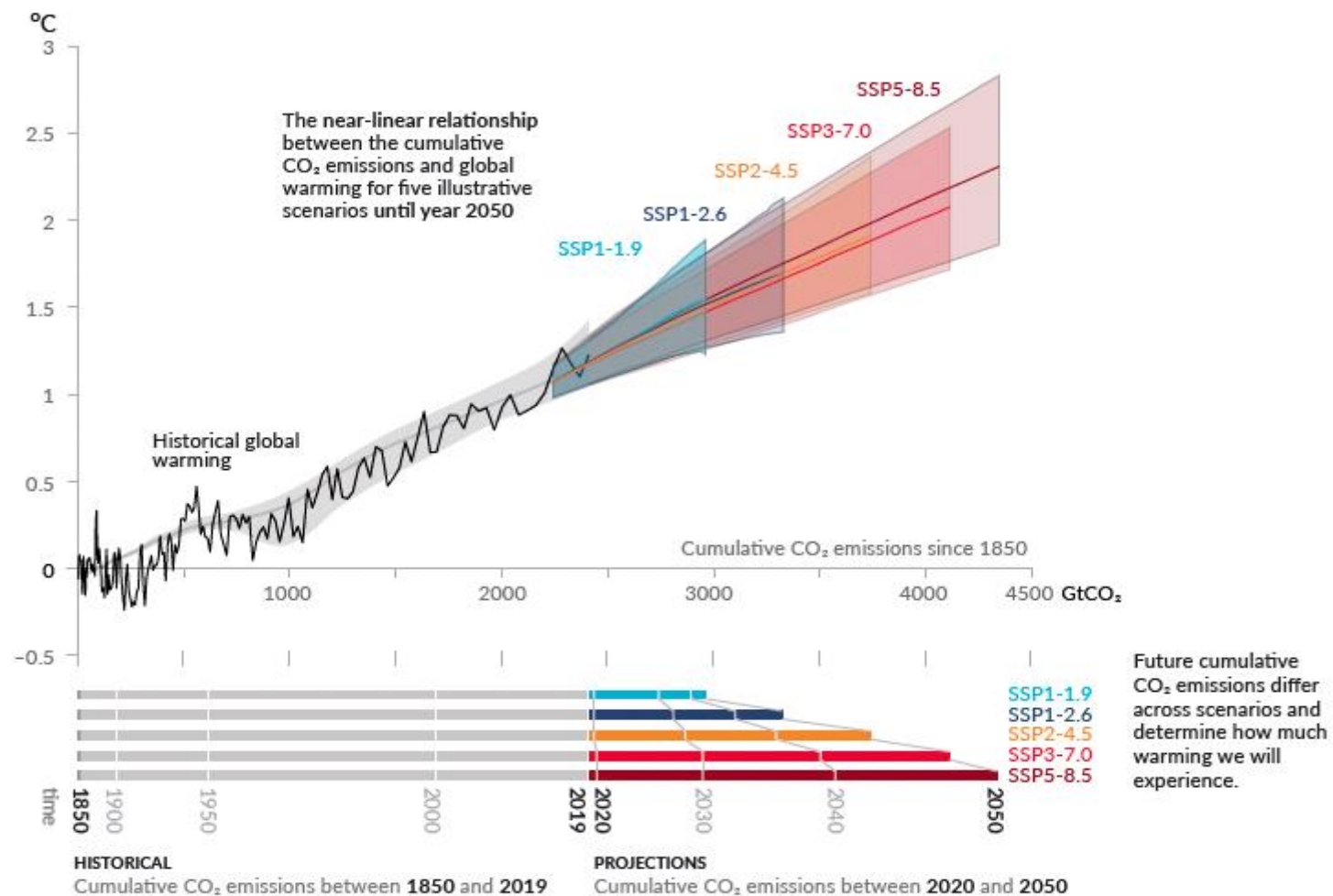
Can we do something about it?

Yes we can add olivine or lime.



Every tonne of CO₂ emissions adds to global warming

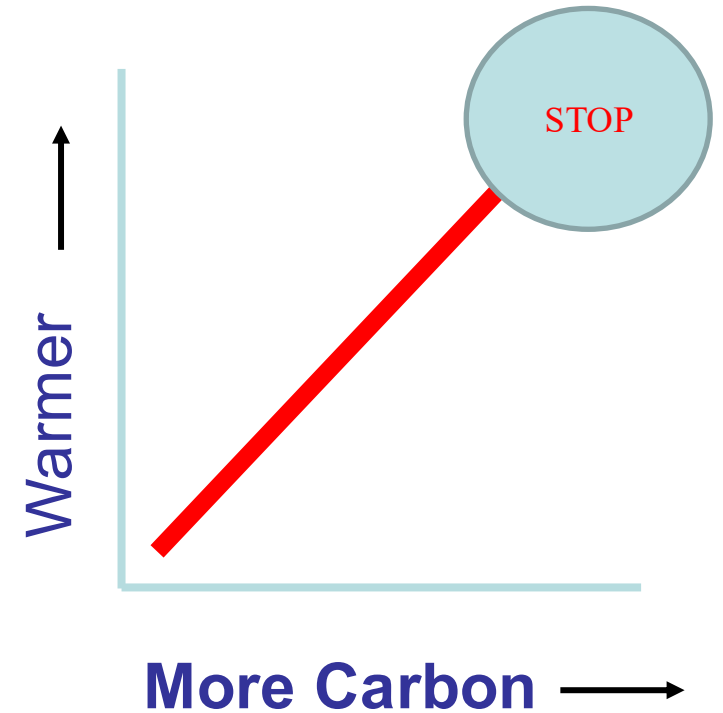
Global surface temperature increase since 1850–1900 (°C) as a function of cumulative CO₂ emissions (GtCO₂)



What can we do?

Less or no emission (mitigation, carbon neutral, deep carbonisation, energy transition,)

- Technical limits (airplanes, cement production....)
- Geopolitics (Global South, drill, drill, baby)

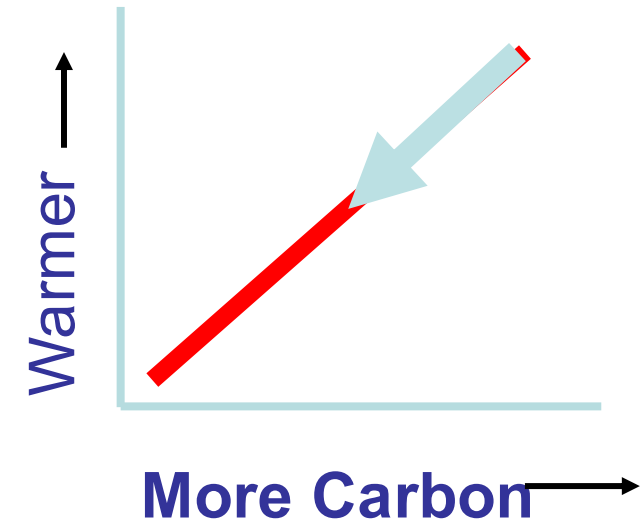
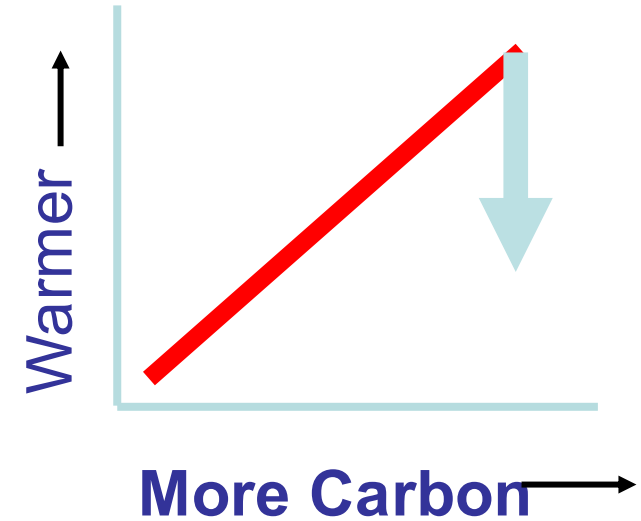


What can we do?

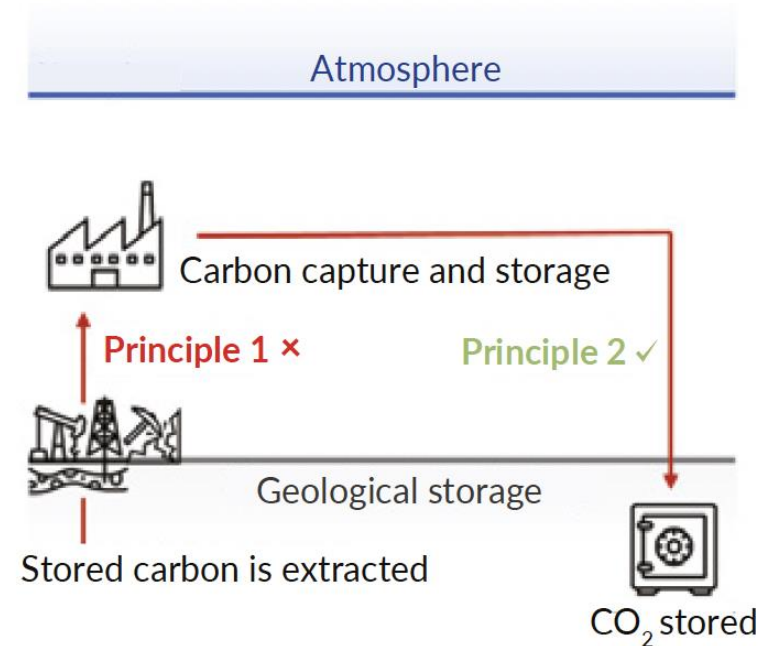
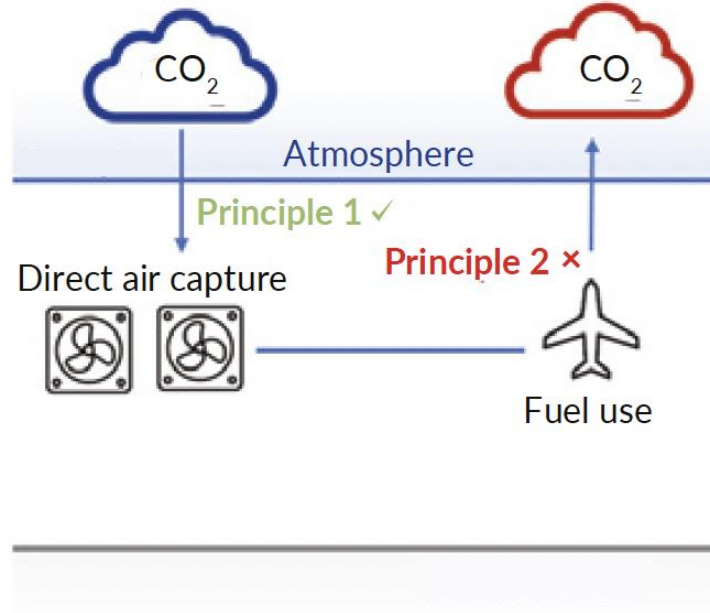
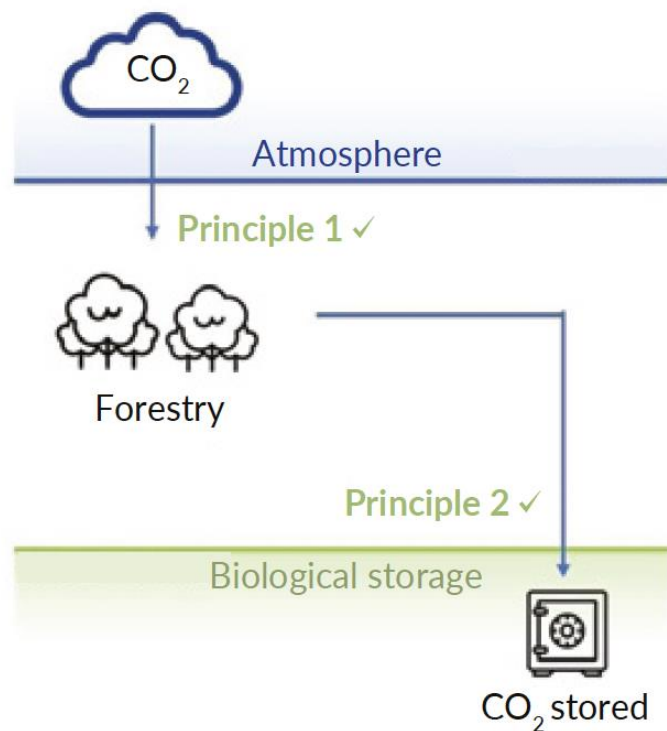
- **Solar Radiation Management**
to lower Temperature



- **Carbon dioxide Removal/**
Negative Emission/
active carbon dioxide removal

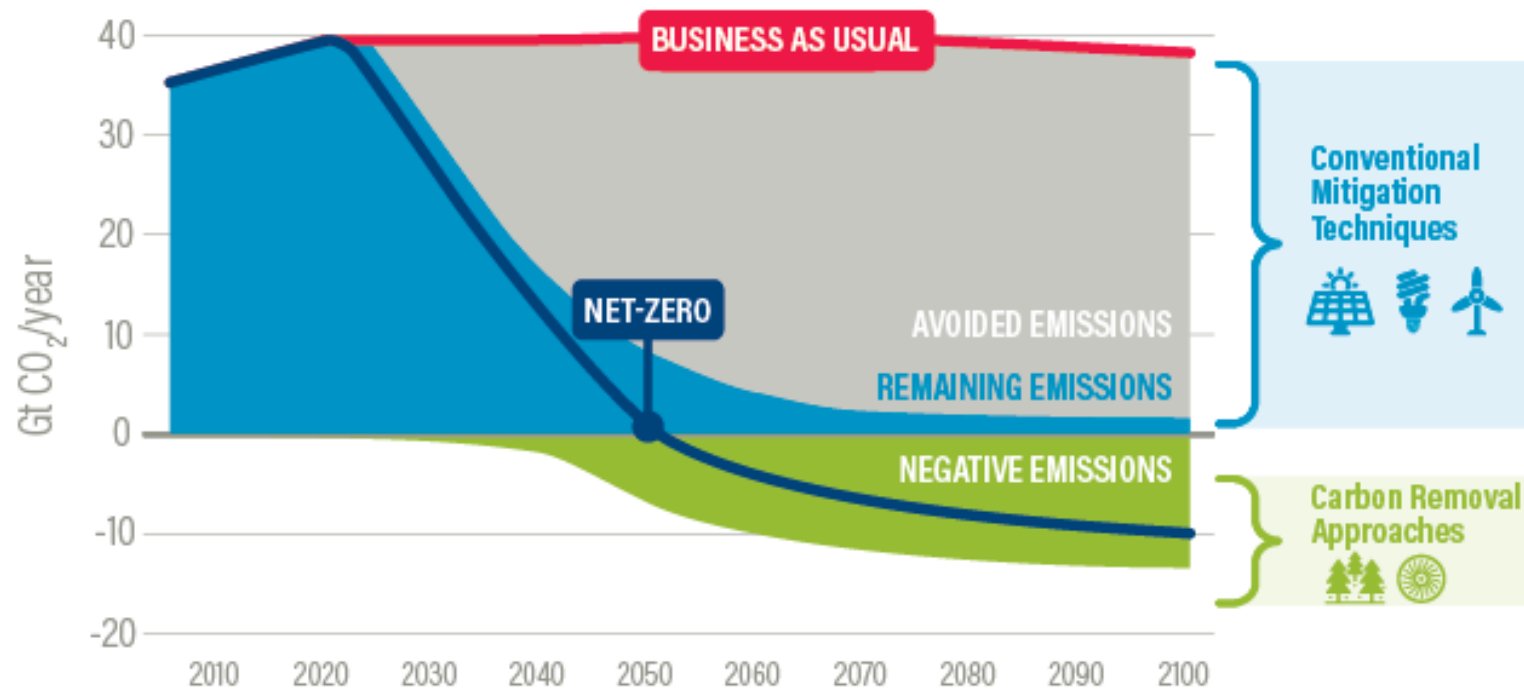


Carbon Dioxide Removal vs Carbon Neutral: Capturing and Storage required

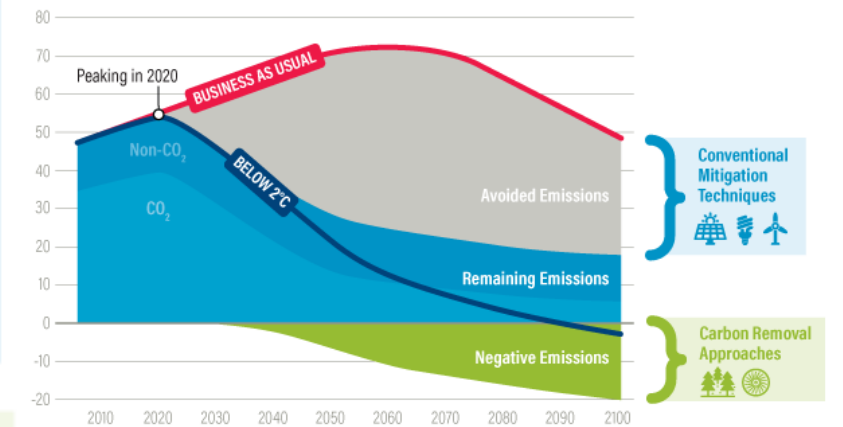


CDR is already in the 1.5 and 2 degree scenario's

Staying Below 1.5 Degrees of Global Warming



Staying Below 2 Degrees of Global Warming



This is a notional scenario consistent with an at least 66 percent chance of limiting global warming to below 2°C. Some residual gross greenhouse gas emissions (both CO₂ and non-CO₂) will remain at the end of the century even with ambitious climate action because they are intractable or costly to remove entirely. Once negative emissions exceed those that remain net zero emissions is reached.

Adapted from a visual in The UNEP Gap Report 2017 (Figure 72)

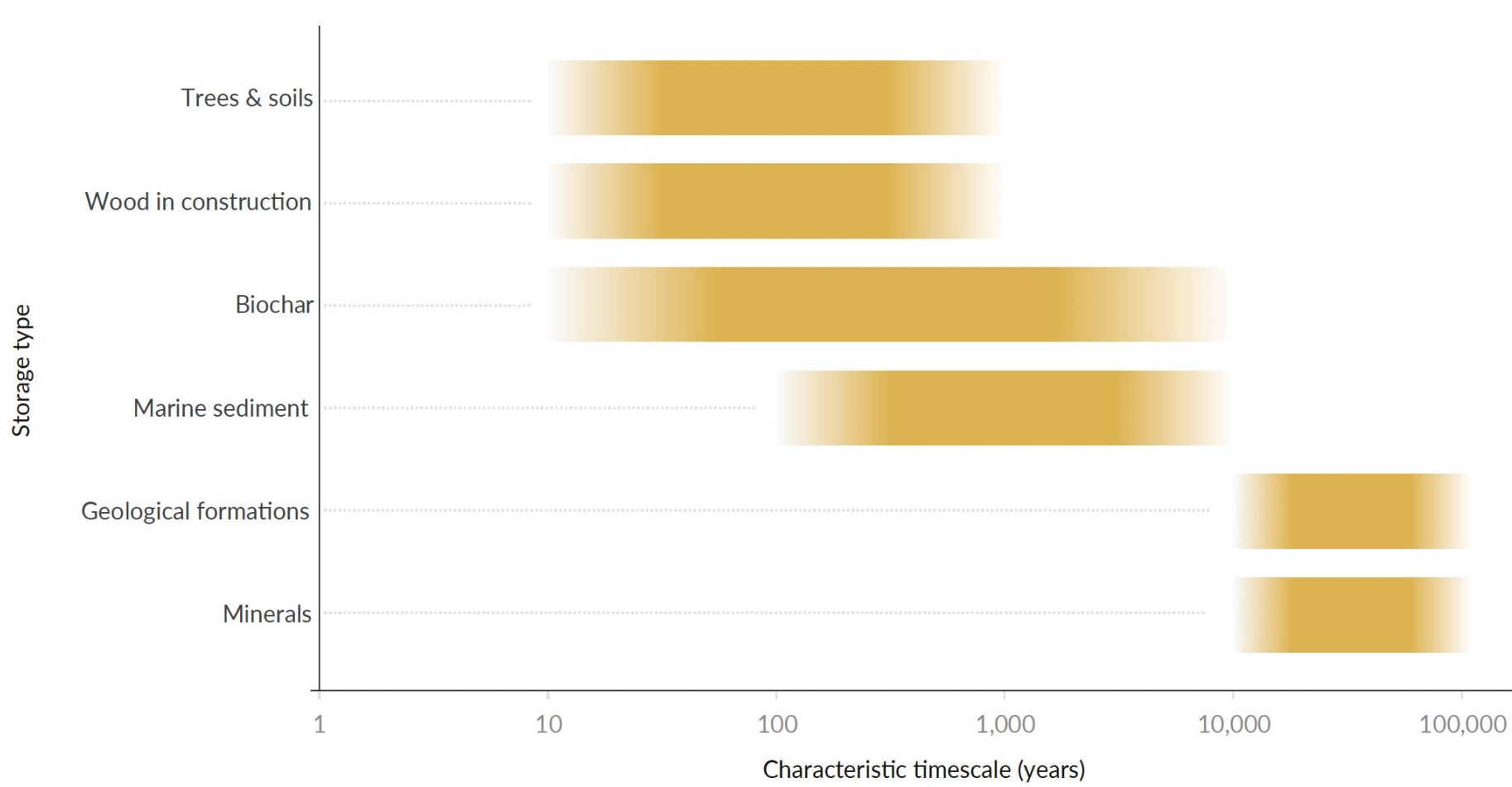
 WORLD RESOURCES INSTITUTE

Source: Adapted from IPCC 2018.

Carbon Dioxide Removal: How?

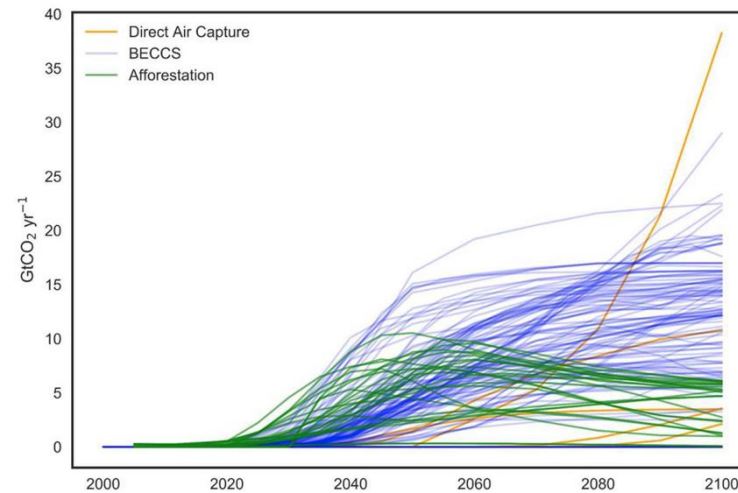
- **Concentrate** (how to get 420 molecules out of one million)
- **Convert/transform and transport**
 - in organic matter (reforestation, blue carbon)
 - in minerals (accelerated weathering)
 - In liquid (base in ocean, liquid carbon dioxide)
- **Store** (for at least a couple of hundreds of years)
 - Limestone
 - Dissolved in (deep) ocean
 - Physical isolation (subsurface)

How Long is the Carbon Stored?



IPCC and policy?

- BECCS (biomass combustion capture and storage)
- (re)forestation
- DAC (direct air stripping) with storage
- Storage subsurface (on land and in the sea)

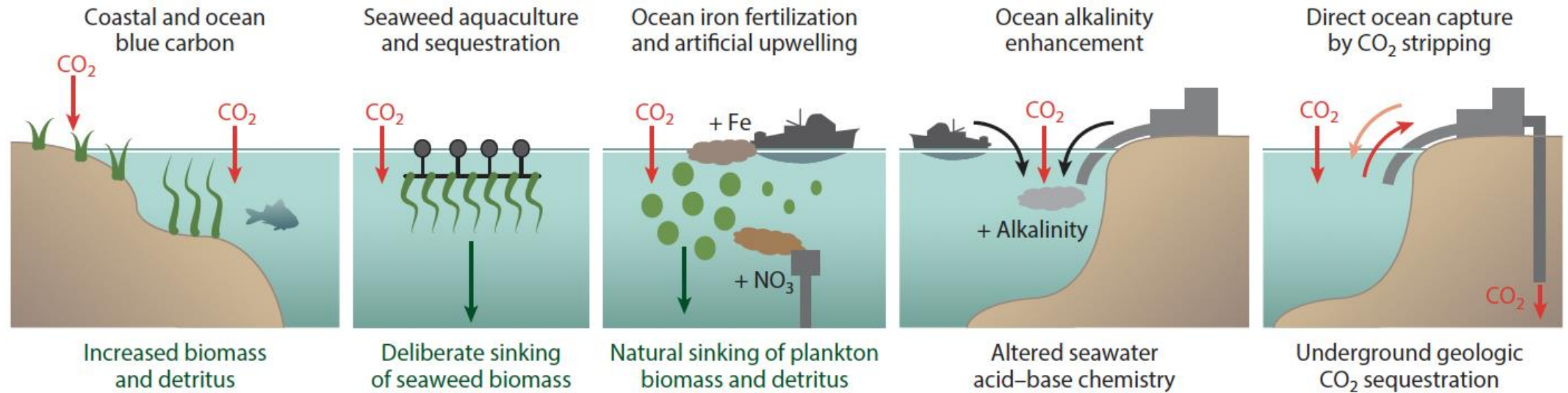


Fuhrman et al., 2019



Ocean Solutions?

a Marine CDR approaches



Misconceptions

- Biology plays a major role in Anthropogenic Carbon uptake by ocean
- Restoring Coastal Blue Carbon ecosystems is an effective climate solution
- Bivalve farming is a potential carbon sink
- Whales play a substantial role in long-term carbon storage

Misconceptions of the marine biological carbon pump in a changing climate: Thinking outside the “export” box

Ivy Frenger , Angela Landolfi , Karin Kvale, Christopher J. Somes, Andreas Oschlies, Wanxuan Yao, Wolfgang Koeve

First published: 08 January 2024 | <https://doi.org/10.1111/gcb.17124> | [VIEW METRICS](#)

Carbon Removal Using Coastal Blue Carbon Ecosystems Is Uncertain and Unreliable, With Questionable Climatic Cost-Effectiveness



Phillip Williamson ^{1*}



Jean-Pierre Gattuso ^{2,3,4}

Bivalve Farming Is Not a CO₂ Sink: From Myth to Legend, Where Is Science?

Fabrice Pernet , Sam Dupont, Jean-Pierre Gattuso, Marc Metian, Frédéric Gazeau 

First published: 17 July 2025 | <https://doi.org/10.1111/raq.70067> | [VIEW METRICS](#)

Volume 10 - 2023 | <https://doi.org/10.3389/fmars.2023.1117409>

Do whales really increase the oceanic removal of atmospheric carbon?



Jan-Olaf Meynecke ^{1,2,3,4*}



Saumik Samanta ⁵